The Curse of Connectivity: Evidence from Village Resettlement in Indonesia

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Abstract

Does connecting a village to markets make it richer? Indonesia's Transmigration program created new villages that differed in their initial access to markets. Otherwise similar villages that began with better market access are poorer today: a one standard deviation increase in initial market access lowers GDP per capita decades later by roughly 4 percent. Better market access moves residents out of agriculture and into local non-farm work. I argue that the move lowers income because agriculture has a scale externality: farmers are more productive when more of their neighbors farm. Using separate variation in each village's initial settlement size, I find evidence of a positive agricultural scale externality. A spatial general equilibrium model of Indonesia reproduces the negative effect of market access only when it includes the externality. Ignoring the externality overstates the aggregate gains from improving the road network by 17 percent and from raising rural agricultural productivity by 64 percent. For the same fiscal cost as the road improvements, a subsidy to agricultural earnings delivers roughly twice the rural gains.

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1 Introduction

Does access to markets make a place richer? A long tradition in economic geography argues that it does (Krugman, 1991; Redding and Venables, 2004). Whether greater market access raises long-run development, however, is theoretically ambiguous. Greater market access generates the standard gains from trade and feeds the agglomeration forces that drive growth. But it can also hinder development when it pulls labor away from activities that carry productivity growth (Matsuyama, 1992). In the literature on transport infrastructure, the worry has been that connected peripheral locations lose economic activity to the core (Faber, 2014; Baum-Snow et al., 2020). Which force dominates over the long run is ultimately an empirical question.

Answering this question is difficult because where places are located, and how connected they are, is endogenous. People and productive resources accumulate where access is high, and settlements and cities often form on productive land. Comparing places with different levels of market access confounds access with composition and fundamentals. What one would like instead is a setting in which otherwise comparable places were allocated different levels of market access.

Indonesia's Transmigration program comes close to such a setting. Between 1952 and 1995, the program resettled over 2 million voluntary migrants into over 1,000 newly constructed villages, placed with little regard for the productive potential of their sites. Officials, under pressure to meet resettlement targets, sited villages wherever land was available, rarely weighing connectivity or the potential of a location. Aside from location, the villages started out nearly identical; settlers had no control over which new village they were placed in, received uniform 2-hectare plots by lottery upon arrival, and were each given the same initial endowment of agricultural inputs (tools, seeds, and fertilizer). Previous work finds no sorting of migrant characteristics across destination villages (Bazzi et al., 2016, 2019), and the migrants themselves were overwhelmingly asset-poor (Kebschull, 1986). Here, market access varied at settlement while the people did not. The program removes much of the confounding that comes from people and institutions accumulating where access is high. I use it to show that villages that began with greater market access are *poorer* today than otherwise similar villages.

I estimate that moving from the 25th to the 75th percentile of initial market access lowers long-run village GDP per capita by about 5.5 percent, and that one standard deviation of initial access lowers it by about 4 percent. These differences are measured 17 to 67 years after settlement. The results are robust to alternative satellite-derived GDP estimates. Resident wages and a novel video-based measure of village material welfare in a small subsample point in the same direction, although the wage estimates are imprecise. Additionally, villages that started more connected moved more labor out of agriculture. I argue that this shift out of agriculture is why they are poorer. Agriculture in these villages is subject to a scale externality, so farm productivity declines as workers leave agriculture.

The externality operates across farms rather than within them. The productivity of all farms

rises with the scale of the local agricultural sector, through channels such as knowledge spillovers across farmers, shared capital and equipment rental markets, and the cooperative management of irrigation (Foster and Rosenzweig, 1995; Conley and Udry, 2010; Bassi et al., 2022; Caunedo et al., 2026; Ostrom and Gardner, 1993). A worker who leaves agriculture therefore lowers the productivity of those who remain, a loss they do not internalize. Consistent with this mechanism, more connected villages use fewer tractors per agricultural worker. Because agriculture is the dominant sector in these village economies, the decline in agricultural productivity lowers village income. The workers who leave move mostly into small-scale services and retail sold to nearby customers, which show no sign of scale economies of their own.

I use a second source of program variation to measure the externality directly. Villages were settled with different initial populations, which I show is unrelated to the characteristics of the sites. Because initial settlement size predicts the size of a village's agricultural when I observe agricultural production directly decades later, comparing agricultural output across villages settled at different scales allows me to identify how output responds to the size of the workforce. I find that output rises with the workforce by much more than the private return to labor would predict, and the gap between the social and private returns is the externality I then embed into my quantitative model.

To formalize the mechanism and quantify its implications, I develop a spatial general equilibrium model of the Indonesian economy. In the model, connectivity pulls workers out of agriculture because rural services are more costly to trade than agricultural goods, so villages in more connected locations experience higher relative demand for their non-agricultural output. This increases the private return to non-farm work and pulls workers out of the agricultural sector. In a simplified version of the model that can be solved in closed form, I show that market access harms a village when the externality large enough that the income lost from this reallocation outweighs the standard gains from trade. The externality point estimates exceed this threshold by a wide margin, while the most conservative readings of the values of the measure fall below the threshold. The threshold value depends on the rest of the model: how readily workers move between sectors, how costly services and food are to trade, and how much of village income comes from agriculture. Under some plausible alternatives, such as workers who are more alike in their abilities, it is substantially lower.

The simplified model makes a statement that is more general than the transmigration setting. Connectivity lowers income when the reallocation it causes moves labor out of the sector in which the social return to labor exceeds the private return, and by enough to outweigh the gains from trade. Both ingredients are needed: without the externality every channel is a gain from proximity, and unless services are effectively harder to trade than food, connectivity does not pull labor out of agriculture in the first place. Structural transformation is therefore not sufficient for growth; what matters is where scale economies reside relative to where connectivity pulls labor, not which sector pays more privately.

In the full model, every location in Indonesia carries the same two-sector economy. The externality is based on the settlement-size estimates, and trade costs and spending shares are disciplined by survey data. The transmigration villages enter with common fundamentals and differ only in where they sit on the map, so the relationship between market access and income among them is a prediction of the model rather than a target. The model reproduces about 40 percent of the negative effect in the full sample and 80 percent excluding the most isolated 5 percent of villages. Removing the externality reverses the prediction. An externality-free model can fit the observed incomes only by assuming connected villages had systematically worse agricultural fundamentals, which the observed site characteristics do not support.

Other explanations are possible, but the evidence does not support the conventional ones. More connected villages experience *more* reallocation of labor out of agriculture and have more firms, weighing against explanations based on agricultural specialization or competition from the core (Matsuyama, 1992; Faber, 2014). I find no evidence that out-migration or selective migration explains the results. Food prices do not fall to offset the lower incomes; if anything, they are higher. Connectivity does not improve non-road public goods, so lower output is unlikely to be compensated by better measured amenities.

I then use the model to demonstrate that the externality changes who benefits from connectivity. A road targeted to an isolated village can make that village poorer by drawing workers out of agriculture. Consistent with the simplified model, the loss in income is largest for villages with substantial employment in both sectors; almost entirely agricultural villages can still gain.

The externality also matters for economy-wide policies, such as improving the whole road network or raising rural agricultural productivity, which are large enough to move the national agricultural employment share. The baseline model has homothetic demand, so the composition of national spending does not respond to these shocks and even large shocks cannot generate *aggregate* structural transformation. I therefore extend the model to incorporate the non-homothetic demand system of Boppart (2014) and Fan et al. (2023), estimating its parameters in household expenditure data. Income growth shifts spending away from food, pulling labor out of agriculture and lowering productivity through the externality. A 10 percent cut in travel times looks 17 percent more valuable in the externality-free economy, and a 5 percent rise in rural agricultural productivity looks 64 percent more valuable.

The bias runs the other way for policies that keep labor in agriculture. An agricultural earnings subsidy at the road program's fiscal outlay raises national income by 0.20 percent and delivers roughly twice the rural gains of roads, while the same subsidy in the externality-free economy is a small distortionary loss. These exercises show that agricultural scale economies change the ranking of policy instruments and the magnitude of aggregate welfare effects.

This paper relates to the literature on the effects of market access in historical developed countries (Donaldson and Hornbeck, 2016; Hornbeck and Rotemberg, 2024; Redding and Sturm, 2008) and developing countries (Morten and Oliveira, 2024; Gertler et al., 2024; Gebresilasse, 2023;

Kaboski et al., 2024). The central finding of this work is that greater market access raises local economic performance; to my knowledge, this paper is the first to find that it *lowers* long-run real income per capita. The canonical results come from economies where the sectors that gain from connectivity, such as manufacturing, drive growth, while in the villages I study the main alternative to farming is low-scale services and retail, which are persistently small, informal, and unproductive (La Porta and Shleifer, 2014; Lagakos, 2016). Much of this literature treats population or land prices as a sufficient statistic for welfare, but measuring income directly, I find that more connected villages are poorer despite slightly higher populations and land prices. Faber (2014) finds that connecting peripheral Chinese counties lowered GDP growth through competition from the core; losing production to the core still allows gains from cheaper imports, whereas losing agricultural scale is a pure productivity loss. In India, a new road moves workers out of agriculture with no detectable change in consumption (Asher and Novosad, 2020). My paper supplies a mechanism for such reallocation without income growth: over the long run, the workers who leave agriculture lower the productivity of those who remain.

My estimates are also inconsistent with surplus labor in agriculture. In the dual economy tradition of Lewis (1954) and Ranis and Fei (1961), labor can be pulled out of agriculture at little or no cost to output, while Schultz (1964) argued that this is not the case, and recent experimental evidence finds labor rationing on short-run margins (Breza et al., 2021). My settlement-size instrument provides a long-run test at the village scale, and the estimated elasticity of agricultural output with respect to the agricultural workforce is far from the surplus labor benchmark of zero. Two points reconcile this finding with the modern evidence. First, my elasticity is a long-run object, and the forces behind it would not appear in short-run experiments or within a few years of a new road. Second, the experimental evidence measures the private return to agricultural labor, while my estimates capture the social return.

To the literature documenting agricultural externalities in developing economies, my contribution is to show that these externalities matter for infrastructure and trade policy. Prior work finds spillovers in the adoption and use of new technologies (Foster and Rosenzweig, 1995; Conley and Udry, 2010; Bandiera and Rasul, 2006; BenYishay and Mobarak, 2019), gains from equipment rental markets and from collective use of capital (Caunedo et al., 2026; Brownstone, 2025; Bassi et al., 2022), and benefits from the cooperative management of irrigation (Ostrom and Gardner, 1993). These otherwise local externalities can reverse the local gains from market access, alter who benefits from infrastructure, and change the aggregate effects of structural transformation.

Sections 2–4 describe the program, data, and empirical strategy. Section 5 presents the effects of market access. Section 6 develops a general model framework that can explain the results, and Section 7 examines other explanations. Section 8 estimates the agricultural scale externality. Sections 9–11 fully calibrate a model to the data, compare it with the reduced-form effects I measure, and then perform policy counterfactuals. Section 12 concludes.

2 The Transmigration Program and the Outer Islands

The Outer Islands Indonesia's population has always been concentrated on Java and Bali, the "inner islands." During the period I study, the inner islands held roughly two thirds of the country's population on less than a tenth of its land. The Outer Islands, principally Sumatra, Kalimantan, Sulawesi, and the eastern provinces, were sparsely settled by comparison. Economic activity there was organized around a small number of provincial capitals and port cities, a plantation region of eastern Sumatra, and small pockets of oil, gas, and mining. Between these areas lay large areas with little permanent cultivation. The main roads and trails present around 1950, which I digitize in Section 3, changed little until the 1990s (Gertler et al., 2024), so a location's position relative to markets at the time of settlement was fixed for most of the period. The lowest unit of rural administration is the village (*desa*), a settlement of typically 2,000 to 4,000 people, which sits below the subdistrict (*kecamatan*) and the district (*kabupaten*). The village is my primary unit of analysis.

The program To relieve population pressure on the inner islands and to promote nation-building, the government of Indonesia relocated over 2 million people from Java and Bali to *newly created* villages across the Outer Islands between 1952 and 1995.[1] The program was organized around the five-year development plans of the period (*Repelita*), and the intensity of the program was heavily influenced by the government's ability to pay for it. Transmigration was financed out of central government revenue, and in the 1970s and early 1980s that revenue was dominated by oil and gas; the number of people moved each year tracks the world oil price closely (correlation 0.86), and the collapse of oil prices in 1986 was followed by a sharp contraction of the program (Appendix Figure A2). While activity was concentrated between 1979 and 1989, the average size of villages created each year does not show any trend (correlation −0.02). Villages were created in every year of the period, and I analyze all of them rather than only the 1979 to 1988 cohorts studied in previous work.

As described in numerous sources such as Bazzi et al. (2016), Hardjono (1988), and Kebschull (1986), the transmigration program lacked an organized structure. Provincial governments, which had been delegated the final decision on placement, submitted whatever land was available, and unrealistic resettlement targets forced officials to resettle people as quickly as possible, often without consideration for the location of new villages (Hardjono, 1988). Land suitability maps were abandoned for nearly the entire program, and the ones that existed were unreliable (Hardjono, 1988; Whitten, 1987). What matters for this paper is that a site's position relative to the towns, ports, and roads of the Outer Islands was not what officials were actively considering; they were placing villages where land could be found and settled quickly. Appendix A.1 collects contemporary

[1] The Transmigration program actually originated in 1905 under the Dutch colonial government. The program greatly expanded after Indonesia became independent, and it is still active today. However, the data I have cover between 1952 and 1995.

accounts of the process from 1960 to 1990.

How a settlement was built Each settlement was built from the ground up on cleared land. A settlement unit typically housed a few hundred families, laid out around a village center with a school, a health clinic, and a place of worship; each was a village in its own right, administered by the program until handover to the regional administration, on average about 5 years after settlement (World Bank, 1988; Bazzi et al., 2016). Every household received the same allocation: a house and house lot, a 2-hectare field plot often assigned by lottery, and the same package of tools, seeds, and fertilizer, with food support for the first year (World Bank, 1988; Bazzi et al., 2016). Most settlements were food-crop settlements, in which settlers grew upland or wet rice, maize, and cassava on their plots; from the mid-1980s a growing share of settlements were built around a central company plantation, with settlers growing tree crops, principally oil palm and rubber, for the plantation's mill alongside their food plots (World Bank, 1988). Settlers had no control over which new village they were placed in (Kebschull, 1986; Bazzi et al., 2019). Aside from where they were located, the villages therefore started out with the same institutions and infrastructure and, on average, the same endowments of land and inputs per household, which is rarely possible for settlements that formed on their own.

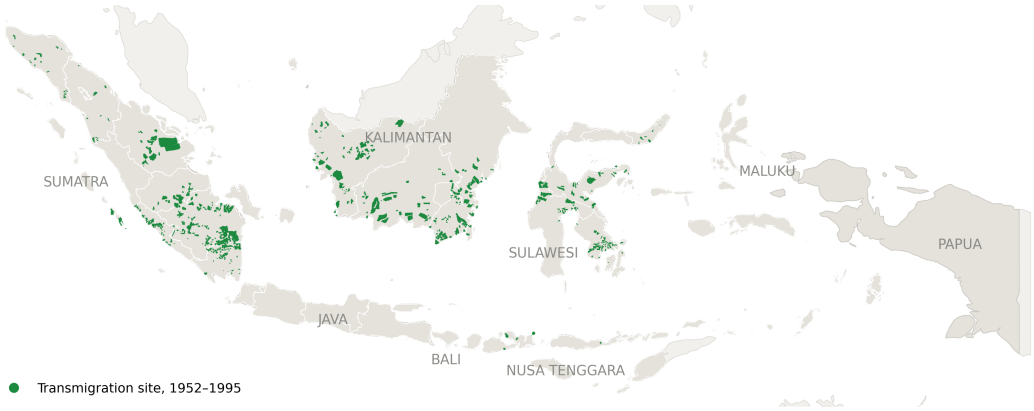
Who the settlers were Transmigrants were volunteers recruited on Java and Bali. Based on surveys of transmigrant households immediately before migration from Kebschull (1986), they were overwhelmingly asset-poor: most reported no savings, and most held far less land than the plot they received, so few can have had experience operating farms of that size (Appendix Figure A1). Previous work on the program, which studies how the skills settlers brought with them and the mix of ethnic groups within a village affected later outcomes, finds no sorting of migrant characteristics across destination villages (Bazzi et al., 2016, 2019).

The settlement register I identify transmigration villages, their location, the year they were settled, and the number of individuals initially placed in them from the Transmigration Census, a register of settled transmigration sites compiled by the program's directorate. Matching the register to the 2000 administrative village boundaries gives 1,152 villages settled between 1952 and 1995, which is the sample for the rest of the paper. Figure 1 maps them by settlement period. There is substantial variation in where villages were placed across the Outer Islands, which I quantify in the next section.

3 Data

Appendix Table B1 lists every data source and Appendix B describes the construction of each variable in detail.

Figure 1: Transmigration Villages by Settlement Period



Note: The figure maps the 1,152 transmigration villages in the analysis sample, settled between 1952 and 1995. White lines mark provincial borders as of 2000. The lighter gray areas are neighboring countries, principally Malaysia, Papua New Guinea, East Timor, and the Philippines.

3.1 Initial Market Access

Following Donaldson and Hornbeck (2016), I measure the market access of every location o at the time of the program as

$$MA_o = \sum_d \tau_{od}^{-\theta} N_d, \tag{1}$$

the population N_d of every other location d in Indonesia, discounted by the cost τ_{od} of reaching it. A village that is more connected to a large population center or many nearby smaller settlements will have relatively higher market access. To construct this measure, I measure the bilateral travel costs τ_{od}, the population distribution N_d, and use an external estimate of the trade elasticity θ.

Travel costs Travel costs require observing the road network of the Outer Islands before the program began in the 1950s. I construct it by hand-digitizing the maps of Indonesia compiled by the US Army Map Service around 1950, which record every road, classified by type from main road down to trail, and every railway in the country across 27 map sheets held in the Perry-Castañeda Library Map Collection.; Appendix B.11 describes the process and shows an example sheet. Because waterways carry goods and people across much of the Outer Islands, I add all navigable waterways from the Humanitarian OpenStreetMap Team.

The digitized network becomes a grid of cells, each assigned the speed of the fastest network type that crosses it. Speeds by mode come from the 1993 part of the Indonesia Family Life Survey (IFLS), where each village head reports the distance to nearby towns, the mode of travel, and the time the trip takes. I use the implied speeds for travel by road, by water, and by rail, and a walking speed for cells with no network. Travel times between every pair of locations then follow from a fast marching algorithm over the grid, as in Allen and Arkolakis (2014). As a robustness check I also compute costs under the monetary cost parameters of Donaldson and Hornbeck (2016), which

Figure 2: Construction of Market Access



(a) Original Image (b) Digitized Road Network (c) Constructed Market Access

Note: Panel (a) shows about one eighth of one original map sheet, panel (b) the digitized travel network for the same area, and panel (c) the resulting market access. The figure maps log initial market access, Equation (1), for the 1,152 transmigration villages, over the digitized 1950 road and waterway network. Darker shading indicates higher market access.

follow Fogel (1964), modified to allow travel by automobile; the two measures of market access have a correlation of 0.89. Appendix B.11 reports the parameters.

Populations Destination populations N_d come from the 1975 layer of the Global Human Settlement Layer (GHS), a gridded estimate of population built from census counts and satellite-detected built-up area, which I aggregate to village boundaries. I use 1975 because it is the earliest year the layer covers. Because villages were settled between 1952 and 1995, the 1975 weights are strictly pre-settlement only for the later cohorts; for early cohorts they could partly reflect growth induced by earlier program activity. My results are robust to restricting the sample to the 1979 to 1988 villages studied in previous work, for which the weights are unambiguously predetermined (Appendix Table C2). The 2010 GHS layer tracks the 2010 census population closely at the village level, which validates the aggregation of the gridded layer to village boundaries; no village-level population source exists for 1975 against which the earlier layer could be benchmarked directly (Appendix Table B2).

The elasticity and the resulting measure I set θ to 8.22, based on Donaldson and Hornbeck (2016). The results are robust to alternatives, including $\theta = 1$, which is essentially a distance-weighted sum of all other locations' population, and to alternative definitions of connectivity.[2] Market access at the time of the program is also strongly correlated with a measure built the same way from Indonesia's modern highway network and a road roughness index from Gertler et al. (2024). Figure 2 illustrates the construction for part of one map sheet, from the original map to the digitized network and the resulting market access. In the analysis I use the log of market access, which has a standard deviation of 7.3 log points and an interquartile range of 9.5 across my sample of transmigration villages.

[2]These include market access computed only over destinations within 2 hours of the origin, with and without the elasticity, and market access built with an alternative fast marching implementation in QGIS.

3.2 Outcomes

My primary outcome is village GDP per capita, for which I use the satellite-derived estimates of Rossi-Hansberg and Zhang (2025). These are gridded estimates of GDP that combine nightlights, population, land cover, and other remotely sensed inputs, trained to predict subnational GDP where it is observed, and they perform substantially better at predicting local GDP than nightlights alone. I aggregate them to the 2000 administrative boundaries of each village and observe them annually from 2012 to 2019, so the outcome is measured 17 to 67 years after settlement depending on the cohort. As an alternative, I build GDP per capita myself from harmonized nightlights (Li et al., 2020) and GHS population, converted to GDP using the elasticities estimated by Henderson et al. (2012), Hu and Yao (2022), and Beyer et al. (2022).[3] Section 5.2 adds two other direct outcomes: the wages that formally employed residents report in household surveys, and a material-welfare index constructed from video footage of a subset of villages.

3.3 Other Sources and Controls

The remaining outcomes come from five sources: the Population Census of 2000 and 2010 (sectoral employment, firm ownership, and migration), the village potential census (*Potensi Desa*, PODES) of 1993, 2000, and 2003 (infrastructure, public goods, land use, and crop production), the National Socioeconomic Survey, SUSENAS (food purchases and wages), the 2016 Economic Census (medium and large enterprises), and the 2003 Agricultural Census (household landholdings, which the model of Section 9 uses as land endowments). Appendix Table B1 and Appendix B describe each.

The controls are of two kinds: time-invariant geographic and agro-ecological characteristics of each village location from Bazzi et al. (2016), including the agroclimatic similarity between a village's migrant origins and its new location, and district characteristics from the 1980 Population Census (Appendix B). The village characteristics describe each site rather than what the settlers made of it: potential crop yields are modeled by FAO-GAEZ from climate, soil, and terrain rather than observed from harvests, soil characteristics come from the FAO-UNESCO Soil Map of the World, compiled in 1971–1981, and rainfall and temperature are averages over 1948–1978 (Bazzi et al., 2016). The 1980 measures are predetermined for villages settled after 1980 but not for the earlier cohorts, for which they could reflect the program's own effects; Section 5 reports the estimates with and without them.

[3] I use the Rossi-Hansberg and Zhang (2025) estimates rather than nightlight proxies as the primary measure because the latter rest on an estimated elasticity of light to income, which Gibson et al. (2021) and Bluhm and McCord (2022) have called into question.

4 Empirical Strategy

4.1 Specification

For an outcome y of village v in region r, settled in year c, observed in year t, I estimate

$$y_{vrct} = \beta_1 \ln MA_{vrc} + \beta_2 E[\ln MA]_{vrc} + X'_{vrc}\Gamma + \eta_r + \gamma_c + \alpha_t + \epsilon_{vrct}, \tag{2}$$

where $\ln MA_{vrc}$ is log initial market access from Equation (1), $E[\ln MA]_{vrc}$ is the expected market access control based on Borusyak and Hull (2023) described below, X_{vrc} is the vector of time-invariant village characteristics and 1980 district characteristics of Section 3, and η_r, γ_c, and α_t are region, settlement-cohort, and outcome-year fixed effects, the last included only for outcomes observed in more than one year. All right-hand-side variables are fixed at settlement, so for outcomes observed in several years the panel dimension enters only through the outcome-year fixed effects and the within-village correlation of the errors. My primary specification adopts islands for the relevant region. The coefficient of interest is β_1, the effect of a one log point increase in initial market access.

The island fixed effects absorb the large differences between the islands in their fundamentals and their development. The cohort fixed effects do two jobs. Villages settled in the same year were built under the same targets, budgets, and administrative practice, and some predetermined characteristics, including some crop-potential measures and the size of settlements, trend across cohorts (Appendix Table A1). And because every outcome is measured in a fixed calendar year, villages settled in the same year have had the same time to develop. All comparisons in the primary specification are therefore among villages placed on the same island and settled in the same year.

Because market access varies smoothly over space and each outcome is observed in several years, the errors of nearby villages and of the same village over time are potentially correlated. I therefore report standard errors following Conley (1999), with a Bartlett kernel in the distance between villages that declines to zero at 100 kilometers and full weight on every pair of years within a village, so the within-village component is the same as clustering by village. Appendix Table C2 reports cutoffs of 50, 250, and 500 kilometers, clustering by district, and a permutation test that makes no assumption on the error structure.

4.2 Identification

The assumption under which β_1 is causal is that, among villages placed on the same island and settled in the same year, and conditional on expected market access and the village and district characteristics in Equation (2), a village's initial market access is unrelated to the unobserved determinants of its long-run income. The assumption does not require that villages were created in random locations.

Relative to the leading designs in the market access literature, this assumption is stronger in

one respect and weaker in another. Those designs use changes in the access of existing places, which difference out fixed fundamentals (Donaldson and Hornbeck, 2016; Redding and Sturm, 2008); these villages have no period before settlement, so my assumption must hold for the level of access. But it does not have to contend with composition, which in existing places is jointly determined with access. Here I rely on the assignment mechanism established in previous work on the program: settlers did not choose their village or sort across villages by their characteristics, and every village was built to the same plan (Bazzi et al., 2016, 2019). Any potential differences in the composition of villagers observed after a village was settled are consequences of access rather than a potential confounder.

The main remaining threat is that the preexisting network of towns are endogenously formed. Towns likely formed on productive land, and a village placed near one may share some of that productivity. This could introduce a positive correlation between market access and a measure of productivity *even if* transmigration villages were placed randomly. Appendix C shows in a simulation, with villages placed at random on a non-random network of towns, that this can bias the estimate toward positive effects of access, the opposite of what I find, and that the village, district, and expected market access controls help account for it.

A second channel could push the other way. Town boundaries are themselves endogenous, so good land near a town may already have been absorbed into it, leaving worse land for a village placed nearby. This requires villages to sit at a town's edge, and few do: the median village is 62 kilometers from the nearest city boundary, and 95 percent are more than 3 kilometers away. Excluding the 1, 5, or 10 percent of villages closest to a city leaves the estimate essentially unchanged (Appendix Table C1). The channel would also leave connected villages on worse land, yet none of the potential crop yields, which settlement could not have affected, is significantly related to access in the balance tests below.

4.3 Controlling for Expected Market Access

Because provincial and local governments often had a say in where new villages went, some of the variation in initial market access reflects which part of the country a village was assigned to rather than anything about its own site. I strip out that component with a recentering in the spirit of Borusyak and Hull (2023). I define potential sites as every transmigration village settled in the same province and every "Planned Development Area" that the program's planning maps designated but never actually settled, so they describe where a village could have gone rather than only where villages did go. Expected market access is the average log market access of the 20 potential sites nearest to a village, excluding the village itself, and I include it as a control. I do not have to claim that a village could have been placed anywhere in its province, only that, among the nearby sites to which it could have been assigned, the particular site it received is uncorrelated with its productive potential. Inverse-distance and exponentially distance-weighted averages over all potential sites in the province give the same results (Appendix Table C2).

Province fixed effects would absorb every province-level component of access at once. I use the expected access control instead because it keeps the variation in access between provinces that reflects the geography of the Outer Islands, which province fixed effects would discard, and because it is built from where villages could have gone and not only where they went. Section 5.1 reports the estimates with province and province-by-cohort fixed effects in place of the control; they are smaller, and negative and significant throughout.

4.4 Balance

I test whether characteristics likely to be associated with a location's fundamental productivity vary with initial market access, estimating for each characteristic X

$$X_{vrc} = \beta_1 \ln MA_{vrc} + \beta_2 E[\ln MA]_{vrc} + \eta_r + \gamma_c + \epsilon_{vrc}, \tag{3}$$

Equation (2) with a characteristic in place of the outcome and without the controls. Table 1 reports each coefficient without and with the expected access control.

The characteristics of the sites themselves are balanced. With the expected access control, none of the potential crop yields, of the village or of its district, is significantly related to access, and of the twelve measures of the site's potential yields, soil, terrain, and climate, only temperature is significant, at the 10 percent level, no more than chance would produce. No site coefficient exceeds 0.07 standard deviations per standard deviation of access. The control does what it is designed to do: it removes the imbalances in rainfall and in the agroclimatic similarity between settlers' origins and the site, and it cuts the average absolute site coefficient from 0.039 to 0.023 standard deviations.

What imbalance remains is concentrated in how developed a village's district was in 1980: its households' access to piped water, sewage, modern fuel, electricity, and television, and its service employment share, with coefficients of 0.06 to 0.11 standard deviations. Connected villages sit in more developed districts, which are richer, so this would bias the estimate toward positive effects of access; these measures are controls in the main specification, and the estimate is unchanged with district fixed effects in their place (Section 5.1). Connected villages were also settled with somewhat fewer people, a program choice rather than a feature of the site, which is controlled for throughout.

5 The Long-Run Effects of Market Access

5.1 GDP per Capita

Table 2 reports estimates of Equation (2) for log GDP per capita, observed annually from 2012 to 2019. Panel A uses the primary measure. Column 1 includes only the island fixed effects and the village controls; column 2 adds the cohort fixed effects; column 3 adds the 1980 district controls;

Table 1: Village Characteristics Across Initial Market Access

Standardized characteristic	Coefficient on standardized ln(MA)	
	Without $E[\ln(MA)]$	**With** $E[\ln(MA)]$
Wetland Rice Potential Yield	0.024 (0.033)	0.015 (0.032)
Palm Oil Potential Yield	-0.015 (0.048)	0.011 (0.047)
Coffee Potential Yield	-0.089 (0.104)	-0.051 (0.085)
Cassava Potential Yield	-0.027 (0.039)	-0.019 (0.032)
Vector Ruggedness Measure	-0.050 (0.037)	0.011 (0.034)
Individuals Placed	-0.111*** (0.039)	-0.091** (0.041)
Agrosimilarity Score	0.115** (0.047)	0.076 (0.046)
Share of Land with Poor Drainage	0.027 (0.061)	0.026 (0.057)
Share of Land with Good Drainage	-0.001 (0.057)	0.005 (0.055)
Organic Carbon (%)	0.045 (0.078)	0.020 (0.049)
Topsoil Salinity (Elco) (dS/m)	0.021 (0.016)	0.016 (0.018)
Topsoil Base Saturation (%)	0.024 (0.042)	0.011 (0.041)
Avg. rainfall, 1948–1978	0.083** (0.039)	0.028 (0.038)
Avg. temp (Celsius), 1948–1978	0.059 (0.041)	0.067* (0.038)
ln(District Population, 1980)	-0.038 (0.036)	-0.051 (0.038)
District Own Electricity, 1980	0.069** (0.035)	0.062* (0.036)
District Own Radio, 1980	0.081** (0.039)	0.050 (0.041)
District Own TV, 1980	0.077** (0.035)	0.067* (0.037)
District Literacy, 1980	-0.025 (0.035)	-0.023 (0.039)
District Avg Years Schooling, 1980	0.005 (0.035)	-0.008 (0.037)
District Agricultural Employment Share, 1980	-0.048 (0.041)	-0.057 (0.041)
District Wetland Rice Potential Yield, 1980	-0.020 (0.029)	-0.006 (0.033)
District Palmoil Potential Yield, 1980	0.029 (0.035)	0.030 (0.036)
District Coffee Potential Yield, 1980	-0.059 (0.039)	-0.048 (0.041)
District Cassava Potential Yield, 1980	-0.025 (0.032)	-0.011 (0.035)
District Mining Employment Share, 1980	0.005 (0.039)	0.019 (0.040)
District Manufacturing Employment Share, 1980	-0.018 (0.037)	-0.019 (0.038)
District Trade Employment Share, 1980	0.025 (0.050)	0.057 (0.043)
District Service Employment Share, 1980	0.066* (0.035)	0.067* (0.037)
District Wage Worker Share, 1980	0.002 (0.035)	0.015 (0.037)
District Share HHs with Piped Water, 1980	0.107*** (0.038)	0.105*** (0.039)
District Share HHs with Sewage, 1980	0.072** (0.035)	0.069* (0.037)
District Share HHs with Modern Fuel, 1980	0.088** (0.036)	0.082** (0.038)
District Share HHs with Modern Floor, 1980	-0.064* (0.039)	-0.038 (0.034)
District Share HHs with Modern Roof, 1980	-0.003 (0.035)	-0.020 (0.037)

	Without $E[\ln(MA)]$	**With** $E[\ln(MA)]$
Omnibus test	$F(35, 284) = 2.152$	$F(35, 283) = 1.464$
Omnibus-test p-value	< 0.001	0.050
Observations	1214	1213

Note: Each row reports the coefficient on standardized log initial market access from two regressions, following Equation (3), of the standardized characteristic listed in the row on initial market access. The first column excludes expected log market access, $E[\ln(MA)]$, while the second column controls for it; both specifications include island fixed effects and cohort fixed effects. Standard errors, clustered at the subdistrict (kecamatan) level, are reported in parentheses. The omnibus F-statistics are from corresponding regressions of standardized log initial market access on all of the listed characteristics jointly, without and with $E[\ln(MA)]$, conditional on the fixed effects, and test their joint significance. The reported omnibus p-values are based on the corresponding cluster-robust Wald F-tests; joint orthogonality tests with many covariates and clustered errors tend to over-reject (Kerwin et al., 2024). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

and column 4, my preferred specification, adds the expected market access control. Higher initial market access significantly lowers long-run GDP per capita in every specification, and the estimate barely moves as controls are added. Adding controls that themselves predict income does not shift the estimate, which is the first indication that market access is not capturing a relationship between a location's productivity and income.

The effect is economically large. Moving from the 25th to the 75th percentile of initial market access lowers GDP per capita by about 5.5 percent, and one standard deviation of access lowers it by about 4 percent. The interquartile effect is nearly 20 percent of the interquartile spread in GDP per capita across the villages. The outcome is measured 17 to 67 years after settlement, so these are long-run differences in the level of income, estimated among villages settled in the same year.

Robustness

The estimate is robust on three margins, reported in Appendix Table C2. The first is the fixed effects. Replacing the island fixed effects with province fixed effects gives -0.0034, smaller than the baseline but significant at the 1 percent level, and province-by-plan-period and province-by-cohort fixed effects give -0.0023 and -0.0033. The province-by-cohort specification is demanding, since the median cell contains 2 villages and over a third of cells contain a single village, yet the estimate remains significant at the 1 percent level.

The second measure is inference. Increasing the spatial cutoff of the standard errors to 500

Table 2: Effects of Initial Market Access on GDP per Capita

	(1)	(2)	(3)	(4)
Panel A. ln GDP per capita, Rossi-Hansberg and Zhang (2025)				
ln(Market Access)	−0.0060***	−0.0059***	−0.0057***	−0.0058***
	(0.0022)	(0.0019)	(0.0019)	(0.0020)
Effect of 1 s.d. of ln(MA)	−0.0438	−0.0431	−0.0416	−0.0423
DMP breakdown fraction	0.714	0.643	0.643	0.857
Villages	1,152	1,152	1,152	1,152
Panel B. ln GDP per capita from nightlights, Henderson, Storeygard, and Weil (2012) elasticity				
ln(Market Access)	−0.0279***	−0.0219***	−0.0169***	−0.0145**
	(0.0068)	(0.0061)	(0.0060)	(0.0067)
Effect of 1 s.d. of ln(MA)	−0.2037	−0.1599	−0.1234	−0.1059
Villages	1,141	1,141	1,141	1,141
Panel C. ln GDP per capita from nightlights, Hu and Yao (2022) elasticity				
ln(Market Access)	−0.0315***	−0.0253***	−0.0203***	−0.0182***
	(0.0068)	(0.0061)	(0.0060)	(0.0067)
Effect of 1 s.d. of ln(MA)	−0.2300	−0.1847	−0.1482	−0.1329
Villages	1,141	1,141	1,141	1,141
Panel D. ln GDP per capita from nightlights, Beyer, Hu, and Yao (2022) elasticity				
ln(Market Access)	−0.0358***	−0.0294***	−0.0244***	−0.0227***
	(0.0069)	(0.0061)	(0.0060)	(0.0067)
Effect of 1 s.d. of ln(MA)	−0.2613	−0.2146	−0.1781	−0.1657
Villages	1,141	1,141	1,141	1,141
Expected market access control	No	No	No	Yes
Village controls	Yes	Yes	Yes	Yes
District controls (1980)	No	No	Yes	Yes
Cohort fixed effects	No	Yes	Yes	Yes
Island fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes

Note: Each entry is the coefficient on log initial market access from Equation (2), with the controls and fixed effects listed at the bottom of the table. Panel A uses the satellite-derived GDP per capita estimates of Rossi-Hansberg and Zhang (2025), observed annually from 2012 to 2019. Panels B to D use GDP per capita built from harmonized nightlights and GHS population, converted to GDP with the elasticity of the study named in the panel heading. The effect of 1 s.d. of ln(MA) multiplies the coefficient by 7.3, the standard deviation of log initial market access. The DMP breakdown fraction is the share of village controls for which the Diegert et al. (2025) breakdown point exceeds the control's own explanatory importance (Appendix C.5). Standard errors, in parentheses, allow for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

kilometers barely changes the uncertainty of the estimates, and the estimate is significant at 1 percent when I instead cluster by district. A permutation test makes the same conclusion without any assumption on the error structure: none of 1,000 random reassignments of market access across villages within province-by-cohort cells produces a relationship as strong as the one I observe in the data.

The third is sensitivity to omitted variables. Following Diegert et al. (2025), I calculate how strongly an unobserved confounder would have to be correlated with market access, relative to the observed controls, to drive the estimate to zero. For over 85 percent of the village controls, the confounder would have to matter more than the control does. This is a higher share than in the application found in Diegert et al. (2025) where they judge a result robust (Appendix C.5).

A final concern is that villages with higher initial market access might have been more exposed to aggregate shocks throughout this period. Given the growth of urban areas during this period, it seems more plausible that higher market access villages were actually more exposed to positive shocks, which would make the negative effects I find be biased towards zero. However, I also examine two particular shocks that could be a threat to my identification argument. The Asian financial crisis of 1997 impacted urban areas most strongly, which connected villages are closer to. The subsequent devaluation of the rupiah then raised the returns to export crops. The palm oil boom of the 2000s increased the incomes of plantation economies, which were more likely located in the more isolated areas where agriculture was more prevalent. I find no evidence that potential exposure to these shocks explains the negative effects of market access on income (Appendix C.4). Using nightlight measures that are from before 1997, I find a similarly large negative effect of market access on estimated GDP per capita (Appendix Table C3). I also find the estimated effect of market access does not vary significantly with a village's potential palm oil yield or whether it grew palm oil in 2003 (Appendix Table C4).

5.2 Alternative Measures of Development

Panels B to D of Table 2 replace the primary measure with GDP per capita built from nightlights under each of three light-to-GDP elasticities. The estimates are negative under all three and *larger* than the primary estimate, by nearly four times under the elasticity of Beyer et al. (2022), where one standard deviation of market access lowers GDP per capita by 15 percent. The primary measure is therefore the conservative one. The nightlight panels also show why the detailed controls matter. Panel A barely moves across specifications, but in Panels B to D the column 4 estimates are on average 42 percent smaller than those in column 1.

I also find similar negative effects of market access on income in household-level survey data. Pooling the 2000 to 2007 waves of the SUSENAS gives wages for formally employed residents of 502 villages (Appendix Table D1). The point estimates for wages are negative in every specification, about 2 percent per standard deviation of access, about half the effect on GDP per capita, with no difference in hours. They are imprecise, since the sample covers formal workers alone, a minority

of rural employment.

Finally, I construct a measure of village material well-being from an independent source: publicly available documentary and vlog footage of 47 transmigration villages in Kalimantan, whose frames I score against a fixed rubric of visible indicators of material welfare using a vision language model (Appendix E, where Figure E1 shows two scored frames). Regressing the resulting index on market access,[4] gives a negative and significant slope for the composite index. Based on the distribution of market access in my full sample of transmigration villages, the estimate implies that one standard deviation of access lowers village welfare by 0.35 standard deviations. Because of the limited sample size, I read this as corroboration from a source that shares nothing with the satellite or administrative data rather than as an estimate of comparable precision.

5.3 Productivity

Lower GDP per capita in more connected villages also appears in measures of village productivity. Table 3 reports the effects of market access on proxies for productivity inside agriculture (crop yields, and tractors per resident and per agricultural worker, from PODES) and outside it (the ratio of employees to employers in the Population Census, a measure of the size of the typical non-farm establishment).

Villages with higher market access are less productive on every measure. One standard deviation of access lowers tractors per thousand agricultural workers by 0.52 and tractors per thousand residents by 0.47, about a quarter of the sample mean in each case. The same variation lowers the yield of a village's top 5 crops by roughly 7 percent and the yield of rice by about 10 percent, significant at the 10 and 5 percent levels respectively, while the palm oil yield is negative but imprecise. Outside agriculture, the ratio of employees to employers falls by about 13 percent of its mean, so the typical firm is smaller. I return to the lower agricultural yields and mechanization rates in Section 8, since both of these effects are what a scale elasticity in the agricultural sector would produce.

5.4 Structural Transformation

Table 4 reports the effects of initial market access on the sectoral composition of employment. Column 1 reports the agricultural employment share in the 2000 and 2010 Population Census, and column 2 its change between the two. A one standard deviation increase in market access lowers the agricultural employment share by 2.7 percentage points, and the share falls faster between 2000 and 2010 in more connected villages. Columns 3 to 5 show which sectors workers move in to. Manufacturing, services, and retail all see increases in the employment share, but services and retail account for nearly 70 percent of the reallocation, and they are also the largest non-agricultural sectors in these rural economies. Market access therefore spurs structural transformation out of

[4]Due to the limited sample I am not able to include the controls of the main design.

Table 3: Effects of Initial Market Access on Productivity

	Employees per Employer	Tractors per 1000 Individuals	Tractors per 1000 Ag Workers	ln(Productivity of Top 5 Crops)	ln(Rice Productivity)	ln(Palm Oil Productivity)
ln(Market Access)	-0.6881**	-0.0650*	-0.0715*	-0.0106*	-0.0145**	-0.0015
	(0.3124)	(0.0361)	(0.0418)	(0.0060)	(0.0058)	(0.0177)
Effect of 1 s.d. of ln(MA)	-5.0231	-0.4745	-0.5219	-0.0774	-0.1058	-0.0109
Expected MA Control	Yes	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes
District Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Island FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	No	No	No	No
Unique Villages	1107	1035	1035	980	591	302

Note: Each column reports the coefficient on log initial market access from Equation (2). Column 1 is the ratio of employees to employers among working residents in the 2000 and 2010 Population Census. Columns 2 and 3 are tractors per thousand residents and per thousand agricultural workers, from the 2000 PODES and the 2000 Population Census. Columns 4 to 6 are log yields per hectare for the 5 most planted crops, for rice, and for oil palm, from the 2000 PODES; the samples differ because not every village grows rice or oil palm. The effect of 1 s.d. of ln(MA) multiplies the coefficient by 7.3, the standard deviation of log initial market access. Standard errors, in parentheses, allow for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

agriculture while lowering output per capita and productivity. Given how often structural transformation is promoted as a route to growth in developing countries, this combination is the central fact the paper explains, and it is exactly the pattern a model in which agricultural TFP depends on the scale of the agricultural workforce predicts.

Table 4: Effects of Initial Market Access on Structural Transformation

	Agriculture Share	Change in Ag Share	Processing Share	Services Share	Trade Share
ln(Market Access)	-0.0037***	-0.0027***	0.0012***	0.0016***	0.0010***
	(0.0009)	(0.0008)	(0.0003)	(0.0004)	(0.0003)
Effect of 1 s.d. of ln(MA)	-0.0270	-0.0197	0.0088	0.0117	0.0073
Expected MA Control	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes
District Controls	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes
Island FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	Yes	Yes
Unique Villages	1151	1151	1151	1151	1151

Note: Each column reports the coefficient on log initial market access from Equation (2), using the 2000 and 2010 Population Census. Column 1 is the agricultural employment share, pooled across the two census years; column 2 is its change between 2000 and 2010. Columns 3-5 are the manufacturing, service, and retail employment share, respectively. The effect of 1 s.d. of ln(MA) multiplies the coefficient by 7.3, the standard deviation of log initial market access. Standard errors, in parentheses, allow for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The shift is toward small-scale services. In these villages, 69.3 percent of rural retail workers are self-employed, against 53.0 percent in agriculture and 19.8 percent in manufacturing. The non-farm sector these villages transform into is not industry but small-scale services sold to nearby customers.

6 A Framework with Agricultural Scale Economies

The empirical results establish three facts. Villages with higher initial market access have lower GDP per capita, they reallocate more labor out of agriculture, and their farms are less productive. In this section I develop a model that ties these facts together and ask how large an agricultural scale externality has to be to reverse the standard gains from market access. The rest of this sections introduces the key model ingredients and solves a one-village version in closed form to provide intuition behind the mechanism. Section 7 then considers other explanations, Section 8 estimates the externality directly, and Sections 9 and 10 take the full model to the data.

6.1 Ingredients

Two features separate the model from a standard spatial model. First, agricultural productivity rises with the scale of the local agricultural workforce, the externality estimated in Section 8. This effect on agricultural productivity is not internalized when individuals make their sector choice. Second, a village's non-farm service good is traded across space. The provider travels to the buyer or the buyer to the provider, so a sale requires a trip. Importantly, trade costs of services rise more steeply with distance than agricultural goods (Section 9.2). This asymmetry concentrates the demand for non-farm work in high-market-access locations.

The rest of the model is standard, or motivated by empirical evidence. Labor is immobile across locations, since migration does not respond to market access (Section 7) and migration frictions in Indonesia are large and well documented (Bryan and Morten, 2019). The non-farm sector produces with constant returns and its productivity does not depend on its size, which is the sense in which the small informal services that absorb labor in these villages do not carry growth. Workers sort between the two sectors within their village, and trade in food and city goods delivers the usual gains from market access.

Placing the externality in agriculture rather than in the non-farm sector is a key feature of the model. Figure I1 shows that rural wages rise with the size of a village's agricultural sector and are flat in retail. This relationship holds even after including village and demographic controls (Appendix Table I1), and retail services in developing countries show low productivity and no evident scale economies (Lagakos, 2016). The non-farm work that connectivity creates here has the same character: most of it is in services and retail, most retail workers are self-employed, and non-farm establishments are smaller in more connected villages (Sections 5.3 and 5.4), consistent with evidence that these sectors are persistently small, informal and unproductive (La Porta and Shleifer, 2014; Lagakos, 2016).

Environment

Locations are either rural villages or urban locations, and each one carries the same two-sector economy. There are three goods. Food is homogeneous, traded, and the numéraire. Rural services

and the urban good are differentiated by origin location and traded, and both are aggregated by buyers with elasticities σ_n and σ_u, respectively.

Locations are separated by travel times over the network of Section 3. The cost factor between locations ℓ and o is $\tau_{\ell o} = 1 + \text{hours}(\ell, o)$, and for objects that depend on a village's access to urban markets I write τ_v for the travel cost to the village's nearest urban location. Each good faces a sector-specific travel cost, with food shipped at τ^{κ_a}, services at τ^{κ_n}, and the city good at τ^{κ_u}. The distinction that matters throughout is that it is less costly to ship food than services, and the difference in trade costs is what drives the labor reallocation.

Within a location, workers possess sector-specific abilities and sort freely between its two sectors, so the sectoral allocation responds to the relative return to work in each sector. Each location is endowed with L workers and H units of land; land works only in agriculture, and the service and city-good sectors use labor alone, which reflects the land-use evidence of Section 7.

Agriculture is the same technology at every location, rural or urban.[5] Production is constant-returns Cobb–Douglas in effective labor $Z_{a,\ell}$, the location's agricultural workers weighted by their ability in agriculture, and land, with total factor productivity depending on the scale of the location's agricultural workforce, measured in the same effective units as its labor input,

$$A_{a,\ell} = \bar{A}_{a,\ell}\, Z_{a,\ell}^{\varphi}, \qquad \varphi > 0, \tag{4}$$

external to the individual farmer and therefore uncompensated in factor markets. Substituting (4) into the production function,

$$Y_{a,\ell} = \bar{A}_{a,\ell}\, Z_{a,\ell}^{\gamma_a + \varphi}\, H_{\ell}^{1-\gamma_a}, \tag{5}$$

so the location-scale technology is homogeneous of degree $1 + \varphi$ in labor and land jointly (in labor alone, land fixed, returns are $\gamma_a + \varphi$, 0.98 at the calibrated values; no restriction either way is imposed), while atomistic firms earn zero profits. The agricultural wage per effective unit is $w_{a,\ell} = \gamma_a p_{a,\ell} Y_{a,\ell} / Z_{a,\ell}$, and land rents $(1 - \gamma_a) p_{a,\ell} Y_{a,\ell}$ accrue to local households. The externality (4) is the same object estimated in Section 8, whose estimating equation is the log of (5) with $\beta = \gamma_a + \varphi$ the coefficient on agricultural labor.[6] The possibility of multiple equilibria that the unrestricted returns create is handled by the selection protocol described in Section 9.1.

The non-agricultural sector is also the same technology everywhere, but its output differs by location type. Rural locations produce a differentiated service, urban locations a differentiated city good, and both are produced with labor alone,

$$Y_{k,\ell} = A_{k,\ell}\, Z_{k,\ell}, \qquad k = n \text{ (rural)},\ k = u \text{ (urban)}, \tag{6}$$

[5] The choice to include agriculture as a sector choice in urban locations is motivated by the fact that the agricultural employment share of urban locations is 0.29.

[6] The data measure agricultural labor as a headcount rather than in effective units. Under the model's sorting the two move nearly one for one with the instrument, by 0.96 to 0.98 at the calibrated externality, and the same 2SLS run on data generated by the model recovers $\gamma_a + \varphi$ to within 2 to 4 percent (Appendix N.1).

so the gate price of a location's variety is tied to its wage, $p_{k,\ell} = w_{k,\ell}/A_k$.

6.2 A Closed-Form Characterization

Before turning to the full economy, I add enough structure to characterize the contest between the externality and the gains from proximity in closed form. Consider a single village at distance τ from a city with the structure described above. The only addition is that agricultural ability is common across workers and non-agricultural ability is drawn from a Pareto distribution. This allows me to write the response to a connectivity shock only in terms of parameters and the service employment share π_n, which I can solve for in closed form under these assumptions. Appendix G contains the derivation, and Appendix F solves a many-village version with the full sorting structure numerically. The quantitative model of Section 9 instead places this economy at every location in Indonesia, with Fréchet sorting, and must be solved numerically.

The village sells food and services at the gate prices p_a and p_n, and because the occupational margin depends on the two producer prices only through their ratio, the relative price of services $\rho = p_n/p_a$ governs sorting. I write $\eta = d\ln\pi_n/d\ln\rho$ as the elasticity of the service employment share with respect to the relative price of village services. An increase in the relative service price pulls workers out of agriculture, and the reallocation lowers agricultural TFP due to the externality. The income lost through this channel is given by,

$$\mathcal{E} \equiv s_a\,\varphi\,\frac{\pi_n}{1-\pi_n}\,\eta, \tag{7}$$

which I call the externality drag. A 1 percent rise in the relative service price reduces effective labor in agriculture by approximately $\frac{\pi_n}{1-\pi_n}\,\eta$ percent. Through the externality, this lowers agricultural TFP by φ times that reallocation, and then the resulting income loss is weighted by agriculture's share of village income s_a.

Proposition 1 (Sign reversal of the connectivity gradient)**.** *In the simplified model of this subsection, derived in Appendix G, the relative price of services falls with distance from the city whenever* $\kappa_n\frac{\sigma_n-1}{\sigma_n} > \kappa_a$*, and real village income rises with distance (proximity is a curse) if and only if the externality loss outweighs the two market-access gains,*

$$\underbrace{\mathcal{E}\left|\tfrac{d\ln\rho}{d\ln\tau}\right|}_{\textit{externality loss}} \;>\; \underbrace{(s_n-\alpha_n)\left|\tfrac{d\ln\rho}{d\ln\tau}\right|}_{\textit{terms-of-trade gain}} \;+\; \underbrace{\alpha_u(\kappa_a+\kappa_u)}_{\textit{cheaper-city-goods gain}},$$

equivalently if and only if φ exceeds a threshold φ^* available in closed form (Appendix G, equation (G14)). Here s_a and s_n are the village's sectoral income shares, π_n its service employment share, α_n and α_u its service and city-good consumption shares, and ρ and η are as defined above. For $\varphi < \varphi^*$ proximity raises real income and the standard gains from trade prevail.
Proof: Appendix G.

The proposition describes the effect of a connectivity shock term by term. Proximity delivers three standard gains. The village sells the services it exports at a better relative price, sells its food at a better farm-gate price, and buys city goods at lower delivered prices. The effect of the agricultural scale externality operates in the opposing direction. At $\varphi = 0$ this term disappears and every channel is a gain from proximity, so any curse in this model must come from the externality. Of note, the mechanism itself requires the sector-specific trade cost asymmetry. The relative service price falls with distance only when $\kappa_n(\sigma_n - 1)/\sigma_n$ is greater than κ_a. This condition guarantees that services are effectively harder to trade than food. Reallocation vanishes only at the knife-edge case where $\kappa_a = \kappa_n(\sigma_n - 1)/\sigma_n$; with equal frictions, $\kappa_a = \kappa_n$, the relative service price would instead rise with distance and proximity would move labor toward agriculture. At the measured values of Section 9.2 this condition holds easily.

The size of the drag depends on how readily workers move between sectors. In the one-village model the structural transformation elasticity is $\eta = 1/(A + B\,\pi_n/(1 - \pi_n))$, where $B = 1 - \gamma_a - \varphi$ and A falls as non-agricultural ability becomes less dispersed (Appendix G). When workers are more alike, a given rise in the relative service price moves more of them out of agriculture, the drag is larger, and the threshold φ^* is lower. The many-village version of Appendix F shows the same force under Fréchet sorting: making workers more homogeneous ($\theta = 30$) more than doubles the curse, while high dispersion in comparative advantage ($\theta = 4$) reduces it, and the threshold falls from 0.45 in the baseline to 0.33 with near-homogeneous workers and rises to 0.59 with high dispersion (Table F1). When comparative advantage is more dispersed, sectoral shifts translate to smaller movements in effective labor than headcount, which reduces the bite of the productivity loss.

The framework makes a statement that is more general than the transmigration setting. Connectivity lowers income when the reallocation it causes moves labor out of the sector in which the social return to labor exceeds the private return, and by enough to outweigh the gains from trade. In these villages that sector is agriculture, while the services that absorb the workers leaving it show no sign of scale economies of their own. If the sector gaining workers carried the externality instead, the same reallocation would add to the gains from trade rather than subtract from them. The sections that follow determine the quantitative strength of each mechanism.

7 Other Explanations

The framework of Section 6 provides one potential mechanism that can explain the empirical results I estimate, and this section asks whether other explanations fit the evidence as well. I examine channels that theory and previous work suggest could drive results: that connected villages specialized into the agricultural sector, that they lost activity via competition from the nearby locations, that they experienced higher rates of out-migration or commuting, that lower income is compensated by better amenities, and that lower measured income is offset by lower prices. None

of these explanations explains my results in this context. What the evidence shows instead is that connected villages moved *more* labor out of agriculture, into more numerous and smaller non-farm firms, while their agricultural productivity fell, which is the combination that an agricultural scale externality produces, and which Section 8 then estimates directly.

Specialization

One explanation for lower output in connected villages is that exposure to competition led productive industry to leave and the village to specialize in agriculture. If better access raised the return to farm output and exposed local industry to competition from more industrial locations, market access would raise the agricultural employment share, consistent with the theoretical arguments of Matsuyama (1992) and the empirical findings of Baum-Snow et al. (2020) and Faber (2014).[7] Table 4 shows the opposite: more connected villages have a lower agricultural employment share, and it falls faster between 2000 and 2010 (Section 5.4).

Firm entry and size

Connectivity could reduce the number of local firms through competition or raise it through the demand that nearby population brings. Panel A of Table 5 shows that in this setting it raises it. One standard deviation of market access raises the number of residents who report operating a firm, formal or informal, in the census by 10.4 per thousand residents, and it raises the number of firms in the village by about one, although that estimate is imprecise. Among the 128 villages with medium or large firms, average employment per firm is about 19 percent lower per standard deviation of access, and about 16 percent lower within sectors, the same pattern as the lower ratio of employees to employers in Table 3, and consistent with entry-driven selection in which the marginal entrant is smaller. The evidence also points away from the competition channel of Faber (2014). If market access mainly exposed transmigration villages to more productive external firms, I would expect firm exit and fewer local firms; instead, connected villages have more firms. I cannot rule out compositional shifts within the firm population, but the absence of net exit, together with the price evidence below, does not support a competition channel behind the negative effects.

Migration and commuting

If initially connected villages offered their residents better opportunities elsewhere, part of the negative effect on GDP per capita could reflect people leaving the village rather than lower resident incomes. I measure out-migration two ways. Because transmigrants came from Java and Bali, the residents of a transmigration village born on the inner islands are, to a close approximation, the original settlers and later arrivals from the same region, so their number and share in the 2000 and

[7] In Faber (2014) increased market access leads to large declines in output in sectors *except* in agriculture, suggesting that perhaps resources were shifted there in response to increased competition.

Table 5: Effects of Initial Market Access: Other Mechanisms

	(1)	(2)	(3)	(4)
Panel A. Firm prevalence and size	Employers per 1,000	Number of firms	Employment (log)	Employment (log)
ln(Market Access)	1.4266**	0.1449	−0.0289**	−0.0244**
	(0.5651)	(0.0941)	(0.0130)	(0.0102)
Effect of 1 s.d. of ln(MA)	10.4142	1.0578	−0.2110	−0.1781
Year fixed effects	Yes	No	No	No
Sector fixed effects	No	No	No	Yes
Villages	1,151	947	128	128
Panel B. Migration	Inner islanders (log)	Inner-islander share	Inner islanders minus settlers	Work elsewhere (0/1)
ln(Market Access)	−0.0050	−0.0009	−2.8469	0.0006
	(0.0059)	(0.0011)	(5.0745)	(0.0012)
Effect of 1 s.d. of ln(MA)	−0.0365	−0.0066	−20.7824	0.0044
Year fixed effects	Yes	Yes	Yes	No
Villages	1,140	1,151	1,151	947
Panel C. Public goods	Full index	Excluding roads	Roads only	
ln(Market Access)	0.0032	0.0019	0.0092**	
	(0.0025)	(0.0027)	(0.0040)	
Effect of 1 s.d. of ln(MA)	0.0234	0.0139	0.0672	
Year fixed effects	No	No	No	
Villages	1,151	1,151	1,151	
Panel D. Food prices, quantities, and expenditure	Price index^{Ind}	Quantity index^{Ind}	Expenditure (log)	
ln(Market Access)	0.0074**	0.0011	0.0154**	
	(0.0034)	(0.0059)	(0.0077)	
Effect of 1 s.d. of ln(MA)	0.0540	0.0080	0.1124	
Year fixed effects	Yes	Yes	Yes	
Villages	84	84	84	
Households	1,234	1,234	1,234	

Cohort and island fixed effects and expected MA control included in every regression.
Village and district controls included except in Panel A, columns 3–4.
Standardized effects multiply the coefficient by 7.3, the standard deviation of log market access.

Note: Entries are coefficients on log initial market access from separate regressions following Equation (2), with specifications indicated above. Panel A: column 1 uses the 2000 and 2010 Population Census (employers per thousand residents); column 2 combines the 2003 PODES and 2016 Economic Census; columns 3–4 use the latter's firm-level employment, and column 4 adds sector fixed effects. Panel B: columns 1–3 use the same population censuses; inner islanders are residents born on Java or Bali, and column 3 subtracts initial settlers. Column 4 indicates residents leaving to work elsewhere (2003 PODES). Panel C uses the 2000 PODES index of 18 standardized public goods measures and its non-road and road components. Panel D uses household food purchases from the 2002 and 2011 SUSENAS; indices weight log unit values and log quantities by household expenditure shares (Ind). Standard errors in parentheses follow Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2010 Census fall when settlers leave and proxy for permanent out-migration. The 2003 PODES also records whether residents have left the village to work elsewhere, a direct if coarse indicator. Panel B of Table 5 shows no relationship between market access and any of these measures. The number and share of inner-island-born residents, their number net of the initial settlers, and the PODES indicator are all unrelated to initial access, and the point estimates are small: one standard deviation of access lowers the number of inner-island-born residents by under 4 percent and their share by less than one percentage point.

Commuting is unlikely to explain the result either. The concern is that some of the output of commuters is recorded outside the village, so that high-commuting villages appear poorer than they are. Two facts weigh against it. First, rural commuting is uncommon: in the 2005 Intercensal Population Survey, most districts have rural commuting rates below 2 percent of the working population. Second, commuting can depress measured local production but is unlikely to lower measures such as nightlight-based measures of GDP per capita, wages, or the material-welfare index built from village footage (Section 5.2). Any additional travel for work in connected villages is itself part of the structural transformation documented in Section 5.4.

Amenities

Lower measured income need not imply lower welfare if connected villages offer amenities that compensate their residents. I test this with a public goods index built from the 2000 PODES. The index, used in Bazzi et al. (2026), contains 18 measures public goods, including health care availability and quality, school presence and staffing, safe drinking water, waste collection and sanitation, and road maintenance and quality, each standardized and averaged. Panel C of Table 5 reports the effect of market access on the full index and on the index split into its road measures and all other categories. Market access significantly improves the road measures, raising the road component by 0.07 per standard deviation of access, which highlights that the index can detect differences across villages. However, it does not improve the full index, and on the non-road measures connected villages are no better provided, with a small and insignificant effect of 0.01. The index cannot cover every dimension households might value, but the estimates show no compensating improvement in public goods not directly related to roads.

Prices and expenditure

Lower GDP per capita would overstate the negative effects of market access if connected villages also faced lower prices. The consumption modules of the 2002 and 2011 SUSENAS record the quantity and expenditure for each food item a household buys. The two waves cover 84 transmigration villages and 1,234 households. I construct a price index from the log unit values, weighting items by the household's own expenditure shares. Under the assumption that quality does not vary systematically with market access within an island (Deaton, 1988), I interpret the effects on unit values as an effect on prices.

Panel D of Table 5 shows that households in more connected villages pay more for food, not less. The price index rises with market access by 0.0074 per log point, or about 5.5 percent per standard deviation, significant at the 5 percent level. Food expenditure rises as well, but there is reason not to interpret this as evidence of higher consumption. Importantly, market access has a small, insignificant effect on the quantity index. This suggests that simply measuring changes in expenditure would obscure the fact that the increase is being driven by higher prices for the same items rather than increased quantities. Also, the module records purchases much more reliably than the consumption of own-produced food. As households leave agriculture and then turn to buying the food they previously grew, which occurred more in higher market access locations, recorded spending on food increases even if total food consumption is unchanged.

Housing, the largest non-food expenditure category in this setting, shows similar effects. Households in more connected villages spend substantially more on housing overall and on a price per square meter measure, though the two components are individually imprecise (Appendix Table H1). Taken together, I find no evidence that the lower measured production per capita is offset by a lower cost of living in more connected locations. If anything, the evidence suggests that any nominal measures of income understate the real effects of market access in this context.

Land

Land-use records from PODES point to the same squeeze from the supply side. The amount of land under cultivation in more connected villages is lower and decreasing over time, while the land devoted to housing and firm activity does not expand to accommodate the additional non-agricultural activity (Appendix Table H2). Converting land between uses appears difficult in this context, so structural transformation operates through labor rather than land. This is how the model of Section 6 treats land: it works only in agriculture, and non-agricultural activity absorbs labor, not land.

Summary

Taken together, the reduced-form results paint a consistent picture. More connected transmigration villages have lower GDP per capita, lower agricultural productivity, more structural transformation out of agriculture, more firm entry but smaller firms, no sizable migration response, no improvement in non-road public goods, and higher local prices. The evidence does not favor any of the standard channels, agricultural specialization, competition, labor drain, or compensating differentials, as the explanation for the negative effects of connectivity in this setting.

The combination of faster structural transformation and lower agricultural productivity is what the agricultural scale externality of Section 6 predicts, and the next section estimates that externality directly.

8 Estimating the Agricultural Scale Externality

The results from Section 5 say that connected villages move more labor out of agriculture and farm less productively. This section estimates the link between the two, the elasticity of agricultural productivity with respect to the size of the agricultural workforce, using a second source of variation from the program: villages were settled with different numbers of people. The estimate is the parameter that the model of Section 6 uses for an external scale elasticity.

8.1 Framework and Identification

Let agricultural production of crop c in village v be

$$Q_{vc} = A_{vc} \left(L_v^{ag}\right)^{\gamma_a} \left(H_{vc}\right)^{\gamma_H}, \tag{8}$$

where L_v^{ag} is the village's agricultural labor, H_{vc} is the area planted to crop c, γ_a is the private labor share, the elasticity of a farm's output with respect to its own labor, which under competition is the share of output paid to labor, and γ_H is the corresponding elasticity with respect to land. Village-level productivity depends on the scale of agricultural labor through an externality,

$$A_{vc} = \bar{A}_{vc} \left(L_v^{ag}\right)^{\varphi}, \tag{9}$$

which no individual farm internalizes, where $\bar{A}_{vc}$ is the village's underlying productivity in crop c. Writing $\ln \bar{A}_{vc} = \delta_c + \varepsilon_{vc}$, with δ_c a crop effect and ε_{vc} the village's unobserved deviation from it, such as the quality of its soil and water, its farmers' skill, and the weather in the survey season, and substituting gives the estimating equation

$$\ln Q_{vc} = \delta_c + (\gamma_a + \varphi) \ln L_v^{ag} + \gamma_H \ln H_{vc} + \varepsilon_{vc}. \tag{10}$$

The coefficient $\beta \equiv \gamma_a + \varphi$ on $\ln L_v^{ag}$ is the object of interest: it is how much a crop's output rises with the size of the village's entire farming workforce. With a separate measure of the private share γ_a, the externality is $\varphi = \beta - \gamma_a$. Ordinary least squares would not recover β, because $\mathrm{Cov}(\ln L_v^{ag}, \varepsilon_{vc}) \neq 0$: better soil and water keep more residents in farming, while a village with few other opportunities keeps more of them on poor land.

The data are the 2003 PODES village-by-crop records of production and area planted in the most recent season; because I observe quantities, price differences across locations do not enter. Agricultural labor L_v^{ag} is the number of residents working in agriculture in the Population Census, treated as contemporaneous with the 2003 outcomes. PODES does not record labor by crop, so each crop's equation uses the village's total agricultural labor, and the coefficient is a pure scale effect as long as the share of the workforce allocated to a crop does not move with the instrument. The crop effects absorb permanent differences in labor intensity across crops, the planted-area

control absorbs differences in the scale of each crop, and initial settlement size does not shift the share of agricultural workers growing cash crops rather than staples in the census, a coarser check since the census records crop types rather than crops.

I instrument $\ln L_v^{ag}$ with $\ln W_v$, the number of individuals initially settled in village v by the program, which was determined decades before the 2003 outcomes. The instrument is relevant because larger initial settlements have larger agricultural workforces decades later: the first-stage coefficient is 0.18 in the baseline specification (Appendix Figure I2). In the estimation sample, and conditional on island and cohort fixed effects, initial settlement size is uncorrelated with the village-level geographic and agro-ecological characteristics, with market access, and with expected market access, so the instrument varies scale independently of connectivity (Appendix Table I2). In the full sample, larger settlements sit in slightly less connected places (Table 1), so I control for market access and expected market access in every specification. Several 1980 district characteristics, concentrated among measures of infrastructure and urbanization, are correlated with settlement size, and I control for them throughout. Settlement sizes were set by how many families a cleared site could hold and how many were in the pipeline, not by the quality of the land (Section 2), and Bazzi et al. (2019) present evidence consistent with this.

The exclusion restriction, $\text{Cov}(\ln W_v, \varepsilon_{vc}) = 0$ conditional on the controls, fails if larger settlements attracted infrastructure, traders, or services that raise crop productivity directly. Two features of the program limit this concern. First, because villages were created with the same household land allocation and internal infrastructure, differences in land, inputs, or infrastructure per farm are unlikely to arise. Second, the downstream channels a larger settlement might have attracted, a permanent market, a bank, or schools, are for the most part unrelated to settlement size in the 2003 PODES, and conditioning on them, which is not a fix since they are themselves mediators, does not substantially change the estimated scale elasticity (Appendix Table I3).

The second stage at the village-by-crop level is

$$\ln Q_{vc} = \beta \widehat{\ln L_v^{ag}} + \gamma_H \ln H_{vc} + X_v'\Gamma + \delta_c + \eta_i + \gamma_k + \varepsilon_{vc}, \tag{11}$$

where $\ln L_v^{ag}$ is instrumented by $\ln W_v$, δ_c are crop effects, η_i and γ_k are island and cohort effects, and X_v holds the village and district controls of Section 3, log market access, expected market access, and the village's number of tractors, which measures its agricultural capital. Standard errors follow Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. The private share γ_a is measured from a farm cost-structure survey as labor's share of the value of output, $\gamma_a = 0.28$ (Appendix J); my measure is similar to the results others have found in Indonesia using different methods (Fuglie, 2003, 2010). With it, I infer $\varphi = \beta - \gamma_a$.

This approach follows the same logic as the macroeconomic literature on returns to scale, which asks whether output rises with inputs by more than their cost shares imply (Caballero and Lyons, 1992; Basu and Fernald, 1997), and as Bartelme et al. (2025), who recover sector-level scale

economies from variation in sector size unrelated to productivity. It differs in three ways. The program shifts the size of the farm workforce directly, rather than through cyclical or demand-driven variation. I observe physical output rather than inferring prices. And because land is a fixed factor, I net out the private return to labor explicitly, whereas Bartelme et al. (2025) require that there be no diminishing returns internal to the sector.

Because the regression controls for planted area and capital, β is the return to labor at given land and capital. Agriculture in the model uses no capital. If it did, capital would leave agriculture along with labor, and the same externality would do more damage, much as variable capital utilization lowers the returns to scale needed for indeterminacy in Wen (1998). Adding rented capital to the simple model of Appendix F, with β held at its calibrated value, deepens the effect of connectivity on income and lowers the externality needed to produce it, so the model without capital is conservative.

8.2 Results

Table 6 reports the estimates. Column 1, the baseline on the main crops with the village and district controls, gives $\beta = 1.25$; the other columns add the non-agricultural village controls, interact the crop effects with island or cohort effects, or use yield per hectare as the outcome. Across the five specifications the point estimates of $\beta = \gamma_a + \varphi$ run from 1.20 to 1.42, and every one exceeds the private labor share of 0.28 by a wide margin: the implied externality $\varphi = \beta - \gamma_a$ runs from 0.92 to 1.14.

Table 6: Estimation of the Agricultural Scale Externality

	(1) ln(Production)	(2) ln(Production)	(3) ln(Production)	(4) ln(Production)	(5) ln(Yield)
$\ln(L_v^{ag})$	1.248***	1.413***	1.415***	1.420***	1.202***
AR 95% CI	[0.31, 6.65]	[0.35, 8.74]	[0.34, 8.65]	[0.38, 11.27]	[0.28, 6.42]
Effective F (Olea-Pflueger)	6.44†	5.89†	5.92†	5.26†	6.44†
Implied φ ($\gamma_a = 0.28$)	0.968	1.133	1.135	1.140	0.922
Implied φ lower bound (AR, $\gamma_a = 0.28$)	0.027	0.068	0.064	0.097	-0.001
Crop Sample	Main	Main	Main	Main	Main
Agriculture Controls	Yes	Yes	Yes	Yes	Yes
District Controls (1980)	Yes	Yes	Yes	Yes	Yes
Non-Ag Controls	No	Yes	Yes	Yes	No
FE Type	Crop + Isle + Cohort	Crop + Isle + Cohort	Crop X Isle + Cohort	Crop X Cohort + Isle	Crop + Isle + Cohort
Observations	1487	1487	1487	1487	1487
Villages	602	602	602	602	602

Note: 2SLS estimates. Inference allows for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. Every column controls for planted area, the village's number of tractors, log market access, expected market access, the village's potential yields, soil, drainage, climate, and agroclimatic similarity, and the 1980 district characteristics; columns 2 to 4 add the non-agricultural village controls (schooling, public school density, banks, and a permanent market). Significance stars reflect the Anderson-Rubin test of $H_0 : \beta = 0$ (weak-IV robust), not the Wald t-test. $\ln(L_v^{ag})$ is instrumented by ln(individuals initially placed). **Effective** F is the Montiel Olea and Pflueger (2013) first-stage statistic under the same spatial covariance; † flags values below 16.38, the threshold above which conventional Wald inference using $|t| > 1.96$ is approximately valid. **AR 95 percent CI** is the Anderson and Rubin (1949) confidence set, recommended by Andrews et al. (2019) as the primary weak-IV-robust inference object for just-identified models; it is reported in brackets beneath the point estimates. Implied externality $\varphi = \widehat{\beta} - \gamma_a$ at $\gamma_a = 0.28$, the private labor share measured in Appendix J. The *lower bound* on φ is computed as (AR lower bound on β) $- \gamma_a$ and reflects the smallest value of the externality elasticity not rejected by the data at the 5 percent level under weak-IV-robust inference.

Because the instrument is only moderately strong, with an effective F-statistic of Montiel Olea

and Pflueger (2013) between 5.3 and 6.4, I rely on the Anderson and Rubin (1949) confidence sets recommended by Andrews et al. (2019). They are bounded in every specification, though wide above. The data reject $\beta = 0$ in all five specifications and reject no externality, $\beta \leq \gamma_a = 0.28$, in four of five, where the implied lower bound on φ is 0.03 to 0.10. The model of Section 9 adopts $\varphi = 0.70$, below every point estimate.

Two further checks are in the appendix. Running the same 2SLS on data generated by the calibrated model, where the true coefficient is known, recovers $\gamma_a + \varphi$ to within 2 to 4 percent, and recovers γ_a when the externality is shut off (Appendix N.1). And planted area does not drive the result. Larger settlements were allocated more land, so planted area rises with settlement size, which is why the baseline controls for it; omitting it lowers the coefficient in column 2 from 1.41 to 1.09, which still implies an externality of 0.81 (Appendix I).

8.3 Magnitude

The externality implies that agricultural output scales roughly one-to-one with the size of the agricultural workforce in a village: even though land is a fixed factor, agriculture behaves as if it has constant returns to scale in labor at the village level. There is no estimate of scale economies in agriculture at this level to compare against; the nearest benchmark is the sector-level estimates of Bartelme et al. (2025) across manufacturing industries, which average 0.21 and range from 0.11 to 0.31, three to five times smaller than what I find. A potential reason for the gap is that my variation comes from relatively small movements in the size of villages, while that work uses variation at the country-by-sector level. If the externality operates through mechanisms with a minimum viable scale, such as equipment rental markets, the cooperative maintenance of irrigation, or learning from other farmers (Bassi et al., 2022; Foster and Rosenzweig, 1995; Conley and Udry, 2010; Caunedo et al., 2026; Ostrom and Gardner, 1993), then in villages of a few thousand people the marginal worker can matter for whether those mechanisms exist at all rather than how intensively they are used. Appendix K sketches where an externality of this size could come from and measures some potential channels directly in my setting.

The estimate is also large relative to the macroeconomic literature. The calibrated externality implies village-level returns to scale of 1.7 in labor and land jointly (0.98 in labor alone), while a typical two-digit U.S. industry has approximately constant returns (Basu and Fernald, 1997) and local agglomeration elasticities are typically 0.04 to 0.07 (Combes and Gobillon, 2015). Beyond the scale argument above, two features can account for part of the gap: the estimate is a long-run object, like that of Bartelme et al. (2025), rather than a cyclical one, and scale effects appear larger in poorer economies, where the density elasticity of wages in China is around three times that in developed countries (Combes and Gobillon, 2015). An externality of this size is nonetheless striking, which is why the calibration adopts a value below every point estimate and Section 10.2 reports the model across a wide range of values.

The scale argument also implies that the externality should weaken as the farm workforce

grows. The instrument is too weak to measure this at the village level, so I group neighboring transmigration villages into contiguous groups and into subdistricts, where the first stage is far stronger and agricultural workforces reach ten times a program village's. I find suggestive evidence that the estimated elasticity falls with sector size, although the decline is not statistically significant (Appendix I.1). Section 11.2 solves the aggregate counterfactuals again with an externality that declines at these rates.

9 A Quantitative Spatial Model

This section takes the framework of Section 6 to the data. Section 9.1 writes out the full spatial general equilibrium model of the Indonesian economy, and Section 9.2 describes how its inputs are measured and how it is calibrated.

9.1 Environment and Equilibrium

The full model embeds the structure of Section 6.1 at every location. The economy consists of 69,225 locations: roughly 190 million people across 48,598 rural villages (1,147 transmigration and 47,451 pre-existing) and 20,627 urban locations, on both the inner and outer islands (the program villages themselves are all located on the outer islands). Every location, rural or urban, carries the same two-sector economy, and all trade flows over the measured travel times. Baseline agricultural productivity $\bar{A}_{a,v}$ is common to all transmigration villages, whose fundamentals were assigned by the program; every other location's fundamentals are recovered from its observed outcomes (Section 9.2).

Households

A representative household lives at every location and has Cobb–Douglas preferences over food, a rural-service composite, and a city-good composite,

$$U_\ell = c_{a,\ell}^{s_{a,\ell}} \, Q_{n,\ell}^{s_{n,\ell}} \, Q_{u,\ell}^{s_{u,\ell}}, \qquad s_{a,\ell} + s_{n,\ell} + s_{u,\ell} = 1, \tag{12}$$

where the expenditure shares differ by location type. Rural households spend with shares $(\alpha_a, \alpha_n, \alpha_u)$ and urban households with shares $(\beta_a, \beta_n, \beta_u)$. The service composite aggregates the varieties of all rural origins by CES,

$$Q_{n,\ell} = \Big[\sum_{o \in \text{rural}} \omega_o^{1/\sigma_n} \, q_{n,\ell o}^{\frac{\sigma_n - 1}{\sigma_n}} \Big]^{\frac{\sigma_n}{\sigma_n - 1}}, \qquad \sigma_n > 1, \tag{13}$$

and the city-good composite aggregates the varieties of all urban origins the same way with elasticity σ_u. Every variety arrives at $\tau_{\ell o}^{\kappa_n} p_{n,o}$, and city varieties at $\tau_{\ell c}^{\kappa_u} p_{u,c}$. The variety weights ω_o convert location populations into village equivalents, with $\omega_o = N_o / \bar{s}$, where N_o is location o's population

from the settlement layer and $\bar{s}$ is the average rural village population. A location twice the average size carries twice the variety mass, and for consistency each transmigration village carries $\omega = 1$, but assigning every location one variety instead leaves the estimated model's moments essentially unchanged (Appendix N.4).[8]

Given location income I_ℓ, the sum of factor payments, which by constant returns equals the value of production, demand follows the Cobb–Douglas shares, and the exact cost-of-living index is

$$P_\ell = p_{a,\ell}^{s_{a,\ell}} \, P_{n,\ell}^{s_{n,\ell}} \, P_{u,\ell}^{s_{u,\ell}}, \tag{14}$$

where $P_{n,\ell}$ and $P_{u,\ell}$ are the CES price indices over service and city varieties. Because preferences are homothetic, every weight in (14) is an expenditure share, so income deflated by P_ℓ is the welfare-consistent real income measure used throughout.

Workers draw sector-specific productivities $(z_{i,a}, z_{i,k})$ independently from Fréchet distributions $F_j(z) = \exp(-T_j z^{-\theta})$ with $\theta > 1$ and work in the sector with the higher return, where the non-farm sector is services in a village and the city variety in an urban location. The agricultural employment share and effective labor supplies are

$$\pi_{a,\ell} = \frac{T_a w_{a,\ell}^\theta}{T_a w_{a,\ell}^\theta + T_k w_{k,\ell}^\theta}, \qquad Z_{j,\ell} = L_\ell \, \Gamma\!\left(1 - \tfrac{1}{\theta}\right) T_j \, w_{j,\ell}^{\theta-1} \left(T_a w_{a,\ell}^\theta + T_k w_{k,\ell}^\theta\right)^{\frac{1}{\theta}-1}, \tag{15}$$

and average earnings are equalized across sectors, so employment shares are the same labor-income shares. This property ties the observed level of agricultural employment to the demand structure and makes the employment share the natural target for the sorting scale T_a/T_k.

Trade

Food is homogeneous and traded, and its market price is the numéraire. Every location, rural or urban, grows food and sells it at the market price net of transport to its nearest urban market, $p_{a,\ell} = \tau_\ell^{-\kappa_a}$, so cities receive the market price and a village at travel cost τ_v from the nearest urban location receives the farm-gate price $\tau_v^{-\kappa_a}$. A village's own food demand is satisfied from its own supply at the farm-gate price, and only the excess is exported, so proximity helps agriculture through this channel.[9] The economy is closed, and by Walras' law the national food market's clearing condition is redundant given the others.

A unit of the service is a unit of labor, so delivering variety o to location ℓ means a provider travels at iceberg cost $\tau_{\ell o}^{\kappa_n}$. The city good combines physical goods, which ship like food, and

[8] Scaling variety mass with population is the standard device for carrying locations of very different sizes in a single Armington structure (Ramondo et al., 2016).

[9] For a rural net food importer, the single farm-gate price misstates the relevant price, and the cost-of-living index (14) understates the consumer price of food, by at most the arbitrage band $2\kappa_a \ln \tau_v$, under 3 percent of the index where importers cluster. I use a single price because switching prices village by village adds many nonlinearities to the solver. The model is exact for net exporters and approximate for the importing remainder, which holds 11 percent of the rural population (Appendix M.4).

person-embodied town services, which behave like the rural service, so κ_u is built from the two frictions in proportion to their measured composition and is much flatter than κ_n; buyers at v receive city c's variety at $\tau_{vc}^{\kappa_u} p_{u,c}$. Trade in services does not conflict with immobile labor: every worker is paid their own village's wage, and what the assumption rules out is commuting, which is rare in rural Indonesia (Section 7).

Every buyer, rural or urban, spends the service share of its income (α_n or β_n) on the rural-service composite and allocates it across origins by CES shares. Village v's service revenue is therefore

$$p_{n,v} Y_{n,v} = \omega_v \, p_{n,v}^{1-\sigma_n} \Big[\beta_n \sum_{\ell \in \text{urban}} \tau_{\ell v}^{-\kappa_n(\sigma_n-1)} \frac{I_\ell}{\Phi_\ell} \; + \; \alpha_n \sum_{\ell \in \text{rural}} \tau_{\ell v}^{-\kappa_n(\sigma_n-1)} \frac{I_\ell}{\Phi_\ell} \Big], \tag{16}$$

where $\Phi_\ell = \sum_{o \in \text{rural}} \omega_o \big(\tau_{\ell o}^{\kappa_n} p_{n,o}\big)^{1-\sigma_n}$ is buyer ℓ's competition index over all rural locations, with a rural buyer's own variety included at $\tau = 1$, and incomes I_ℓ are equilibrium objects. City-good revenue has the identical gravity structure over city varieties with (κ_u, σ_u), the buyers' city-good budgets α_u and β_u, and the competition index taken over urban locations.

Equilibrium

Given the measured geography, populations, endowments, fundamentals, and parameters, an equilibrium is a set of wages $\{w_{a,\ell}, w_{k,\ell}\}$ and goods prices $\{p_{a,\ell}, p_{k,\ell}\}$ at every location such that: (i) workers choose the sector that pays them more, so employment and effective labor follow (15); (ii) households maximize (12) given the delivered price of every variety and their income I_ℓ, the sum of wages and land rents; (iii) firms maximize profits taking prices and the externality as given, so the agricultural wage equals labor's private marginal product, $w_{a,\ell} = \gamma_a p_{a,\ell} Y_{a,\ell} / Z_{a,\ell}$, and each non-farm variety sells at unit cost, $p_{k,\ell} = w_{k,\ell} / A_{k,\ell}$; (iv) the food price is the market price, the numéraire, net of transport to the nearest urban market, $p_{a,\ell} = \tau_\ell^{-\kappa_a}$; and (v) the market for every variety clears, service revenue (16) for villages and the analogous variety revenue for cities. The national food market then clears by Walras' law. Because (iii) and (iv) tie every goods price to a location's wages and its distance to market, the equilibrium reduces to a wage pair at every location: given wages, I compute the competition indices and demand sums, and given those aggregates each location's two-equation system delivers its wages.[10]

9.2 Taking the Model to the Data

Most of the model's inputs are measured directly, and the fundamentals of each location are recovered by inverting the model. For every location that is not a program village, I choose the agricultural and non-farm productivities that exactly match its observed income per capita and

[10] Uniqueness cannot be guaranteed because the externality raises returns to scale in agriculture above one in labor and land jointly. I solve at $\varphi = 0$, where the equilibrium is unique under standard gravity conditions, and follow that branch as φ rises to its calibrated value. I verify that solving the model without this continuation, and after perturbing every price, converges to the identical equilibrium at every location (Appendix M).

agricultural employment share, the standard way of calibrating a spatial model, so the baseline economy reproduces the observed Indonesian economy by construction. The program villages instead carry *common* fundamentals, since they were built to be alike (Section 4), and their incomes are solved from the model given only their geography, so the relationship between market access and income among them is a prediction of the model rather than something it was fitted to. Table 7 collects the parameter map, and Appendix L gives the measurement in full.

Geography, endowments, and outcomes

The trade-cost matrix is the travel-time network of Section 3, evaluated between all locations, and the variety weights use the same population layer as the market-access measure. Every program village is assigned the mean recorded labor and land endowment of the program, so that geography alone generates differences among them, and all other land comes from the 2003 Agricultural Census of household landholdings (Appendix L.4). The observed outcomes that the inversion matches are income per capita, from the same village-GDP series as Section 5 normalized to the measured rural income level, and the agricultural employment share, from the 2010 census where locations can be matched and from the 2000 census elsewhere, adjusted for the province-level change between the two waves. I classify as urban every administratively urban location and every location whose centroid falls inside a satellite-defined functional urban area, which raises the urban population share from the administrative 44 percent to just under 65 percent, closer to aggregate statistics; Appendix N.5 shows that the results do not depend on where the boundary is drawn.

Measured parameters

Technology. The private agricultural labor share is $\gamma_a = 0.28$, labor's share of the value of output in the 2017 BPS farm cost-structure survey for paddy in South Sumatra, a transmigration province (Appendix J); alternative treatments of contracted work and family labor put it between 0.25 and 0.31, and it agrees with Fuglie's aggregate estimates for Indonesia (Fuglie, 2003, 2010). The land share $1 - \gamma_a$ covers every input that is not labor. The externality is set to $\varphi = 0.70$, a deliberately conservative reading of Section 8: it lies below every 2SLS point estimate while remaining inside every confidence interval, and Sections 10 and 11 report results across a wide range of values and at $\varphi = 0$. As a check that the estimator measures the model's parameter, I simulate the program villages in the calibrated model, using their recorded settlement sizes, and run the regression of Section 8 on the simulated data, where the true externality is known. The regression recovers $\gamma_a + \varphi$ to within 2 to 4 percent (Appendix N.1).

Trade frictions. I measure the three trade frictions from the 1993 IFLS community survey, which records travel times and fares to nearby destinations, local wages, doctors' fees, and staple prices for 312 communities (Appendix L.2). Because a service is delivered in person, its trade cost is the cost of the trip, in time and fares, as a share of the value of the service. How fast this share rises with travel time gives κ_n: 0.94 when the service is valued at a day's wage for a provider who

travels, and 0.89 when it is valued at a doctor's fee for a patient who travels; I set $\kappa_n = 0.90$. Priced as freight, the same trips give κ_a of 0.10 to 0.12, an upper bound since commercial shipments carry far more per trip, and village rice prices show no relationship with travel time, so I set $\kappa_a = 0.05$ and vary it between half and twice that value (Appendix N.7). The city good is 82 percent physical goods, which ship like food, and 18 percent health and education services, which move with people, so $\kappa_u = 0.82\kappa_a + 0.18\kappa_n = 0.203$. Appendix N.8 shows what happens when the service and goods frictions are set to each other's values.

Demand. I measure rural expenditure shares on outer-island rural households in the 2002 SUSENAS, in value-added terms. Many purchases that appear as local services, such as prepared food and small-scale retail, combine local service labor with purchased food, so I assign a fraction $\ell = 0.48$ of these purchases to service labor and the remainder to food, using Economic Census cost data (Appendix L.1). The survey does not record whether a service was produced within the village or outside it, so the local sourcing share s^{own} that this crosswalk requires is estimated inside the model below; at the estimated value the rural shares are $(\alpha_a, \alpha_n, \alpha_u) = (0.667, 0.103, 0.230)$. The urban shares are implied by market clearing, and the implied urban food share of 0.595 is close to the 0.560 measured independently in the survey; other survey years and samples give similar results (Appendix N.6).

Elasticities. Preferences across food, rural services, and the city good are Cobb–Douglas, a restriction the SUSENAS supports: expenditure shares on city goods do not fall as their delivered price rises with isolation (Appendix L.6). For the sorting elasticity θ, I impose the value of 12.5 (s.e. 1.36) that Bryan and Morten (2019) estimate for Indonesian workers choosing among locations, the closest measured elasticity to the within-village choice of sector that the mechanism runs on. Freeing θ gives a value just above their confidence interval with every prediction essentially unchanged (Appendix N.2), and because more homogeneous workers deepen the curse (Section 6.2), imposing their estimate is conservative. There is no accepted estimate of the substitution elasticity across rural services σ_n; available values range from about 2.8 across service industries (Gervais and Jensen, 2019) to about 9 across establishments within a city (Couture, 2016). I set $\sigma_n = 5$, equal to the city-variety elasticity σ_u; the moment used in the estimation below is nearly flat in σ_n over this range, and Appendix N.14 sweeps σ_u.

Inversion and internal calibration

For each rural location other than a program village, I choose the agricultural productivity and the service-sector fundamental that make the model reproduce its observed income per capita and agricultural employment share; two observed outcomes recover two fundamentals. Urban locations are treated the same way, with an agricultural productivity and a city-good fundamental. The program villages receive common agricultural and service fundamentals, and their outcomes are solved from the model; the one common parameter, the agricultural sorting scale T_a, is chosen so that the model reproduces the program-wide agricultural employment share of 0.648. Appendix M.3

Table 7: Parameter map

Parameter	Value	Source
A. From the literature		
θ	12.5	Bryan and Morten (2019) choice elasticity (s.e. 1.36)
σ_n, σ_u	5	within the range of Gervais and Jensen (2019) and Couture (2016); swept in App. N.14
B. Measured directly		
γ_a	0.28	labor's share of output, BPS farm cost-structure survey, South Sumatra 2017 (App. J)
κ_n	0.90	IFLS 1993 trip costs, two routes (0.94 delivery, 0.89 consumption)
κ_a	0.05	IFLS 1993 freight costs (upper bound 0.10–0.12) and village rice prices
κ_u	0.203	κ_a and κ_n mixed at SUSENAS rural consumption shares (0.82, 0.18)
ℓ	0.48	Economic Census labor content of local services, band [0.42, 0.54]
C. Estimated in this paper		
φ	0.70	2SLS on initial settlement size, $\hat{\beta} - \gamma_a$ (Section 8); below every point estimate
D. Estimated inside the model (Section 9.2)		
s_{own}	0.533	matches the structural-transformation slope b_{ST} among program villages
T_a	5.4×10^{-23}	matches the program agricultural employment share (0.648)
E. Implied by identities or inverted from observed outcomes		
$\alpha_a, \alpha_n, \alpha_u$	0.667, 0.103, 0.230	SUSENAS 2002 expenditure shares through the value-added crosswalk at the estimated s_{own}
β_n, β_u	0.024, 0.381	service and city-good market clearing at observed incomes
β_a	0.595	residual; Engel-implied survey value 0.560 (overidentifying check)
$(\bar{A}, T_n)$ per rural location	fields	observed income and agricultural share, non-program villages
$(\bar{A}, T_u)$ per urban location	fields	observed income and agricultural share, urban locations

Note: The scale of T_a reflects θ in its exponent and has no free interpretation.

gives the algebra and computational details.

One parameter remains to be estimated inside the model, the local sourcing share s^{own}. Through the value-added crosswalk it sets the rural service expenditure share α_n, and so how strongly access to outside service demand pulls labor out of agriculture: a lower s^{own} implies a larger α_n and a stronger shift. I estimate it from that relationship, targeting the coefficient b_{ST} from a regression of the non-agricultural employment share on the same reduced-form measure of market access used in Section 5, among the transmigration villages, which in the data is 0.0053 (s.e. 0.0009).[11] For each candidate value I re-invert the economy, solve the full equilibrium, and run the same regression in the model. The value that matches the data is $s^{\text{own}} = 0.533$, inside the range of 0.471 to 0.904 that the survey's purchase categories allow, and both targeted moments are matched exactly (Appendix Table M1). As an untargeted check, the crosswalk at this value implies an own-village share of service consumption of about 0.35, close to the 0.30 that the spatial model produces in equilibrium.

10 Model Validation

This section asks whether the estimated model reproduces the negative relationship between market access and income among the program villages. The relationship between market access and structural transformation is targeted; the relationship with income is not, and there is no guarantee that the model produces a negative one, let alone one of the magnitude in the data. Section 10.1 reports that comparison, Section 10.2 traces the gradient across values of the externality to locate

[11] Both the data and model regressions include island and cohort fixed effects, with standard errors clustered by village in the data. The coefficient is larger than in Section 5 because I am only using the 2010 Population Census in order to have measurements as close to the 2012 measure of GDP per capita that is the other primary input.

the point at which its sign flips, Section 10.3 asks whether the model needs worse fundamentals in connected villages to fit the data, and Section 10.4 asks whether, as in the model, the negative effects of market access in the data comes from population near the village.

10.1 The Income Gradient

Table 8 compares the slope of log GDP per capita on log market access among the transmigration villages in the data and in the model, estimated the same way in both, with island and cohort fixed effects, on the reduced-form access measure of Section 5. The rows differ only in the sample: every program village, or the sample that excludes the most isolated 5 or 10 percent. The last row reports the model slope with the externality removed.

I find that the model explains between roughly 40 and 130 percent of the effect in the data, depending on the sample of transmigration villages. The model cannot match the high incomes of the most isolated program villages, so I focus on the sample that excludes the most isolated 5 percent, where the model explains just over 80 percent of the relationship between market access and GDP per capita (Figure 3). The premium earned by the most remote villages is a distinct phenomenon: Appendix N.3 traces how the slopes move as those villages are excluded, and Section 10.3 shows that the inversion would load it into fundamentals.

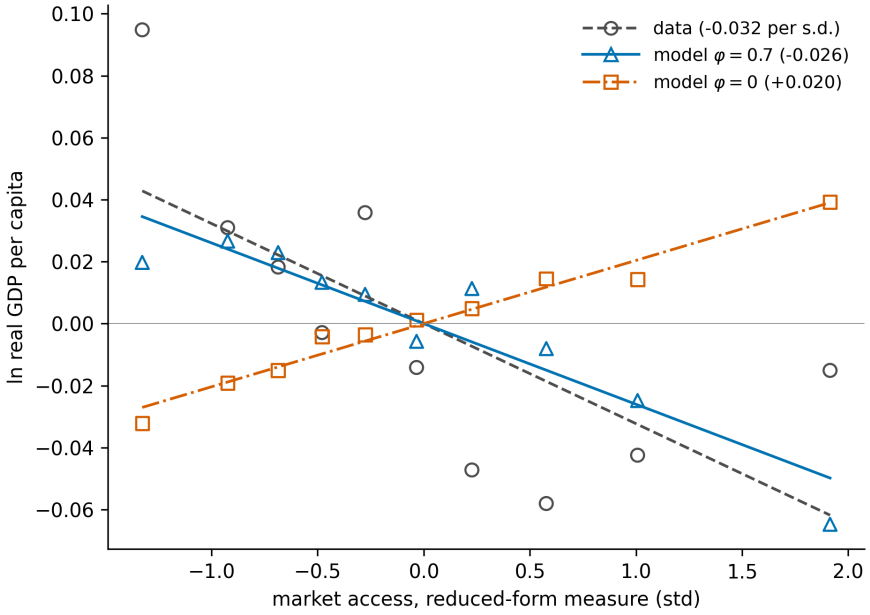
Table 8: Income gradients: model against data

Slope of ln GDP pc	Sample	Data (s.e.)	Model	
reduced-form measure, real	full	−0.0066 (0.0016)	−0.0028	untargeted
reduced-form measure, real	excl. bottom 5%	−0.0047 (0.0017)	−0.0038	untargeted
reduced-form measure, real	excl. bottom decile	−0.0030 (0.0018)	−0.0040	untargeted
reduced-form measure, real, $\varphi = 0$	full	—	+0.0029	untargeted

Note: Regressions include island and cohort fixed effects, district-clustered standard errors. Model slopes are computed in equilibrium on the reduced-form access measure of Section 5, with model income per capita deflated by each village's price index. The $\varphi = 0$ row solves the full economy with the externality removed and the productivity fields re-inverted to reproduce the same observed baseline, with the transmigration agricultural sorting scale re-pinned to match the observed mean program agricultural employment share.

Shutting off the externality reverses the effect, and Figure 3 includes the relationship between market access and income when there is no externality. With $\varphi = 0$ and the world re-inverted to the same observed baseline, the same geography generates a *positive* relationship between market access and income. As the model predicts, connectivity becomes a blessing (better farm-gate prices, cheaper goods, service-export demand) once agricultural productivity no longer depends on agricultural scale. The externality is not one contributor to the curse among several, but the driver of the negative relationship between market access and GDP per capita.

Figure 3: Income and Market Access in the Data and the Model



Note: The figure plots the relationship between income and market access for transmigration villages in the data and in the model, excluding the lowest 5 percent of villages in market access. For both data and model, it plots 10 quantile bins of the island-and-cohort fixed-effect residuals for the 1,089 villages, with the fitted slope per standard deviation of log access in the legend, with and without the externality.

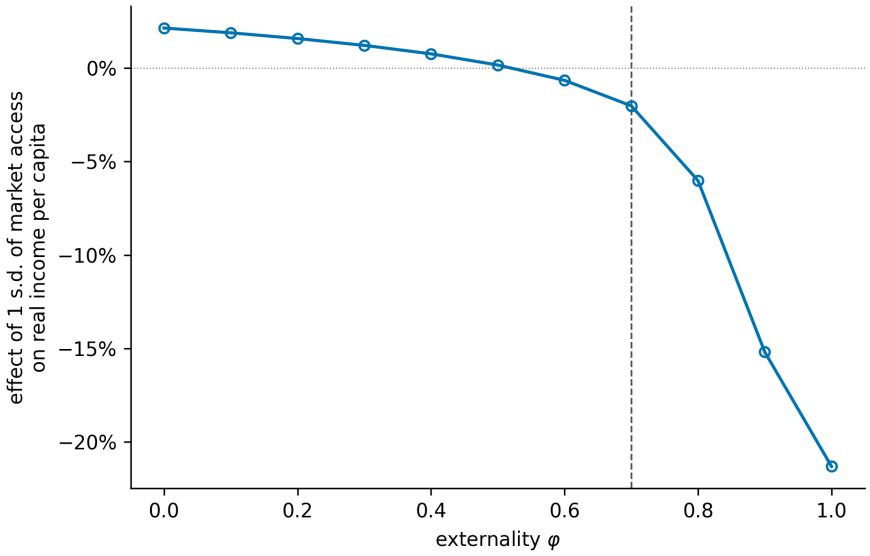
10.2 The Externality Threshold

Proposition 1 says that proximity becomes a curse once the externality exceeds a threshold, and the full model locates that threshold. Figure 4 traces the model's cross-sectional access gradient, the slope of program-village log real income on log market access estimated as in Table 8, across values of the externality. Each point re-solves the whole economy at that φ, with T_a re-pinned and the non-program productivity fields re-inverted so that every φ reproduces the same observed baseline (above φ of roughly 0.8, only approximately; Appendix N.13). The gradient reverses between $\varphi = 0.5$ and 0.6, and the calibrated $\varphi = 0.70$ sits inside the curse region. This is the quantitative counterpart of the sign-reversal thresholds of the simplified models, about 0.38 in the closed-form case once the measured frictions are included (Appendix G) and 0.33 to 0.65 across the variants (Appendix F). The point estimates and the calibrated value clear the threshold; the most conservative readings of Section 8, the weak-instrument-robust lower bounds of 0.03 to 0.10 in four of five specifications, do not.

A contest between the size of an externality and a threshold is familiar from the macroeconomic literature on increasing returns. In the one-sector model of Benhabib and Farmer (1994), indeterminacy requires aggregate returns to scale of at least about 1.4, in the upper range of available estimates (Benhabib and Wen, 2004; Schmitt-Grohé, 1997), and later work brought the requirement

down to about 1.1 by adding margins along which labor responds more elastically (Benhabib and Farmer, 1996; Wen, 1998). The threshold here behaves the same way: it falls as workers become more homogeneous (Section 6.2), and estimating the sorting elasticity inside the model raises it from 12.5 to 16.3 (Appendix N.2), the direction that lowers the threshold.

Figure 4: The income gradient across externality values



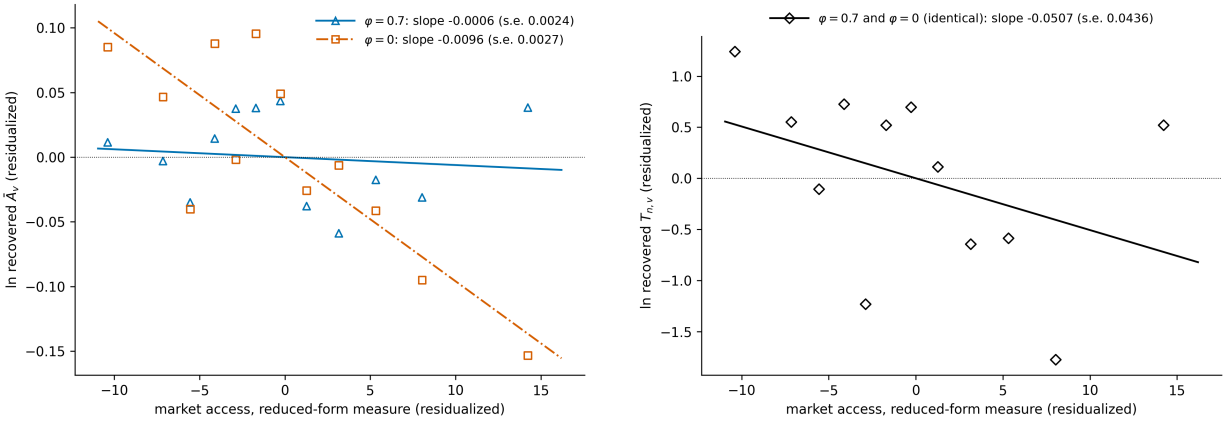
Note: The figure plots the model's income gradient, the effect of one standard deviation of log initial market access on real income per capita, in percent. It is the slope of log real income per capita on log initial market access among the 1,147 transmigration villages, with island and cohort fixed effects, on the reduced-form access measure of Section 5, multiplied by 7.3, the standard deviation of log initial market access. Each point re-solves the whole economy at the stated φ, re-pinning the program villages' agricultural sorting scale so that the baseline reproduces the observed program agricultural employment share. The dashed line marks the calibrated value, $\varphi = 0.7$. Above φ of roughly 0.8, some program villages stop farming altogether and the re-solved economy reproduces the observed baseline only approximately (Appendix N.13).

10.3 Structural Balance

The balance tests of Section 4 find no evidence that observable determinants of income vary with initial market access, and the model permits a structural analog of the same test. Inverting each transmigration village with the same process used for every other location recovers the fundamentals that reproduce its observed employment share and income exactly, and I ask whether the estimated model needs those fundamentals to vary with market access to fit the data. It does not. Figure 5a shows no relationship between a transmigration village's recovered agricultural productivity and market access, and Figure 5b shows the same null for the service fundamental. This null is a property of the model *with* the externality only. The same exercise in the $\varphi = 0$ world uncovers a strong negative relationship between market access and baseline agricultural productivity: without

the scale externality, only worse fundamentals can rationalize connected villages being poorer. Since I observe no relationship in the data between market access and many measures that capture agricultural productivity, a model without the externality is at odds with the data.

Figure 5: Recovered Productivity and Market Access



(a) Agricultural productivity

(b) Service fundamental

Note: Each panel plots a recovered program-village fundamental against market access, residualized on island and cohort fixed effects, in 12 quantile bins with the linear fit, with the slope and district-clustered standard error in the legend. Panel (a) plots agricultural productivity with the externality ($\varphi = 0.7$) and without it ($\varphi = 0$). Panel (b) plots the service fundamental, which the inversion recovers without reference to φ and which is therefore the same in both worlds.

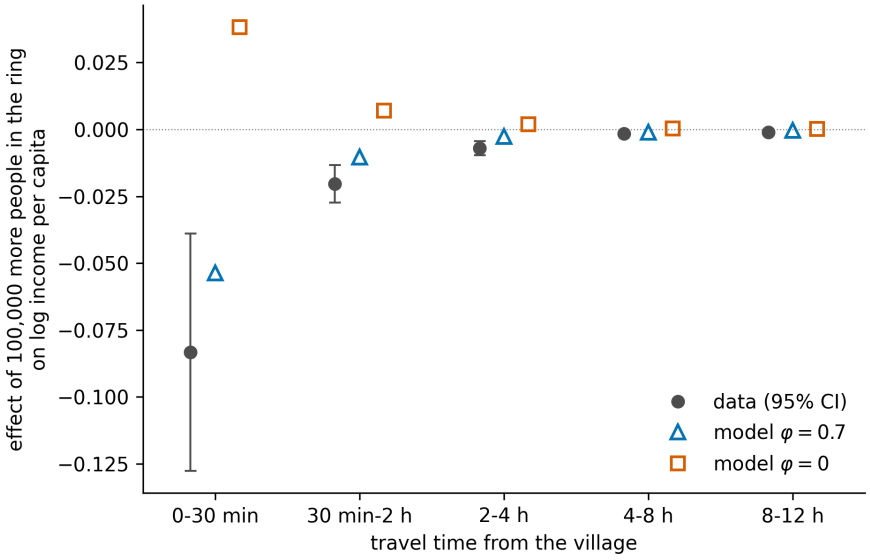
10.4 Near and Distant Market Access

In the model, connectivity lowers a program village's income because nearby population pulls its workers into services. Intuitively, the model then attributes the negative effects of market access mainly to areas within a few hours of the village. Whether the data show the same pattern is an untargeted check. Figure 6 regresses log income per capita on the 1980 population within each of five travel-time bands around a program village in the data and in the model. Per person, the effect is concentrated locally in both the data and model. In the data, 100,000 more people within 30 minutes of a village lower its income per capita by 8 percent, those 30 minutes to 2 hours away by 2 percent, and those 8 to 12 hours away by 0.1 percent. The model with the externality reproduces this general pattern. While without the externality, the model has the opposite sign at every distance.

11 Counterfactuals

This section uses the estimated model to evaluate policy. Every aggregate experiment re-solves the full joint general equilibrium under the counterfactual shock, so the baseline is the observed economy. The single-village experiments of Section 11.1 instead solve the treated village against

Figure 6: Income and Population by Travel Time



Note: Each point is the coefficient from a regression of log income per capita among program villages on the 1980 population, in 100,000s, whose travel time from the village falls in the band on the horizontal axis, with no other band in the regression. Data: log GDP per capita, with the expected market access control, the village and 1980 district controls, and island, cohort, and year fixed effects, for 1,151 villages; the 95 percent confidence intervals use Conley standard errors with a Bartlett kernel and a 100 kilometer cutoff. Model: log real income per capita in the calibrated model ($\varphi = 0.7$) and in the re-inverted externality-free world ($\varphi = 0$), for 1,147 villages, with island and cohort fixed effects.

the fixed rest of the economy, which is the appropriate frame for a shock to one of 69,225 locations. Fundamentals are held fixed at their baseline inverted values throughout, and every experiment is also solved at $\varphi = 0$, in the externality-free world re-inverted to the same baseline, to quantify what the externality changes.

11.1 A Better Road Lowers a Program Village's Income

The first experiment improves a single transmigration village's travel times by 10 percent and nothing else, for each of the 1,147 villages in turn. With or without the externality, improved connectivity leads to reallocation out of agriculture and into rural services. With the externality in place, improving the connectivity of transmigration villages *lowers* the improved village's real income by 2.5 percent on average, with 60 percent of villages made worse off, while without the externality every village is made better off (Figure 7a). Table 9 decomposes the effect by channel. In both economies villages benefit from cheaper access to goods, but the increase in external demand for services is the largest channel, and it pulls workers out of agriculture regardless of whether there is an externality. With the externality, the erosion of agricultural productivity that comes with that reallocation outweighs the gains from this and every other channel. Observing structural transformation is therefore not a sufficient condition for improved welfare.

Table 9: The targeted road, by channel and by world

	$\varphi = 0.7$ (measured)		$\varphi = 0$	
Channel	Δ ln real income	Δ agric. share	Δ ln real income	Δ agric. share
service-demand access alone	−4.2%	−4.3 pp	+0.5%	−2.1 pp
import competition, cheaper services	+1.0%	+0.6 pp	+0.4%	+0.4 pp
farm-gate price	+0.3%	+0.2 pp	+0.1%	+0.1 pp
urban-goods price	+0.5%	0.0 pp	+0.5%	0.0 pp
full shock	−2.5%	−3.5 pp	+1.4%	−1.6 pp
share of villages losing	60%		0%	

Note: Population-weighted means across the 1,147 single-village experiments. Each experiment cuts one village's excess travel times by 10 percent while holding all other locations fixed. Each channel row applies the same shock to one trade friction at a time, holding the others at baseline. The $\varphi = 0$ columns run the identical experiments in the externality-free world. Figure 7a plots the distribution of effects across villages.

Rural roads have had similar effects elsewhere. Asher and Novosad (2020) find that India's rural road construction program moved workers out of agriculture without raising incomes, and the model supplies a mechanism for that pattern: the road shifts workers toward service production, and the uncompensated externality turns that reallocation into a loss in income.

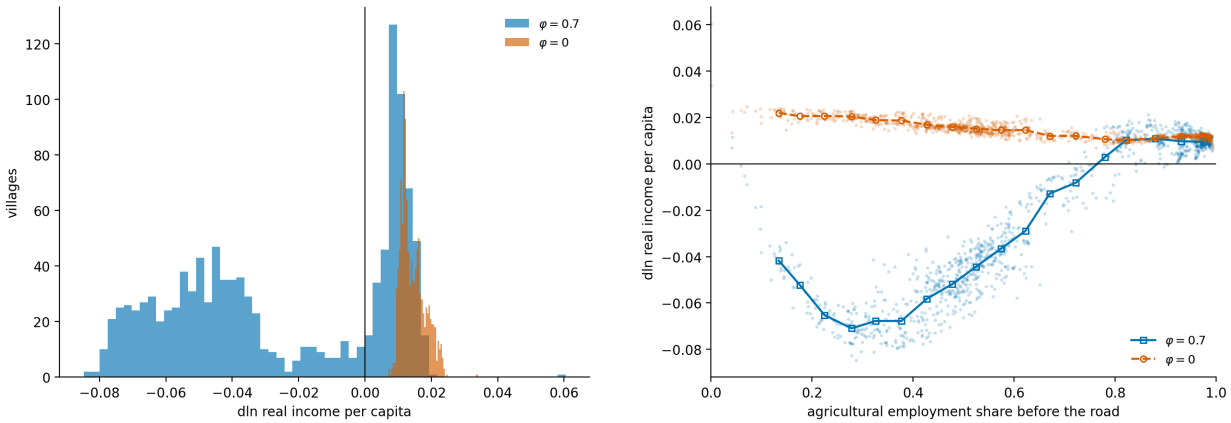
The same experiment also answers a design question about the program. The program placed its villages in isolated areas, with the median transmigration village sitting below the twentieth percentile of the access distribution. Improving every program village's travel times by 10 percent at once lowers the settlers' real income by 1.8 percent with the externality and raises it by 1.3 percent without it, with a national effect near zero either way. Under the externality, the program's haphazard remoteness sheltered the settlements rather than harming them.

The model also predicts which villages a road hurts. Figure 7b plots the change in real income from the single-village road against each village's agricultural employment share before the shock. Without the externality every village gains between 1 and 2 percent.[12] With it, nearly every village with a share between roughly 0.1 and 0.75 loses, most of all near 0.3, where the loss approaches 7 percent, while villages above roughly 0.8 still gain. As Proposition 1 predicts, a road does the most damage where both sectors employ many workers, since there the reallocation it causes is largest, and little where nearly everyone farms, since there is little non-farm employment for the road to expand.

In the model, nine in ten program villages start with an agricultural share above 0.3, so among them a road hurts the more agricultural villages less. I do not observe how agricultural a village would have been at different levels of market access, but potential crop yields, which all else equal keep workers in farming, offer an imperfect test. Interacting market access with potential yields gives the predicted pattern: one standard deviation above the mean of a potential-yield index, the negative effect of market access on GDP per capita is less than half as large, and one standard deviation above the mean of rice potential yield it is about a quarter as large (Appendix Table H3).

[12] The slightly larger gains for the least agricultural villages are due to the fact they are able to take advantage of the cheaper imports slightly more than more isolated, primarily agricultural, villages.

Figure 7: Effects of a Road Built to a Single Program Village



(a) Distribution of income effects

(b) By initial agricultural employment share

Note: Both panels plot the change in log real income per capita across the 1,147 single-village experiments, each of which cuts one village's excess travel times by 10 percent and re-solves that village against the unchanged rest of the economy, with the externality ($\varphi = 0.7$) and without it ($\varphi = 0$). Panel (a) plots the distribution of effects. Panel (b) plots each village's effect against its agricultural employment share before the shock, with lines connecting means within bins of width 0.05 that contain at least 8 villages.

11.2 Aggregate Programs

The remaining experiments improve the entire road network, subsidize agricultural earnings, and raise rural agricultural productivity, shocks of a size that move the aggregate agricultural employment share in the data. The model of Section 9 has homothetic demand, so the composition of national spending does not respond to shocks and even large shocks cannot generate *aggregate* structural transformation. That shuts down a response these policies should produce, and it is the response the externality attaches a cost to. For these experiments I therefore extend the model with the price-independent generalized linear (PIGL) demand system of Boppart (2014), which Fan et al. (2023) use for India, under which households shift spending away from food and toward services and city goods as they grow richer.

Budget shares take the form $s_{g,\ell} = \omega_{g,\ell} + \nu_g \, x_\ell^{-\varepsilon}$, where x_ℓ is real expenditure per capita, $\omega_{g,\ell}$ is the share a household approaches as it grows rich, anchored so that each location reproduces its observed baseline shares, and the loading ν_g is positive for food and negative for the other goods; homothetic demand is the case $\nu_g = 0$. I estimate the curvature ε with the estimator of Fan et al. (2023) on the same expenditure survey that measures the budget shares and obtain $\varepsilon = 0.34$, nearly identical to their 0.33 for India (Appendix L.7). Because the system is anchored to each location's observed budget, the calibration and every result of Section 10 are unchanged. This is why I keep the simpler homothetic model everywhere else: the extension changes nothing in the estimation or the validation, and only shocks large enough to move national income engage it. On its own the extension does almost nothing: at $\varphi = 0$, switching demand systems changes the aggregate gains

from the road and productivity experiments by about 1 percent of their size. Whatever the income margin contributes is therefore the interaction of the externality with non-homothetic demand.

A uniform road improvement and an agricultural subsidy

A uniform 10 percent cut in all travel times makes every type of location better off, with or without the externality, but the externality shrinks the aggregate gain. National real income rises by 0.73 percent with the externality against 0.86 percent without it (Table 10), so ignoring the externality overstates the gain by 17 percent. As real incomes rise, households spend a smaller share of their budget on food, the fall in demand for agricultural output pulls workers out of agriculture, and the externality lowers agricultural TFP. Nearly all of the difference runs through this income response: under homothetic demand the gains are 0.83 percent with the externality and 0.85 percent without it (Appendix Table O1). Maintaining a network improvement of this size would cost at least 0.68 percent of GDP a year (Appendix O.4), so the improvement is just worth its cost in the favorable case, and if raising speeds requires costlier works the benefits are well under half of the cost.

Given an externality, a natural question is what policies that target it directly do. Agricultural support programs are commonplace in developing countries, and I consider two subsidies on agricultural earnings, each financed *within* the subsidized village by a lump-sum tax on its households. The corrective subsidy applied to the transmigration villages alone raises their real income by 24.2 percent and turns the relationship between market access and income among them positive, but it lowers national income slightly (-0.06 percent), because the subsidized villages expand agriculture and push nearby villages out of it.[13] A national planner has no reason to correct the externality in these villages alone, so the second subsidy spreads the road program's annual budget, 0.68 percent of GDP, across every rural village, a rate of about 8 percent on rural agricultural earnings. At essentially no resource cost, it raises national income by 0.20 percent and delivers roughly twice the road program's gains in rural villages, with a small loss in urban ones (Table 10). In the $\varphi = 0$ world the same subsidy is a distortion and generates a small loss. Agricultural support programs are usually defended as redistribution and criticized as distortions; where agriculture carries scale economies, they can also raise aggregate income.

Agricultural productivity growth

The externality also changes the aggregate effect of the force usually credited with producing structural transformation. I raise the agricultural TFP of every rural location by 5 percent. Table 11 shows that national income rises by 0.74 percent with the externality against 1.21 percent without it, so ignoring the externality overstates the gain by 64 percent. Two things account for the gap. First, rural productivity growth shifts agricultural activity from urban locations to rural ones, urban agricultural employment falls by almost 2 percentage points, and the externality erodes the

[13] The corrective rate is $s^* = (\gamma_a + \varphi)/\gamma_a - 1 \approx 2.5$.

Table 10: Roads and agricultural subsidies compared

	$\varphi = 0.7$						$\varphi = 0$
Instrument	Gross natl.	Cost	Net natl.	TM	Outer rural	Urban	Net natl.
uniform road network	+0.73	0.68	+0.05	+0.6%	+0.9%	+0.6%	+0.18
agricultural earnings subsidy, same budget	+0.20	≈ 0	+0.20	+2.2%	+1.9%	−0.5%	−0.01
memo: full corrective s^*, program villages only	−0.06	≈ 0	−0.06	+24.2%	−2.0%	−0.1%	—

Note: The gross, cost, and net columns are percent of national income per year; the TM, other rural, and urban columns are within-group percent changes in real income. Other rural is every non-program rural village, on the inner and outer islands. The road row is the uniform 10 percent cut in all travel times, solved under non-homothetic demand. The subsidy rows apply a proportional subsidy to rural agricultural earnings, financed within each subsidized village by a lump-sum tax on its own households. The cost column records real resource costs, which arise only for the road. The first two rows operate at a common fiscal outlay of 0.68 percent of GDP, and the memo row applies the full corrective wedge, which requires an outlay of 1.35 percent. The $\varphi = 0$ column re-solves each experiment in the re-inverted externality-free world.

cities' agricultural TFP; this channel operates under either demand system, and with homothetic demand the gains are still roughly 25 percent smaller with the externality. Second, as incomes rise, spending shifts away from food, which moves more labor out of agriculture, and the externality taxes that reallocation. Without the externality the choice of demand system has little effect on the aggregate gains (1.21 against 1.20).

Table 11: Rural agricultural productivity growth

Rural agricultural TFP +5%	TM	Outer rural	Urban	National
Non-homothetic demand				
measured externality ($\varphi = 0.7$)	+4.5	+4.5	−0.8	+0.74
externality off ($\varphi = 0$)	+3.9	+3.9	+0.2	+1.21
Homothetic demand				
measured externality ($\varphi = 0.7$)	+4.6	+4.7	−0.7	+0.89
externality off ($\varphi = 0$)	+4.1	+4.1	+0.1	+1.20

Note: Population-weighted changes in log real income, in percent, and the change in national real income, from a 5 percent rise in the agricultural TFP of every rural location. Other rural is every non-program rural village, on the inner and outer islands. The upper rows solve the model under non-homothetic demand and the lower rows under homothetic demand. The $\varphi = 0$ rows re-solve the identical experiment in the re-inverted externality-free world.

An externality that declines with scale

The calibrated model applies the externality I estimate in program villages to every location, including places with larger farm workforces, but Section 8.3 suggests that it weakens with scale. I therefore solve both experiments again with an externality that is 0.70 in a location the size of a transmigration village and shrinks as its farm workforce grows, recalibrating the model each time (Appendix N.10). In the steepest version, which matches the fastest decline I estimate, the externality falls by 0.28 each time the farm workforce doubles and disappears in locations with about six times the farm workforce of a program village, so its average across the farm workforce falls from 0.70 to 0.30. The road's gain barely moves, staying at 0.73 to 0.74 percent against

0.86 percent without the externality. The road moves farm labor toward the largest locations, so weakening the externality there removes gains as well as costs. The gain from productivity growth, by contrast, rises from 0.74 to 0.91 percent against 1.21 percent without the externality, because that shock moves farm labor out of large, mostly urban locations. Ignoring the externality then overstates the gain by 33 rather than 64 percent (Appendix Table N4).

12 Conclusion

This paper uses the placement of Indonesia's Transmigration villages to study the long-run effects of market access. Villages that began more connected are poorer today and farm less productively, and the standard explanations, agricultural specialization, competition, out-migration, compensating amenities, and lower prices, do not account for the gap. What does account for it is a scale externality in agriculture. Connected villages moved more labor out of farming, and each worker who left lowered the productivity of those who stayed. I estimate the externality from program variation in settlement size, and a spatial general equilibrium model of the Indonesian economy that carries it reproduces the negative effect of market access on income among program villages without targeting it.

The model's counterfactuals show that the externality changes both who gains from connectivity and how much there is to gain. A road built to a single village can make that village poorer, and ignoring the externality overstates the aggregate gains from improving the whole travel network and from rural agricultural productivity growth, while a subsidy to agricultural earnings that would be a distortion without the externality raises national income with it.

More generally, this paper makes the point that structural transformation itself is not sufficient for generating growth. The gaps in output per worker between agriculture and the rest of the economy are measured from private returns, and they overstate what reallocation delivers when the social return to agricultural labor exceeds the private one, as it does in these villages, and when the workers who leave move into sectors with no (or smaller) scale externalities. The policy implication from this paper is not necessarily that policymakers should design programs to stop structural transformation, but that policies that promote externalities in the non-agricultural sector can allow for structural transformation and growth.

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Appendix for
"The Curse of Connectivity"

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A Background

A.1 Direct Qualitative Evidence of Program Implementation

This section presents a collection of quotes from a variety of sources that describe the chaotic nature of the transmigration site selection process during my period of study. Since the transmigration program was divided into different five-year plan periods (Repelita) over this period, I document each period separately as well:

Repelita I-II (1969-1979) and pre-periods

There was transmigration activity observed in the transmigration census even before Repelita I, although limited. In particular, there was little government oversight over transmigration site selection during this period:

- "During the first part of the 1960s the government introduced a policy of permitting mass organizations and political parties to participate in transmigration... Although it was hoped that this policy of non-government involvement in the program would enable greater numbers of people to be moved from Java and that the financial and administrative burden upon the government could thus be reduced, the organizations themselves were much more interested in the political advantages that they could gain from transmigration than in the establishment of economically viable settlements. Transmigrants were simply transported to project locations, most of which were badly selected." Hardjono (1988)

Repelita I (1969-1974) was primarily a planning period for the expansion of the transmigration program. Few sites were actually settled during this period. Activity began to ramp up during Repelita II (1974-1979), along with documented instances of the lack of sufficient care in selecting sites. Hanson (1981) reports that transmigration officials during this period had "limited management skills," while followup research on transmigration sites selected during this period revealed that they were frequently chosen in a haphazard manner (Hardjono, 1988). Repelita II was also when reports emerged that unrealistic resettlement targets set by government officials meant that individuals in charge of resettlement lacked sufficient time or incentives to deeply consider potential new village locations:

- "... the emphasis on reaching the [resettlement] targets rode roughshod over moderation and caution, and over the recommendations of planning guidelines and manuals." Whitten (1987)

Repelita III (1979-1984)

By all accounts this was the most chaotic period during the transmigration program. It also was the Repelita with the most individuals resettled. Reports on the program reference multiple instances of the government issuing statements encouraging the acceleration of the resettlement process,

particularly between 1981 and 1982. Whitten (1987) additionally reports that the information bureaucrats had available to them at the time to help inform site selection decisions was "often based on poor resource data and maps." Hardjono (1988)'s research on this period provides a perfect summary:

- "Despite occasional references in government statements to regional development and the welfare of transmigrants, the aim was once again to move as many people as possible"
- "As a consequence of the focus on numbers, the land-use plans developed during the 1970s were totally abandoned. Transmigrants were placed on whatever land was submitted by provincial governments for settlement purposes."
- "To sum it up briefly, during Repelita III quality was sacrificed to quantity."

Repelita IV (1984-1989)

While there was limited transmigration activity following Repelita IV, the global decline in oil price and other political concerns led to a drastic reduction in transmigration activity following Repelita IV (Bazzi et al., 2016). Although the transmigration authority attempted to develop new, more accurate land use maps to aid in the selection process of new sites, this only began during the end of this period, and progress was slow (Whitten, 1987). Despite the reported issues that came with setting unrealistic resettlement targets during Repelita III, the government actually increased the relocation target by 50 percent relative to Repelita III[14]. In addition to the haphazard placement process described above in Repelita III, there were additional reports about inter-agency coordination issues that increased the chaotic nature of the program:

- "Due to the rapid expansion [of the transmigration program], a number of program activities were taken from the Directorate General of Transmigration (DGT) and delegated to separate government agencies to speed up the settlement process. Inter-agency coordination problems made it more difficult to carefully match transmigrants... to their outer island settlements." Bazzi et al. (2016)

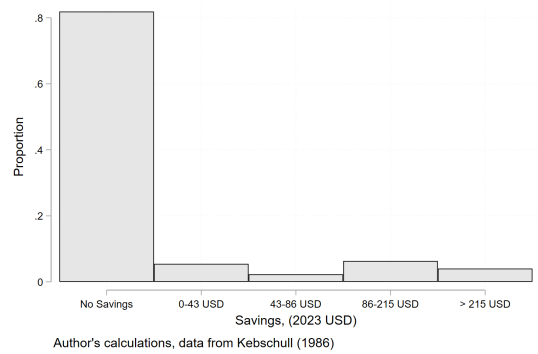
Two things should be said about what this record establishes. It documents that site selection was rushed, poorly informed, and driven by resettlement targets rather than economic potential, which makes systematic selection of sites on their productive fundamentals implausible. It cannot by itself rule out subtler channels, such as provincial governments choosing which land to submit for settlement; the empirical design of Section 4, with its recentering, balance tests, and controls, attempts to address this. The qualitative record is best read as motivating rather than proving the identification assumptions.

[14]The target was 500,000 households in Repelita III, and 750,000 households in Repelita IV. Neither target was actually reached.

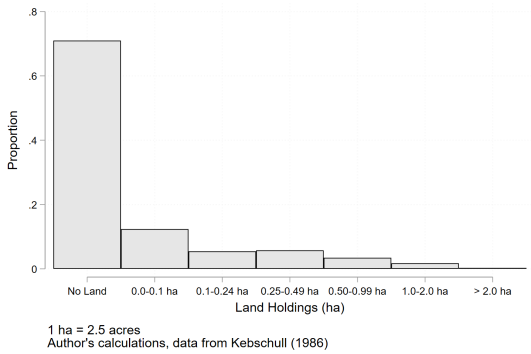
A.2 Settlers, Program Activity, and Village Characteristics over Time

Figure A1 reports the savings and landholdings of transmigrant households before migration from the survey of Kebschull (1986). Figure A2 plots the annual number of transmigrants and the average settlement size against the world oil price. Table A1 asks whether the villages settled in different years differ in their predetermined characteristics. Most of the underlying geographic characteristics of villages show no trend across creation years, though a handful of variables, including some crop-potential measures and the size of settlements, do display statistically significant linear trends, on the order of 0.4 to 0.7 standard deviations over the four decades of the program. These trends are one motivation for the cohort fixed effects used throughout the paper.

Figure A1: Survey Measures of Migrant Characteristics Prior to Relocation



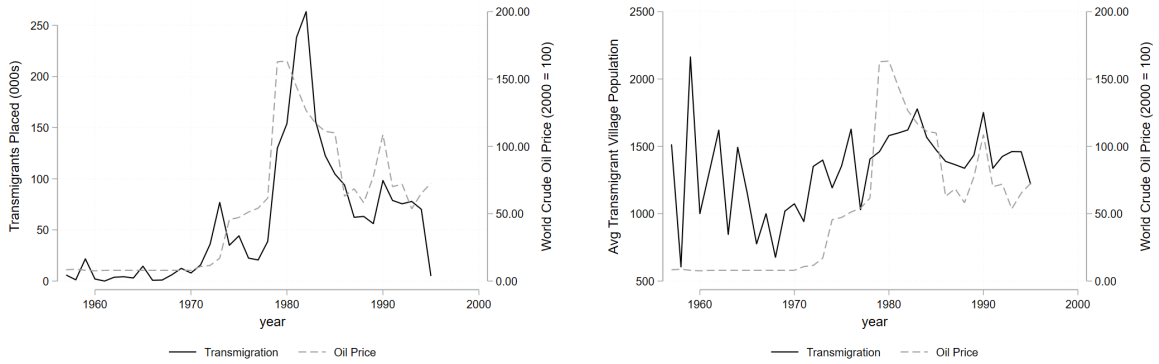
(a) Migrant Savings Immediately Before Migration



(b) Migrant Landholdings Before Migration

Note: Both panels present results from a pre-transmigration survey of transmigrant households by Kebschull (1986), digitized by the author. Panel (a) plots the distribution of savings immediately before transmigration while panel (b) plots the distribution of household landholdings prior to transmigration.

Figure A2: Transmigration and Oil Prices



(a) Total Transmigration by Year

(b) Average Transmigration Village Size by Year

Note: Both panels include world crude oil prices from Bazzi and Blattman (2014). Panel (a) also plots the total number of individuals resettled as a result of the transmigration program annually. Panel (b) also plots the average settled transmigration village population after winsorizing village population measures at the 1st and 99th percentiles.

Table A1: Village Characteristics Across Creation Time

Dependent variable (std): Year Created	
Wetland Rice Potential Yield	-0.011** (0.005)
Palm Oil Potential Yield	-0.008 (0.006)
Coffee Potential Yield	-0.010** (0.004)
Cassava Potential Yield	-0.013** (0.005)
Vector Ruggedness Measure	0.010 (0.006)
Individuals Placed	-0.016* (0.009)
Agrosimilarity Score	-0.008 (0.007)
Share of Land with Poor Drainage	-0.001 (0.008)
Share of Land with Good Drainage	-0.002 (0.007)
Organic Carbon (%)	0.007 (0.008)
Topsoil Salinity (Elco) (dS/m)	-0.004 (0.003)
Topsoil Base Saturation (%)	-0.007 (0.006)
Avg. rainfall, 1948–1978	0.008 (0.006)
Avg. temp (Celsius), 1948–1978	0.007 (0.009)
ln(District Population, 1980)	0.001 (0.005)
District Own Electricity, 1980	0.003 (0.005)
District Own Radio, 1980	-0.005 (0.005)
District Own TV, 1980	0.004 (0.005)

Dependent variable (std): Year Created	
District Literacy, 1980	-0.002 (0.005)
District Avg Years Schooling, 1980	-0.005 (0.005)
District Agricultural Employment Share, 1980	-0.005 (0.005)
District Wetland Rice Potential Yield, 1980	-0.006 (0.005)
District Palmoil Potential Yield, 1980	0.005 (0.005)
District Coffee Potential Yield, 1980	0.002 (0.006)
District Cassava Potential Yield, 1980	-0.005 (0.005)
District Mining Employment Share, 1980	-0.000 (0.004)
District Manufacturing Employment Share, 1980	0.005 (0.004)
District Trade Employment Share, 1980	0.003 (0.005)
District Service Employment Share, 1980	0.004 (0.005)
District Wage Worker Share, 1980	0.009* (0.005)
District Share HHs with Piped Water, 1980	-0.000 (0.004)
District Share HHs with Sewage, 1980	0.001 (0.005)
District Share HHs with Modern Fuel, 1980	0.004 (0.005)
District Share HHs with Modern Floor, 1980	0.002 (0.005)
District Share HHs with Modern Roof, 1980	-0.006 (0.005)

F-stat of omnibus test	2.319
Observations	1219

Note: Each row reports the coefficient from a regression of the standardized dependent variable listed in the row on the year a transmigration village was created, controlling for island fixed effects. Standard errors, clustered at the subdistrict (kecamatan) level, are reported in parentheses. The omnibus F-statistic tests joint differences across village creation year for all dependent variables, conditional on island fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Data Appendix: Sources and Construction

This appendix provides detailed descriptions of the data sources summarized in Table B1 and used throughout the analysis, followed by the digitization of the road network and the construction of travel times.

B.1 Local Measures of GDP per Capita

My primary measure of local GDP per capita is the high-resolution estimates of Rossi-Hansberg and Zhang (2025), which combine a diverse set of inputs beyond population and nightlights and perform significantly better at predicting local GDP than previous measures. Figure B4a shows that the measure varies substantially across transmigration villages. The alternative measure combines nightlights from Li et al. (2020), converted to GDP with the elasticities in Beyer et al. (2022), Hu and Yao (2022), and Henderson et al. (2012), with population from the Global Human Settlement Layer. It is subject to some of the issues raised in Rossi-Hansberg and Zhang (2025) but is built from fewer inputs. Throughout my analysis I classify village areas according to their administrative boundaries in 2000.

B.2 Measures of Village Level Productivity

To supplement the satellite-derived measures of GDP per capita above, I construct several proxies for village level productivity across multiple sectors. I measure agricultural productivity directly using village level data from the 2000 and 2003 waves of PODES on agricultural area devoted to different crops along with reported per-crop yields. The top-five-crop productivity measure is the total production of the five most planted crops in the village divided by the total area planted to those crops (tonnes per hectare, in logs), an area-weighted average of the crop-specific yields; because the identity of the top crops can differ across villages, the crop-level analysis of Section 8, which conditions on crop fixed effects, is the specification that isolates within-crop variation. I construct the same measure for rice and palm oil separately given their importance in rural agricultural economies. Figure B4d shows that village agricultural productivity is very heterogeneous. I also measure the number of tractors per capita and tractors per agricultural worker as a proxy for uptake of more productive agricultural practices. Outside of agriculture I use data from two waves (2000 and 2010) of the Indonesia Population Census where working individuals report if they are an employee or employer to construct the ratio of employees per employer. A lower ratio typically implies less productive firms given their smaller average size.

I complement these measures with individual-level wage data from the 2000 to 2007 waves of the SUSENAS, which each year samples individuals within a random set of villages across Indonesia. No transmigration village is guaranteed to be sampled in a given year, but pooling the waves gives wages for formally employed individuals in roughly one third of my villages, which I convert to their 2007 equivalents. Because I observe wages only in this subset of villages and only for

Table B1: Data Description

Sample	Description	Level of Observation	Year(s)	Outcomes
Rossi-Hansberg and Zhang (2025) (Global)	Measures of local GDP per capita from Rossi-Hansberg and Zhang (2025) and population per cell	Village	2012–2019	GDP per capita
Other Satellite Data (Global)	Nightlights harmonized across satellites from Li et al. (2020)	Village	1975–2020	Population, nightlight intensity (combined to construct an alternative GDP per capita measure)
Population Census (Complete Count)	Population census carried out every 10 years that collects basic demographic and socio-economic information	Individual	2000, 2010	(All aggregated to village level) sectoral employment shares, employment status/type, province of birth and province of residence 5 years prior
PODES (∼1148 Village Heads)	Survey of village heads that collects information on a wide variety of economic and social measures of the village	Village	1993, 2000, 2003	Sectoral employment shares (1993); agricultural productivity and tractors (2000); amenities/public-goods index (2000); migration and commuting patterns (2003); small-industry and shop counts (2003); land-use patterns and changes (2000 and 2003); village-by-crop production, area, and yield (2003)
SUSENAS – Prices (∼1234 Households)	Detailed consumption and expenditure module administered at the household level	Household	2002, 2011 (food only)	Quantity and unit price of food items consumed, individual-expenditure-weighted and rural-reference-weighted food price levels, total expenditure, expenditure by non-food category, housing value per square meter
SUSENAS – Wages (∼5781 Individuals)	General socio-demographic survey administered annually	Individual	2000–2007	(For formally employed only) monthly labor earnings, hours worked
Economic Census (Complete Count)	Complete listing of the universe of medium and large firms across Indonesia (plus a sample of micro and small firms in urban areas), with limited firm-level information	Firm	2016	Medium/large firm presence, total employment within the firm
Video-Derived Welfare (111 videos / 47 villages)	Publicly available Indonesian-language documentary and vlog footage of transmigration villages in Kalimantan, with frames scored by a vision language model against a fixed rubric	Village	2021-2022	Composite material-welfare index across seven dimensions of visible welfare (independent validation outcome)
Transmigration Census (Administrative)	Administrative records identifying settled transmigration villages, together with designated Planned Development Areas used to construct expected market access	Village	1952–1995	Village location, year settled, and number of individuals initially settled (instrument for agricultural labor in Section 8)
Geographic Characteristics (per Bazzi et al. (2016))	Time-invariant geographic and agro-ecological characteristics of village locations prior to settlement	Village	Time-invariant	Soil quality, terrain ruggedness, rainfall, temperature, agro-ecological similarity, and location
1980 Population Census (District-level)	District-level characteristics roughly contemporaneous with the transmigration program, used as controls	District	1980	Sectoral employment composition, formal employment rate, housing quality, agricultural potential yield, education and literacy, asset ownership
Historical Road Network (Army Map Service; HOT/OSM)	Hand-digitized maps of Indonesia's pre-program road network (roads by type, railways, trails) compiled by the Army Map Service Corps of Engineers, combined with navigable waterways from the Humanitarian OpenStreetMap Team	Raster / network	∼1950s	Travel-cost layer used to construct bilateral trade costs and pre-transmigration market access
IFLS (1,395 Village Heads)	Survey of village heads that includes a module on travel times to nearby locations along with method of travel and estimated trip time	Village	1993	Travel time by road type (used to calibrate trade-cost parameters)

formal workers, these estimates have less power than the GDP per capita results, but they provide contextual backing to the more abstract satellite measures (Appendix D).

The village-by-crop production data from the 2003 wave of PODES also plays a central role in my estimation of the agricultural externality in Section 8. For each village, PODES reports the production quantity, area planted, and yield for each crop grown in the most recent season, which I combine with census data on agricultural employment to construct the village-by-crop panel used in the instrumental variables estimation.

B.3 Video-Derived Welfare Measure

To complement the satellite and survey-based measures of economic performance, I construct a measure of village material well-being from publicly available video footage of transmigration villages in Kalimantan. I match 111 Indonesian-language documentary and vlog videos from a creator's YouTube playlist to 47 distinct villages in my main analysis sample using location information contained in video titles and captions. Frames sampled from each video are scored by a vision language model against a fixed rubric covering 7 dimensions of visible material welfare, and the frame-level scores are aggregated into a village-level composite welfare index. Because the sample is small and restricted to Kalimantan, this measure serves as a complementary validation of my primary outcomes from an entirely independent data source rather than as a primary outcome. Appendix E provides full detail on the sample construction, scoring rubric, and aggregation procedure.

B.4 Sectoral Employment

Sectoral employment shares come from the 2000 and 2010 waves of the Population Census, which record every individual's demographic information, education, employment status, and sector of employment. Figures B4b and B4c plot the distribution of agricultural and formal employment for the transmigration villages in 2000. While most villages are still primarily agricultural, there is a good deal of variation across transmigration villages, and the most common employment type is informal employment. I supplement these census measures with village-level data on sectoral employment from the 1993 wave of PODES.

B.5 Firm Dynamics

Village-level measures of formal firm presence and employment come from the listing of all medium and large enterprises in the 2016 Economic Census, which also contains a large sample of micro and small enterprises in each urban location and records the owner of the firm, a detailed sector classification, and employment. I collapse this listing to the village level to construct an indicator for whether a village has a medium or large firm and the average employment of those firms. Only 13 percent of transmigration villages have at least one such firm, so I also use the PODES waves, which are administered to village heads, to construct the number and density of small industries

and the number of shops in a village. These measures give a more accurate picture of the smaller scale production and trade present in transmigration villages.

B.6 Local Prices and Expenditure

Prices and expenditure come from the 2002 and 2011 waves of the SUSENAS, which contain detailed consumption modules. In both years I construct a price index for food expenditure,[15] and in 2002 I can also construct total expenditure on non-food categories and the price per square meter of housing. Because only these two waves separate prices from quantities, the sample is limited; I observe the food price index for 84 villages and 1,234 households and non-food expenditure for 70 villages and 1,086 households. Other waves of the SUSENAS record only coarser measures of total consumption expenditure and do not separate prices and quantities, so I cannot construct prices for a larger sample.

B.7 Amenities

The public goods index measures each transmigration village's provision of public goods by local village and regional governments, covering health care availability and quality, school presence and staffing, safe drinking water, waste collection and other sanitation, and road maintenance and quality. I standardize each of the 18 measures and take the average.

B.8 Migration and Commuting

Because the census asks each individual their birth location, I construct the two village-level migration variables used in Section 7, the log number and the share of current residents born on the inner islands (Java and Bali), the sending region of the program. Given the nature of transmigration villages, these shares are often large. I measure village-level commuting from the 2003 PODES, which asks whether individuals in the village commute outside it for work. The survey does not record how many individuals commute, but it does indicate the prevalence of commuting at the village level.

B.9 Land Use and Development

Land use comes from the 2000 and 2003 waves of PODES, which record the amount of land in a village devoted to agriculture, housing, and firms. The 2003 PODES also records *changes* in land use over the previous 3 years, so I can compare land allocations across villages in levels and in changes.

[15] In my sample nearly two thirds of total expenditure is devoted to food consumption.

B.10 Time-Invariant Characteristics

The time-invariant characteristics of transmigration villages are the geographic variables of Bazzi et al. (2016) that describe a village's soil quality, terrain, and location, none of which is an outcome of settlement. They include potential yields from FAO's Global Agro-Ecological Zones for wetland and dryland rice, cocoa, coffee, palm oil, cassava, and maize, which are modeled from climate, soil, and terrain rather than observed; soil texture, drainage, and chemistry from the Harmonized World Soil Database, which for Indonesia draws on the FAO-UNESCO Soil Map of the World compiled in 1971–1981; elevation, slope, and ruggedness from satellite elevation data; and average rainfall and temperature over 1948–1978. An additional control based on Bazzi et al. (2016) is a measure of the average agroclimatic similarity between a village's migrant origins and the new destination. The measure captures how much experience migrants have farming in similar conditions, and is a strong predictor of agricultural productivity. The Transmigration Census identifies the transmigration villages and records their location, the year they were settled, and the number of individuals relocated to each, which gives the settlement cohorts and, in Section 8, the instrument.

District characteristics roughly contemporaneous with the program come from the 1980 Population Census, from which I build the sectoral employment composition, formal employment rates, housing quality, agricultural potential yield, education and literacy, and asset ownership rates of the district each village was placed in.

B.11 Map Digitization and Travel Times

See main text for the location of the maps, and Figure B1[16] for an example of the un-digitized maps. These maps were created by the US Army Corps of Engineers and feature extremely detailed road network information for most of the world circa 1950. Roads are classified as either (from most navigable to least): Main road, secondary road, other road, or trail. The maps also mark all active and under-construction railways. There is additional information on settlement location, waterways, and topographic detail that I do not digitize. The maps are available via pdf or png format and are split into separate blocks. To cover all of Indonesia requires digitizing 27 separate images. The key used to identify different roadways is not amenable to automatic digitization; for instance, both the main road and secondary road are solid red lines of the same color, and the only way to distinguish them is based on small differences in the thickness of the lines. Trails and other roads are also red, but dashed. I was unable to determine how to separately classify each roadway type and connect the network in an automatic way. Additional complications include the fact that railways are the same color as location names and boundaries, and text often obscures sections of roadways that are easily interpolated by human inspection, but cause issues when trying to automate the process.

[16] Due to limitations on the image size in Overleaf, the included image is more blurry than the actual images I digitized.

To digitize these maps, I downloaded a copy of each map segment for each network type (27 images X 5 network types = 135 files) so that I could separately identify each network in the least cluttered way possible. I then hand-traced each network type for the entire landmass of Indonesia using a distinct color[17] so that I could later distinguish each type of network clearly. After this I used the coordinate system on the edges of the maps to geo-reference each image so that I could stitch together the 27 images to make up a unified image of Indonesia. Based on the recorded color value of each network type, I was then able to easily extract the paths from each of the 5 unified maps of Indonesia (by network type) and record them as separate raster files. I then overlapped all five layers together to generate a single raster layer that included the full, detailed, transportation network of Indonesia. Figure B2 shows the resulting layer for Sumatra. I cross-checked roughly half of the sheets for missed segments. A few sheets incorporate limited information from after 1950, but there was little to no road expansion in Indonesia until near the 1990s (Gertler et al., 2024), so the network I digitize is the network the program's villages faced for most of their existence.

To estimate the travel times between origin o and all destinations d I implemented a fast marching method in Matlab as described by Allen and Arkolakis (2014) where each grid point is assigned a travel cost based on the network type. Network types are in turn identified by the color of the raster at that particular pixel/grid point. In instances where two lines overlapped (due to intersections of multiple road types), I assigned those pixels to the fastest transportation method. The value of each grid point is meant to represent the cost of traveling over that point. Based on the projection and the map details, each grid point in my network was calculated to be roughly 730m. Therefore, I conducted a simple conversion to convert either a cost per mile (from Donaldson and Hornbeck (2016)) or speed (from my own estimations from the IFLS 1993) to the pixel cost. Figure B3 displays the travel cost parameters. The speed parameters come from my own calculations based on the Indonesia Family Life Survey 1993 Community Survey. In that survey a village head is asked details about the (i) distance to nearby cities (up to 3 distinct cities) (ii) the method of travel and (iii) the time it takes to arrive to that location. I then split the travel types into travel via car (roads), boats (water) and train (railway) and separately estimated their respective speeds. Due to limited observations of travel via train, I instead inputted the speed estimate from the linked source; however, my own estimate of the speed was only slightly slower (roughly 45 km/hr) than the final parameter used. The cost parameters come straight from Donaldson and Hornbeck (2016) except include the inclusion of travel over roads in automobiles, which I assumed to be slightly cheaper than water and train transport for main highways, but slightly more expensive than those two methods for more minor roads.

After calculating the estimated travel times between each transmigration village centroid and all other locations (either a city "block" or village "desa") centroids, I then estimated a transmigration village's initial market access using Equation 1 and the 1975 GHS Layer population estimates.

[17] I loaded the images in Goodnotes and traced the paths on an iPad, ensuring that roads connected across different image files when needed.

Table B2 compares the GHS layers with Indonesian sources.

Figure B1: Northern Sumatra, Example Map



Figure B2: Digitized Road Network - Sumatra



Note: This figure displays a subset of the travel network as it is input into Matlab to construct travel times and market access. Blue represents ocean and navigable waterways, green is land that has neither water nor any road type. Light blue are railways. Each other color represents different road types from highways to unpaved trails.

Figure B3: Travel Cost Parameters

Type	Code (in QGIS)	Cost (DH) ($/km)	Cost Transform Pixel	Speed Est IFLS (km/hr)	Speed Transform Pixel	Notes
Wilderness (off road)	0	0.3	218.0430107	3.21869	13.54855613	scale down walking speed to 2.5 mph
Main	1	0.004	2.907240142	24.41258	1.786316814	From IFLS 1993
Secondary	2	0.01	7.268100356	24.41258	1.786316814	cont. not separating by road type
Other	3	0.01	7.268100356	24.41258	1.786316814	
Trail	4	0.231	167.8931182	4.82803	9.032380109	assume avg walking speed of 3 mph
Rail	5	0.0063	4.578903224	51.499	0.8467854159	from https://transportgeography.org/conte
Water	6	0.0049	3.561369174	15.8953	2.743490348	From IFLS 1993
Ocean	10	0.01	7.268100356	15.8953	2.743490348	same as water
			Cost*726.8100356		(Speed*(1hr/60min)*(1000m/1km))^-1 * 726.8100356m	
					How many minutes will it take	
					you to travel over the pixel	

Table B2: Validation of GHS Population Measure

	Total Population	Total Population	Total Population	Total Population
Individuals Placed	0.8483	1.2171		
	(0.7184)	(1.0968)		
2010 Census Pop			0.9060***	0.9103***
			(0.1009)	(0.1040)
FE Level	None	Year and Island	None	Year and Island
N	176	176	1015	1015
Unique Villages	176	176	1015	1015

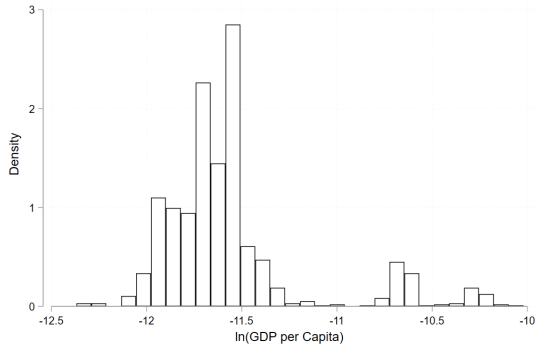
Standard errors in parentheses

Standard errors clustered at the village level

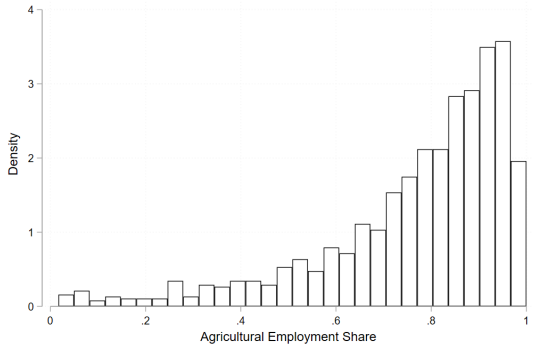
* $p < .1$, ** $p < .05$, *** $p < .01$

Note: This table displays the results from a regression of either the number of individuals placed according to the transmigration census on the GHS predicted village level population in 1975 for transmigration villages created between 1975-1979 (columns 1-2) or the number of individuals within a village from the 2010 population census on the GHS predicted village level population in 2010 (columns 3-4).

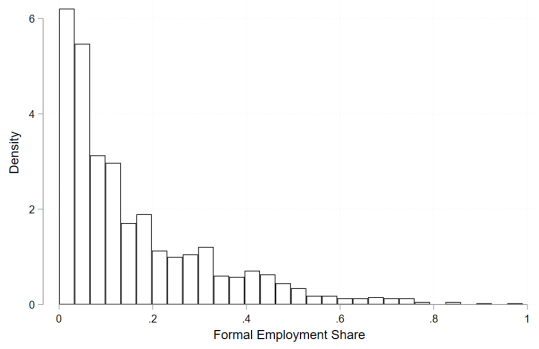
Figure B4: Distribution of Village Level Outcomes



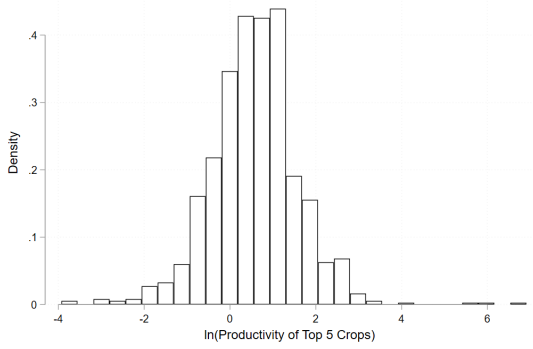
(a) GDP per Capita



(b) Agricultural Employment Share



(c) Formal Employment Share



(d) Agricultural Productivity

Note: Each panel plots the distribution of a different village-level variable among the transmigration villages in the analysis sample. Panel (a) plots the log of GDP (expressed in millions of constant 2017 USD) per capita in 2012. Panel (b) and (c) plot the agricultural and formal employment shares from the Population Census in 2000, respectively. Panel (d) plots the log of the productivity (tonnes of yield per hectare planted) of the top 5 crops in each village.

C Empirical Strategy: Simulation, Robustness, and Sensitivity

C.1 Simulation of the Identification Threat

In this section I provide detail around a simulation that demonstrates the potential for omitted variable bias when estimating the relationship for newly created locations between an outcome variable and market access even if the placement of new villages is as-if-randomly assigned. Additionally, the simulation highlights how leveraging the assignment process of new villages and using the selected controls addresses potential bias concerns.

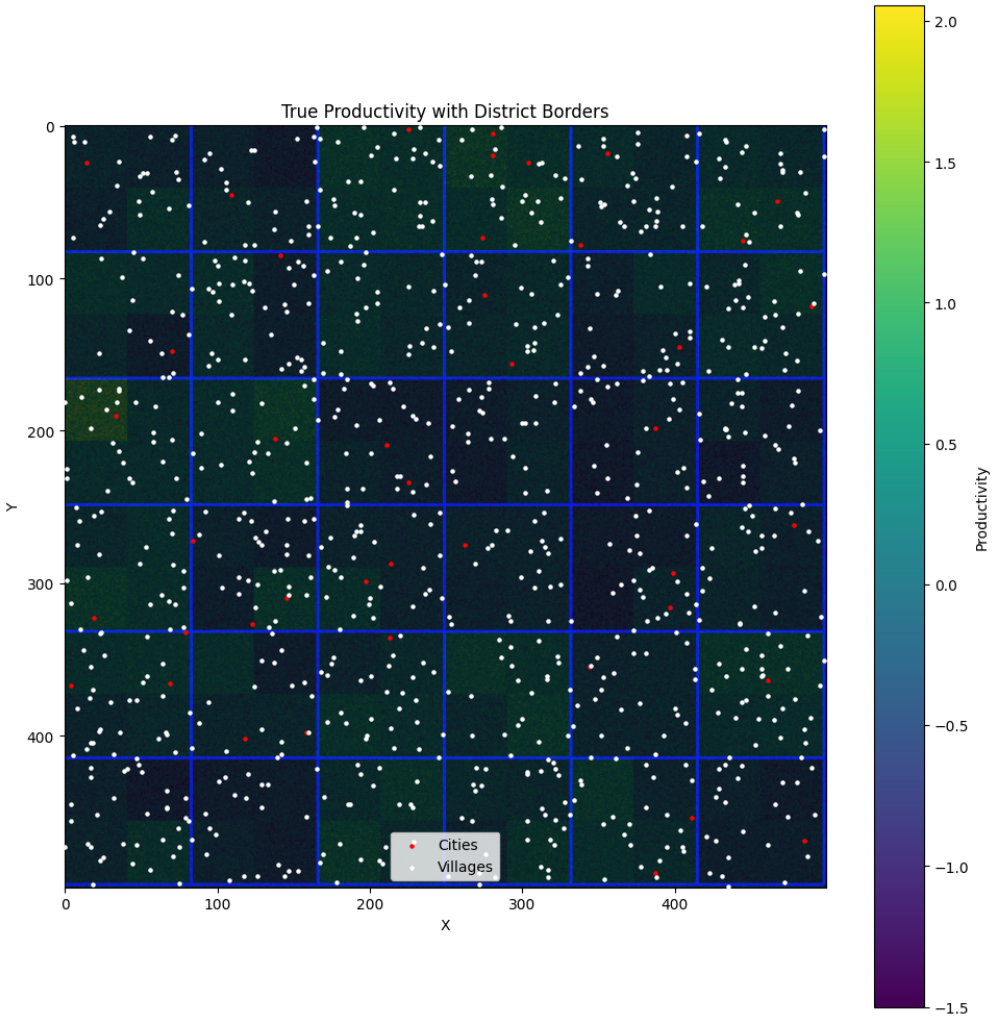
I create a space that is a 500 x 500 grid, then divide it into 9 equally sized provinces, each with 4 districts. Within each district there are then 4 subdistricts. Provinces, districts, subdistricts, and cells each draw a random fundamental productivity value (all the same variance). I specify that provinces, districts, and subdistricts have productivity draws such that there is a correlation between productivities[18]. A cell's total productivity is then the sum of each productivity component. I then create an additional measure I name "observed productivity" that is able to serve as a proxy for the unobserved true productivity[19]. In order to replicate the situation described in the empirical strategy section, I then first place 50 initial "cities" that are more likely to be located in highly productive areas. The potential for bias decreases as you introduce more initial locations, but this choice of initial locations highlights the identification challenge. To replicate the assignment process of the transmigration program, I then randomly place new villages stratified by province. The choice to stratify by province more closely mimics the reality where provincial governments exercise some oversight in the placement process of villages, and the as-if-random assignment occurs even within provinces. Figure C1 displays an example draw of this simulation exercise.

In order to demonstrate the threat to identification that persists even under random assignment of new villages, I then attempt to match the estimation strategy I implement in my actual data. For each new village, I construct its market access to preexisting cities assuming travel costs over each cell are equalized for simplicity. I then estimate the relationship between this measure of market access and the true, unobserved productivity of the new villages. If there was no omitted variable bias, then I would expect a coefficient of 0, meaning that there is no relationship between randomly placed new villages and their fundamental productivity. I also progressively add different controls that are explicitly designed to help control for the potential omitted variable bias present in this setting. I include a control for the average observed productivity of cities within a village's district to proxy for the district level controls I include in the true data. I also construct a measure of expected market access as the province-level average market access of new villages, which corresponds to the province-wide average variant of the empirical control rather than to the preferred closest-20-sites measure. Finally I also include the cell level observed productivity to match the village-level controls

[18] In this simulation example the correlation coefficient is 0.3

[19] The specified R-squared of a regression of observed productivity on unobserved productivity is 0.6.

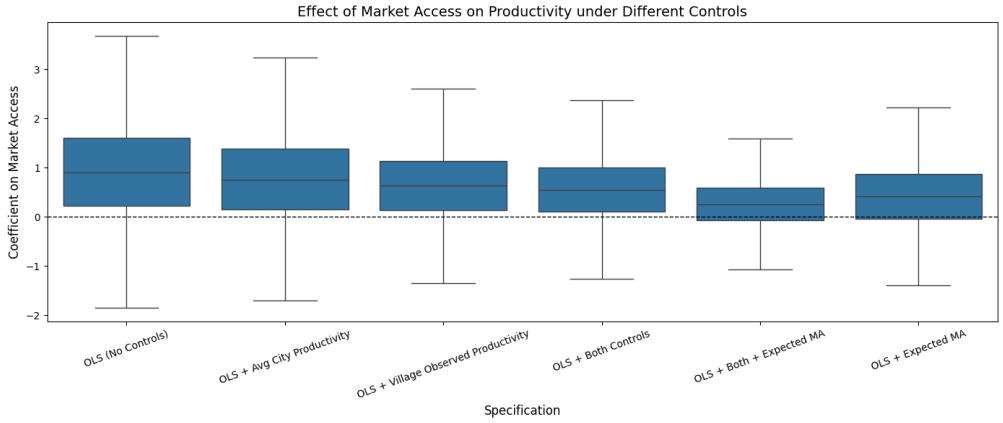
Figure C1: Map of Simulated Space



I include. I simulate this process 1000 times and store the estimated coefficients from each iteration. Figure C2 displays the results from my simulation. Notably, in the simple OLS case, there is a positive association between a new village's market access and its unobserved productivity despite the fact that new villages are placed randomly. Importantly, I also find evidence that in this type of assignment process, each of my different controls helps address this threat to identification. In the specification that most closely matches my preferred estimation strategy in the true data (OLS + Both + Expected MA), the interquartile range of my estimates contains 0.

This simulation is an illustration of the potential threats to identification, and its conclusions are conditional on the assumed placement process, productivity structure, and control quality. Given this, it supports two points. First, the identification threat described in Section 4 is real in that even random placement on a network does not guarantee identification in this setting, and the controls of my preferred specification substantially reduce the threat. Second, the direction of the bias in this environment is likely favorable for the findings of this paper. When preexisting

Figure C2: Simulation Results



locations are more likely to sit in productive places, a randomly placed new village's market access is *positively* related to its fundamental productivity, so omitted variable bias of this form would push my estimates toward finding that higher market access raises GDP per capita. If that is the relevant bias structure, my negative estimates would understate the true negative effects. In a network that is already extremely crowded, which is not the case for the outer islands of Indonesia, this relationship could flip if only the "bad" land is left over in high market access areas; Section C.2 examines the villages closest to existing towns. These observations complement, rather than replace, the balance tests and the Diegert et al. (2025) robustness diagnostics reported in the main text.

C.2 Villages Near Existing Towns

Section 4 raises the possibility that land at the edge of an existing town is worse than average, because good land there has already been absorbed into the town. Table C1 re-estimates the preferred specification after excluding the villages closest to a city.

Table C1: Excluding Villages Closest to a City

	(1) Full sample	(2) Distance measured	(3) Excl. closest 1%	(4) Excl. closest 5%	(5) Excl. closest 10%
ln(Market Access)	-0.0057***	-0.0050***	-0.0045***	-0.0040***	-0.0042***
	(0.0017)	(0.0017)	(0.0015)	(0.0015)	(0.0016)
Effect of 1 s.d. of ln(MA)	-0.0414	-0.0364	-0.0326	-0.0292	-0.0302
Distance cutoff (km)			1.3	10.0	15.3
Villages	1151	959	949	911	863

Note: Each column reports the coefficient on log initial market access from Equation (2) for log GDP per capita, with the controls and fixed effects of column 4 of Table 2. Column 1 is the full sample, and columns 2 to 4 exclude the 1, 5, and 10 percent of villages closest to a city. Distance is measured from each village's [centroid] to the boundary of the nearest [city definition, year]; the median is 62 kilometers and the 5th percentile [3.x] kilometers. Standard errors, in parentheses, allow for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: Robustness of the Market Access Estimate to Fixed Effects and Inference

	ln(GDP per Capita)	ln(GDP per Capita)	ln(GDP per Capita)	ln(GDP per Capita)	ln(GDP per Capita)	ln(GDP per Capita)	ln(GDP per Capita)	ln(GDP per Capita)
ln(Market Access)	-0.0057***	-0.0057***	-0.0057***	-0.0057***	-0.0057***	-0.0034***	-0.0023***	-0.0033***
	(0.0017)	(0.0017)	(0.0017)	(0.0017)	(0.0018)	(0.0009)	(0.0007)	(0.0007)
Expected MA Control	Closest 20	Closest 20	Closest 20	Closest 20	Closest 20	Closest 20	Closest 20	Closest 20
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Geography FE	Island	Island	Island	Island	Island	Province	Prov x Repelita	Prov x Cohort
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Std. Errors	Conley 100 km	Conley 50 km	Conley 250 km	Conley 500 km	Kabupaten	Conley 100 km	Conley 100 km	Conley 100 km
Permutation p (within-cell)	< 0.001							
Unique Villages	1151	1151	1151	1151	1151	1151	1151	1151

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Note: This table presents results from estimating the effect of market access on GDP per capita under alternative approaches to inference and alternative fixed effects. Every column uses the specification of Table 2, with the village and 1980 district controls, the closest-twenty expected market access control, and year fixed effects, on the village-year panel of outcome years 2012 to 2019. Column 1 is the baseline: island and cohort fixed effects with Conley standard errors, computed with a Bartlett kernel that declines with distance and reaches zero at 100 kilometers, admitting all covariances between observations within the cutoff, including those of the same village across outcome years, which enter with full weight. Columns 2-4 repeat the identical specification and sample with the spatial cutoff set to 50, 250, and 500 kilometers; the point estimate is identical across columns 1-4 by construction, and only the standard error changes. Column 5 replaces the spatial standard errors with clustering at the district (kabupaten) level. Columns 6-8 return to the Conley standard errors of column 1 and vary the fixed effects, replacing the island fixed effects with province, province-by-plan-period (Repelita), and province-by-cohort fixed effects. The permutation p-value reports the share of 1,000 random reassignments of market access across villages within province-by-cohort cells that produce an absolute coefficient at least as large as the observed one, and is valid under the sharp null that placement is exchangeable within cells; with 1,000 draws it is a Monte Carlo approximation to the exact permutation distribution. No reassignment does, so the p-value is below 0.001. The sample contains 290 province-by-cohort cells with a median of 2 villages per cell. The 106 cells containing a single village hold 106 of the 1,151 sample villages; they remain in the sample but contribute no identifying variation in column 8 and are unaffected by the permutation reassignments.

C.3 Robustness to Fixed Effects and Inference

C.4 Aggregate Shocks Correlated with Access

This appendix reports the two tests of Section 5.1 for whether an aggregate shock correlated with initial market access, rather than access itself, produces the negative estimate. Each shock predicts a specific pattern. Suppose the 1997–98 Asian financial crisis opened the income gap, by hitting connected villages harder or by raising the returns to the export crops of remote ones. Then the gradient should be absent or weaker before 1998 and steeper afterward. If the palm oil boom of the 2000s opened the gap, the gradient should be concentrated among the villages able to take part in the boom and be weak or absent elsewhere.

The Asian financial crisis. The primary measure of GDP per capita begins in 2012, so this test uses the nightlight measure of Panels B to D of Table 2, which can be built annually from 1992. I estimate a single pooled regression on the village-year panel for 1992 to 2020. In it, log initial market access and the expected market access control are each interacted with indicators for three periods: before the crisis (1992–1997), after it (1998–2013), and the years in which the harmonized series is built from VIIRS rather than DMSP lights (2014–2020). The last period is kept separate because VIIRS detects far more light. The share of village-years with any recorded light is 16 percent before the crisis, 28 percent from 1998 to 2013, and 90 percent from 2014. Lights

enter through their inverse hyperbolic sine, which keeps the unlit village-years in the sample.

Table C3 reports the results. Under all three elasticities the gradient is negative and significant at the 1 percent level before the crisis, between −0.033 and −0.038. After the crisis it is 21 to 31 percent smaller, and the difference between the two periods is significant at the 1 percent level. The gradient predates the crisis, so the crisis cannot have produced it, and if anything the gap narrowed afterward.[20]

Table C3: The Market Access Gradient Before and After the Asian Financial Crisis

	ln(GDP per Cap), HSW (2012)	ln(GDP per Cap), Hu and Yao (2022)	ln(GDP per Cap), BHY (2022)
ln(Market Access) × Pre-Crisis (1992–1997)	-0.0329***	-0.0350***	-0.0375***
	(0.0099)	(0.0098)	(0.0097)
ln(Market Access) × Post-Crisis (1998–2013)	-0.0226**	-0.0257***	-0.0295***
	(0.0094)	(0.0093)	(0.0093)
ln(Market Access) × VIIRS Years (2014–2020)	-0.0150*	-0.0188**	-0.0233**
	(0.0091)	(0.0091)	(0.0091)
p-value: Pre = Post-Crisis	0.000	0.001	0.002
Expected MA Control	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes
District Controls	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes
Island FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Std. Errors	Conley 100 km	Conley 100 km	Conley 100 km
Unique Villages	1066	1066	1066

Note: Each column reports one regression of log GDP per capita on log initial market access interacted with indicators for three periods, on the village-year panel for 1992 to 2020. GDP per capita is built from two inputs. The first is the harmonized nightlights of Li et al. (2020), entered through their inverse hyperbolic sine. The second is GHS population, interpolated log-linearly between the five-yearly GHS epochs. Lights are converted to GDP with the elasticity of the study named in the column heading. The expected market access control is interacted with the same period indicators. The regressions also control for village area and include the village and 1980 district controls and island, cohort, and year fixed effects. The pre-crisis period is 1992 to 1997 and includes 1997, the year the crisis began. The post-crisis period is 1998 to 2013, the last year of DMSP lights, and the VIIRS period is 2014 to 2020. The p-value tests whether the pre-crisis and post-crisis coefficients are equal. The sample excludes the 85 villages with zero GHS population in some year between 1990 and 2020. Standard errors, in parentheses, allow for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff, and admit all years of a village. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The palm oil boom. I measure exposure to the boom in two ways. The first is the GAEZ potential yield of palm oil. It is fixed by soil, climate, and terrain, so it measures where palm oil could be grown regardless of who chose to grow it. The second is whether a village grew palm oil in the 2003 PODES, early in the boom. Of the 1,151 sample villages, 348 did, and the indicator is nearly uncorrelated with initial market access (correlation −0.05). Table C4 interacts log initial market access with each measure, and neither interaction is significant. The interaction with potential yield is positive, so the gradient is, if anything, flatter where palm oil grows best, which is the opposite of what a boom-driven gradient would produce. Among villages that did not grow palm oil, the gradient is −0.0045, significant at the 5 percent level and about four-fifths of the

[20] Using the logarithm of lights instead drops the unlit village-years, which are most village-years before 2014, along with the 5 villages that are never lit. The two DMSP-period estimates are then imprecise and close to zero, between −0.008 and 0.019, and they do not differ from one another (p between 0.18 and 0.21). The VIIRS-period estimates, between −0.015 and −0.021, are close to those in Table C3. Before 2014, then, the gradient in Table C3 comes from which villages are lit at all rather than from how brightly they are lit. In neither version does the gradient steepen after the crisis.

baseline. The point estimate is steeper among villages that did grow it, −0.0082 (0.0026), but the difference is not significant. Estimating the two groups separately gives nearly identical gradients: −0.0057 (0.0019) among the 803 villages that did not grow palm oil and −0.0065 (0.0023) among the 348 that did.[21]

Table C4: Market Access and Exposure to the Palm Oil Boom

	(1) ln(GDP per Capita)	(2) ln(GDP per Capita)
ln(Market Access)	-0.0056***	-0.0045**
	(0.0017)	(0.0021)
ln(Market Access) × Palm Oil Potential Yield	0.0016	
	(0.0013)	
ln(Market Access) × Grew Palm Oil, 2003		-0.0037
		(0.0032)
Expected MA Control	Closest 20	Closest 20
Local Controls	Yes	Yes
District Controls	Yes	Yes
Cohort FE	Yes	Yes
Island FE	Yes	Yes
Year FE	Yes	Yes
Std. Errors	Conley 100 km	Conley 100 km
Unique Villages	1151	1151

Note: This table presents results from estimating Equation (2) for log GDP per capita, with log initial market access interacted with a village's exposure to the palm oil boom and with the controls and fixed effects of column 4 of Table 2. Column 1 interacts it with the GAEZ potential yield of palm oil, standardized over the regression sample. The interaction is therefore the change in the coefficient on log market access per standard deviation of potential yield. The unstandardized potential yield is already among the village controls. Column 2 interacts it with an indicator for villages that grew palm oil in the 2003 PODES (348 of the 1,151) and also controls for the indicator itself. The 116 villages for which the indicator is missing are coded as not growing palm oil. In column 2 the coefficient on log market access is the effect among villages that did not grow palm oil; the effect among those that did is −0.0082 (0.0026). Standard errors, in parentheses, allow for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Sensitivity to Omitted Variables

I assess the likelihood that omitted variables determine the estimated relationship by implementing methods proposed by Oster (2019) and Diegert et al. (2025). I elect not to report "Oster's delta" due to concerns raised by Diegert et al. (2025)[22], but instead report the fraction of times that the "breakdown" point from Diegert et al. (2025) is larger than the ρ_k values for my local covariates. The breakdown point is the degree of selection on unobservables, relative to selection on observables, at which the estimated effect would be driven to zero; ρ_k measures the importance of each covariate k relative to all of the other observed covariates. The higher the fraction of covariates whose ρ_k

[21] The separate regressions drop the topsoil salinity control, which takes a single value among the villages that grew palm oil. Dropping it from the pooled regression in column 2 of Table C4 leaves the estimates essentially unchanged.

[22] Specifically, the method proposed by Oster (2019) can be incorrect when covariates are correlated with the potential omitted variables.

falls below the breakdown point, the less an unobserved confounder that behaved like the observed ones could explain the result. Table 2 reports this fraction for each specification. In the preferred specification with the full control set it is over 85 percent, higher than the same fraction Diegert et al. (2025) report in their empirical application where they conclude the results are robust; it is lower in the specifications without district controls or the expected market access control.

D Resident Wages

This appendix reports the wage results discussed in Section 5.2. The sample pools the 2000 to 2007 SUSENAS waves and covers formally employed workers in the transmigration villages the survey happens to sample, 5,781 individuals in 502 villages, with wages converted to 2007 rupiah; Appendix B describes the survey. Each column of Table D1 estimates Equation (2) at the individual level, with log monthly wage, log hourly wage, or log hours worked as the outcome, individual controls for age, sex, and education, and survey-year fixed effects; columns 4 and 5 restrict the sample to individuals working in agriculture, manufacturing, and services, and column 5 adds sector fixed effects. The point estimates for wages are negative throughout and hours do not move; the estimates are imprecise, and Section 5.2 discusses them more.

Table D1: Effects of Initial Market Access on Wages

	ln(Wage)	ln(Hourly Wage)	ln(Hours Worked)	ln(Wage)	ln(Wage)
(max) ln_MA_full_noself	-0.0027	-0.0027	-0.0003	-0.0022	-0.0038*
	(0.0023)	(0.0027)	(0.0017)	(0.0023)	(0.0023)
Borusyak and Hull	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes
District Controls	Yes	Yes	Yes	Yes	Yes
Individual Controls	Yes	Yes	Yes	Yes	Yes
Sample	Full	Full	Full	Limited	Limited
Sector FE	No	No	No	No	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes
Island FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Unique Villages	502	501	503	501	501
Unique Households	5781	5706	5989	5462	5462

Standard errors in parentheses

Standard errors clustered at the village level

* $p < .1$, ** $p < .05$, *** $p < .01$

Note: Individual-level regressions of log monthly wage, log hourly wage, and log hours worked on log initial market access, following Equation (2) with individual controls (age, sex, education) and survey-year fixed effects, using formally employed workers in the 2000 to 2007 SUSENAS waves matched to transmigration villages. Wages are in 2007 rupiah. Columns 4 and 5 restrict the sample to individuals working in agriculture, manufacturing, and services, and column 5 adds sector fixed effects. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Video-Derived Welfare Measure

E.1 Overview

The main analysis identifies the curse-of-connectivity result using satellite-derived GDP per capita estimates from Rossi-Hansberg and Zhang (2025) and a nightlight-based alternative. As a complementary check from a data source independent of both, I construct a welfare measure based on direct visual evidence of village material conditions in a Kalimantan subsample of transmigration villages. The measure is built from publicly-uploaded video footage scored frame by frame against a pre-specified rubric using a vision language model. The estimated relationship between this video-derived welfare measure and initial market access is negative for every welfare outcome considered, and significant at conventional levels for the composite index. The appendix evidence is directionally consistent with the GDP per capita result in Table 2 and was constructed without reference to any of the satellite, census, or PODES data sources used in the main analysis.

E.2 Sample

The source is a curated YouTube playlist of Indonesian-language documentary and vlog content covering Kalimantan transmigration villages, containing 310 videos at the time of writing, created by user Pie'ie Mejink. For each video I extracted candidate location information from the title and the Indonesian-language auto-generated captions. Of the 310 videos in the playlist, 111 of the videos contained location information that enables a high-confidence match to a village in my primary sample. Multiple villages appeared in more than one video, so my final sample of analysis corresponds to 47 distinct villages. Geographic coverage is concentrated in Kalimantan Barat, with smaller numbers of videos in Kalimantan Tengah, Kalimantan Timur, and Kalimantan Selatan.

E.3 The Welfare Rubric

The rubric scores villages based on 7 indicators of visible material welfare in a village environment. These measures are: housing quality, road and path surface, electrification, agricultural assets, commercial activity, public infrastructure, and clothing condition of individuals. Each indicator is scored on a 1-5 scale. Each indicator may be marked "not visible" for any individual frame if, for example, the frame is an interior shot, a talking-head segment or B-roll of unrelated landscape. Irrelevant frames are then not incorporated into the welfare metric.

In order to match the context from the videos so the scoring exhibits real variation, I made two choices to calibrate the model's behavior. First, the scale value of 3 is anchored to "typical rural Kalimantan," so it is not left up to the agent to set its own reference point. This prevents the rubric from compressing scores in the event it selects a poor reference point to compare villages to. Second, the rubric explicitly instructs the model not to assign a score of 1 unless the frame shows real material deprivation. The rubric was developed on a small sample of 4 hand-picked

videos before the bulk run and held fixed across all subsequent scoring; the full text is reproduced verbatim in Section E.7.

E.4 Construction of the Village-Level Welfare Measure

Each video is downloaded at 480p resolution and frames are extracted at a fixed 11-second interval then sent in batches of 5 to a vision language model (Claude Sonnet 4.5) with the rubric in the system prompt. The model is required to return a structured JSON object conforming to one record per frame, with typed indicators for: `visible`, `score`, and `evidence` fields for each of the 7 measures. This eliminates free-text parsing and ensures every per-frame record is mechanically validated.

The per-video score for each indicator is the mean of the 1-5 scores across frames in which that indicator was marked visible. The per-video composite welfare index is the equal-weighted mean of the per-indicator means, taken over the indicators observed for that video; when an indicator is never visible in any of a village's videos, the composite for that village averages over the remaining indicators, which is why the component columns of Table E1 have slightly smaller samples than the composite. The equal-weighting choice is a decision: each dimension of welfare contributes once to the composite regardless of how frequently it appears on camera in any particular video, so a village walk-through and a market segment do not receive differentially weighted contributions from their dominant scene types. The per-village welfare measure is the simple mean across all high-confidence-matched videos sharing that village identifier.

For the regression reported in Section E.6, the two composite measures and the 7 individual indicator means are each z-scored across the analysis sample of 47 villages before estimation. Slopes are therefore interpreted as standard deviations of welfare per log point of ln(MA).

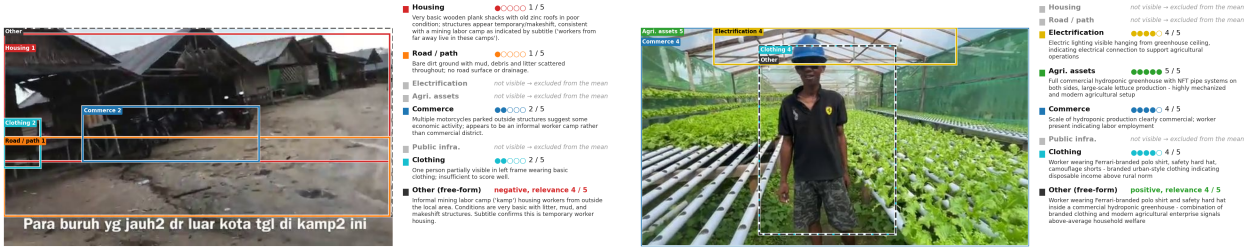
I report two composite measures. *Composite* is the equal-weighted mean of the 7 predefined indicators described above. *Composite (incl.)* additionally folds in a free-form "other" channel that captures visually striking welfare-relevant elements not covered by the 7 categorical indicators: luxury vehicles, abandoned structures, large industrial installations under construction, ceremonies, visible child labor, and similar. The per-frame "other" entries are assigned a signed welfare relevance by the same model call, aggregated to a video-level signed score, and combined with the predefined composite as a single pseudo-indicator on the same 1–5 scale. The two composites are highly correlated; the inclusive measure is reported alongside the predefined measure as a robustness check on the decision to fix the indicator set ex ante.

E.5 Examples of Scored Frames

Figure E1 shows two frames from opposite ends of the scale. In the first, from a video of Goaboma, housing and the road score 1, with plank shacks under old zinc roofs and bare dirt ground with no road surface or drainage given as the evidence, and commerce and clothing score 2. Electrification, agricultural assets, and public infrastructure are not visible, so they are left out rather than scored

low. The free-form observation adds context the seven indicators cannot record: the subtitle identifies the structures as a camp for laborers from outside the area. In the second, from a video of Sungai Deras, a commercial hydroponic greenhouse scores 5 on agricultural assets and 4 on electrification, commerce, and clothing.

Figure E1: Scored Frames at Opposite Ends of the Welfare Scale



(a) Low-Welfare Frame (Goaboma, Mean 1.5)

(b) High-Welfare Frame (Sungai Deras, Mean 4.25)

Note: Each panel is a single frame as scored for the video welfare measure: panel (a) from a video of Goaboma at 544 seconds, panel (b) from a video of Sungai Deras at 374 seconds. The legend to the right of each frame reports, for each of the seven indicators, the model's 1–5 score and its stated evidence, or that the indicator is not visible and is excluded from the mean; the last entry is the free-form "other" observation with its sign and welfare relevance. The frame mean averages the scores of the visible indicators and is shown for illustration; the village measures of Section E.4 average each indicator across frames before averaging across indicators. The colored boxes locate the evidence for each score in the frame. They were drawn after scoring, by the same model, and play no role in any welfare measure.

E.6 Results

I regress each z-scored welfare outcome on ln(MA) at the village level. Each observation is a unique village, with welfare averaged across replicate videos when a village is matched by more than one video. Due to the small sample, I exclude controls or fixed effects, and standard errors are clustered at the village level. Table E1 reports the results for the welfare indicators.

All nine slopes are negative, directionally consistent with the main paper's GDP-per-capita finding. The welfare index is consistently lower in high market access transmigration villages. The magnitudes are economically meaningful as well; based on the distribution of market access from the *full* sample of transmigration villages, moving from the 25th to 75th percentile of initial market access corresponds to roughly 0.5 standard deviations lower measured welfare, a figure that combines a slope estimated on the 47-village Kalimantan sample with the full-sample access spread and is therefore indicative rather than exact. This pattern is found in every welfare category except commerce, where I find a slope of essentially zero. This is consistent with the proposed mechanism in the paper itself. The convergence of two welfare measures constructed from entirely disjoint data sources strengthens confidence that the pattern documented in Section 5 is not an artifact of any specific feature of the satellite or administrative data. The video-derived estimates should be read as descriptive corroboration rather than as independent identification: the sample is small (47 villages) and could be selected through filming, playlist inclusion, and location matching, each

potentially related to connectivity or visible conditions; the specifications carry none of the main design's controls; and the geographic coverage is restricted to Kalimantan.

Table E1: Video-derived welfare measures regressed on initial market access

	Composite	Composite (incl.)	Housing	Road & Path	Electrification	Agri. Assets	Commerce	Public Infra.	Clothing
ln(MA)	-0.048*	-0.051*	-0.025	-0.043**	-0.030	-0.011	-0.002	-0.063	-0.056
	(0.026)	(0.030)	(0.024)	(0.022)	(0.025)	(0.026)	(0.036)	(0.040)	(0.036)
N (villages)	47	47	47	47	45	46	43	42	47

Note: Each column reports a univariate OLS regression of a z-scored video-derived welfare outcome on ln(MA) at the village level. Welfare outcomes are averaged across replicate videos within villages before z-scoring. *Composite* is the equal-weighted mean of the 7 indicator means; *Composite (incl.)* additionally folds in the free-form "other" channel described in Section E.4. Standard errors are clustered at the village level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E.7 Welfare Rubric Reproduced Verbatim

The text below is the system prompt used to score every frame in every video, reproduced word-for-word from the implementation code.

```
You score visible welfare indicators in frames sampled from videos
of Indonesian transmigration villages (mostly Kalimantan).

For each frame, rate each indicator 1-5:
  1 = very poor / severe deprivation
  2 = poor / below regional norm
  3 = average / typical for rural Kalimantan
  4 = good / above regional norm
  5 = excellent / clear prosperity

If an indicator is NOT VISIBLE in the frame (e.g. interior shot, talking head,
B-roll of only sky/forest/water with no village context), set visible=false
and score=0. Be honest: most B-roll frames will have only 1-2 visible indicators.

Indicators:

- housing_quality: walls/roof materials and condition. Tarp/scrap/bamboo < raw
  wood plank < painted wood < zinc roof + plank < concrete with tiled roof <
  multi-storey painted concrete. Note upkeep and signs of recent investment.
- road_path_surface: dirt < graded dirt < gravel < paved (aspal) < paved with
  drainage. Quality of drainage and ditches matters.
- electrification: power lines, transformers, exterior lights, satellite
  dishes, visible appliances/electronics. Score the density of signs, not
```

```
  whether *any* signs exist.
- agricultural_assets: hand tools only < draft animal < small motorized
  (handtractor / pump) < tractor / mechanized rice planter. Visible canals,
  weirs, and pumps count here too.
- commercial_activity: warungs, shops, kiosks, vendors, parked vehicles,
  construction, motorcycle density. Large income-generating builds (swiftlet
  farms, sawmills, brick kilns, logging trucks) push this high.
- public_infrastructure: schools, mosques, churches, clinics (puskesmas),
  government offices, sports facilities. Score on apparent condition.
- clothing_condition: residents' clothing - worn/torn/dirty < clean modest
  village clothing < new/branded/urban-style.

ALSO record an "other" observation per frame: a visually striking element not
covered above that bears on welfare (luxury vehicle, abandoned house,
ceremony, child labour signs, large modern equipment, etc). If nothing
notable, set present=false.

SCENE TYPE: Also tag each frame with the single category that best describes
what the camera is primarily framing. This is purely descriptive | it does
NOT change the 1-5 scoring above. Pick exactly one:
  - residential: homes, yards, residential streets, kampung from the outside
  - commercial: markets, shops, kiosks, vendors, storefronts, commercial buildings
  - public_institutional: schools, mosques/churches, clinics (puskesmas),
    government offices, sports facilities
  - agricultural: paddy fields, plantations, canals, farming activity, livestock
  - infrastructure: roads in transit, bridges, power infrastructure, when these
    are the subject of the frame (not just incidental background)
  - interior_or_talking_head: inside a building, close-up portrait, on-camera
    interview, title cards / intro graphics
  - b_roll_or_landscape: sky, forest, water, generic natural landscape,
    anything else not informative for welfare

CALIBRATION: Anchor 3 to rural Kalimantan, NOT urban Java. A simple wooden
house with a zinc roof and dirt yard is a 3. Do not score 1 unless the frame
shows real deprivation (collapsing structure, residents in rags, etc).

Return ONE entry per frame in the same order as received.
```

F The Simple Model

This appendix presents a simplified version of the quantitative model of Section 9. I include the same three goods, technology and externality, and sorting structure, but on a symmetric stylized geography (forty identical villages on a line, one city) instead of the full Indonesian economy. Additionally, parameters are set to illustrative values rather than estimated. The simple model serves three purposes. First, it highlights the mechanism in a setting that is potentially easier to interpret. Second, it is much easier to solve computationally, so I carry out various stress tests with this model. I relax certain assumptions and show how the sign and magnitude of the effect of market access on income changes. Finally, I confirm numerically, under Roy–Fréchet sorting, the sign-reversal threshold that Proposition 1 establishes in closed form under Pareto sorting (Appendix G).

F.1 Environment

Villages $v = 1, \ldots, V$ ($V = 40$) sit at positions x_v evenly spaced on $[1, 4]$ along a line, with the city at the origin; the gross iceberg cost factor between any two locations is one plus the distance between their positions, so the village-to-city factor is $\tau_v = 1 + x_v \in [2, 5]$. Each village has $L_v = 1$ workers and $H_v = 1$ units of land. There are three goods, as in the main text. Food is homogeneous and shipped at iceberg cost τ^{κ_a}; rural services are differentiated by village of origin and traded among villages and to the city at cost τ^{κ_n} under CES aggregation with elasticity σ_n; and an urban good is produced only by the city with a linear technology, taken as the numéraire at source and delivered to villages at cost τ^{κ_u}. City income I_c is exogenous, and the city spends it in fixed shares $(\beta_a, \beta_n, \beta_u)$ on food, rural services, and its own good.

Technology is that of the main text. All village land works in agriculture,

$$Y_{a,v} = \bar{A}_a \, Z_{a,v}^{\gamma_a + \varphi} \, H_v^{1-\gamma_a}, \qquad Y_{n,v} = A_n \, Z_{n,v}^{g_n}, \tag{F1}$$

with the externality $A_{a,v} = \bar{A}_a Z_{a,v}^{\varphi}$ uncompensated, and the rural service produced with labor alone; $g_n = 1$ is the constant-returns baseline, and $g_n < 1$ (a stress-test variant) makes service production decreasing-returns. Workers draw sector abilities from independent Fréchet distributions with shape θ and scales (T_a, T_n) and choose the sector offering the highest private return, exactly as in the main text. Village households have Cobb–Douglas preferences with weights $(\alpha_a, \alpha_n, \alpha_u)$. Food arbitrage sets the village-gate price $p_{a,v} = \tau_v^{-\kappa_a} p_a$, and the economy-wide food price p_a clears the aggregate food market against city demand.

An equilibrium is a pair of wages in each sector per village and the food price ($2V+1$ unknowns) solving each village's agricultural first-order condition, each village's service-variety market clearing, and aggregate food clearing; the urban-good market clears via Walras' Law and is verified numerically. Because $\gamma_a + \varphi$ approaches one, multiple equilibria are possible; the reported equilibrium is selected by continuation in φ from the unique $\varphi = 0$ benchmark, the same device used at

scale in the quantitative model.

F.2 Calibration

The parameters I choose here are illustrative and chosen to highlight the workings of the model. They are not estimated in the same way as the full model. The technology block matches the main calibration where the object is common: $\gamma_a = 0.28$ and $\varphi = 0.70$. The sorting dispersion is set at $\theta = 7$; the stress test below reports the model at $\theta = 4$ (strong comparative-advantage dispersion) and $\theta = 30$ (approaching homogeneous workers). The quantitative model's imposed $\theta = 12.5$ sits in the middle of this range.

The trade block sets $\kappa_a = 0.01$, $\kappa_n = \kappa_u = 0.60$, and $\sigma_n = 6$. These values were chosen to be relatively close to the calibrated values and such that the behavior of income and distance to the city is roughly monotone at the equilibrium. The demand weights $(\alpha_a, \alpha_n, \alpha_u) = (0.80, 0.10, 0.10)$ and city shares $(\beta_a, \beta_n, \beta_u) = (0.15, 0.10, 0.75)$ were selected so that the average agricultural employment share roughly matches the data (0.652 in the model compared to 0.648 in the data).

F.3 Baseline Results

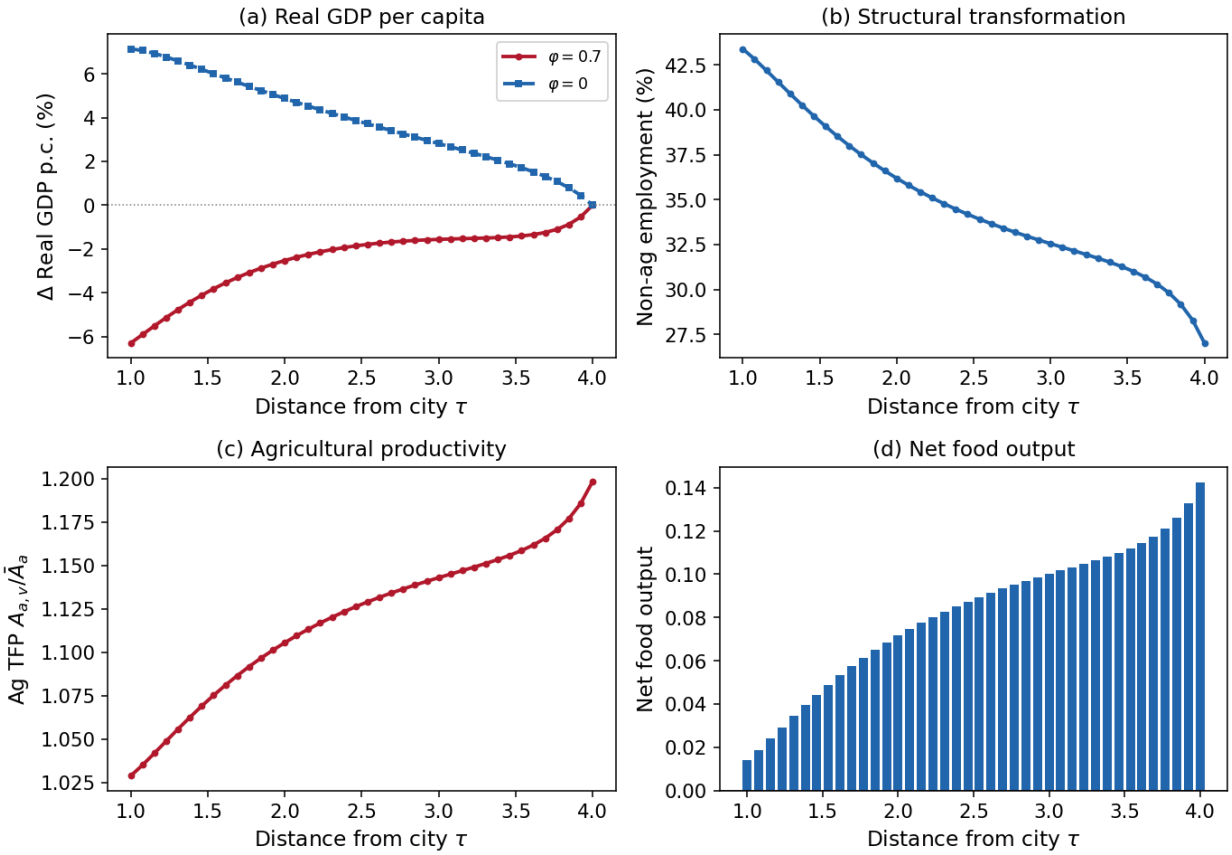
Figure F1 reports the baseline model behavior. Real income per capita falls monotonically toward the city. The closest village is 6.3 percent poorer than the most distant one. The non-agricultural employment share rises toward the city by 16.4 percentage points, and agricultural TFP falls via the effects of the externality. Re-solving the identical economy at $\varphi = 0$ reverses the relationship. With no externality, the closest village becomes 7.1 percent *richer* than the most distant, the standard gains from trade.

F.4 Stress Tests

Table F1 re-solves the simple model under variants that relax, one at a time, assumptions that could influence either the sign or size of the relationship between connectivity and income. Figure F2 plots the corresponding spatial profiles for each test. The table reports the difference in real income with and without the externality, along with the difference in the non-agricultural employment share between the closest and most distant village from the city. Additionally, the table reports the estimated value of φ where the relationship between connectivity and income flips sign; this is the numerical counterpart of the closed-form threshold in the simple model in Proposition 1.

There are several observations worth commenting on from these results. First, the negative relationship between market access and income is present in every stress-test I included, but the magnitude of the relationship varies significantly. There are parameter combinations where the difference in income is positive, but the point of this exercise is to highlight that the general direction of the relationship is robust to a wide range of parameter combinations. Also, there is no

Figure F1: Simple Model: Outcomes by Distance from the City



Note: I plot four outcomes for the 40 villages of the simple model against their distance from the city. Panel (a) plots real income per capita relative to the most distant village, at the calibrated $\varphi = 0.70$ and at $\varphi = 0$. Panel (b) plots the non-agricultural employment share, panel (c) the agricultural TFP multiplier $A_{a,v}/\bar{A}_a$, and panel (d) net food output.

scenario where the difference in income *without* the externality is negative. The externality is the force that makes it possible to generate a negative relationship between connectivity and income.

Second, worker heterogeneity is an important force that determines the magnitude of the relationship between connectivity and income. Making workers more homogeneous ($\theta = 30$) more than doubles the curse, while high dispersion in comparative advantage ($\theta = 4$) reduces the relationship. When comparative advantage is more dispersed, sectoral shifts translate to smaller movements in effective labor than headcount. Since the externality is modeled with respect to effective agricultural labor, this reduces the bite of the productivity loss that arises as villages become more exposed to demand for their non-agricultural output.

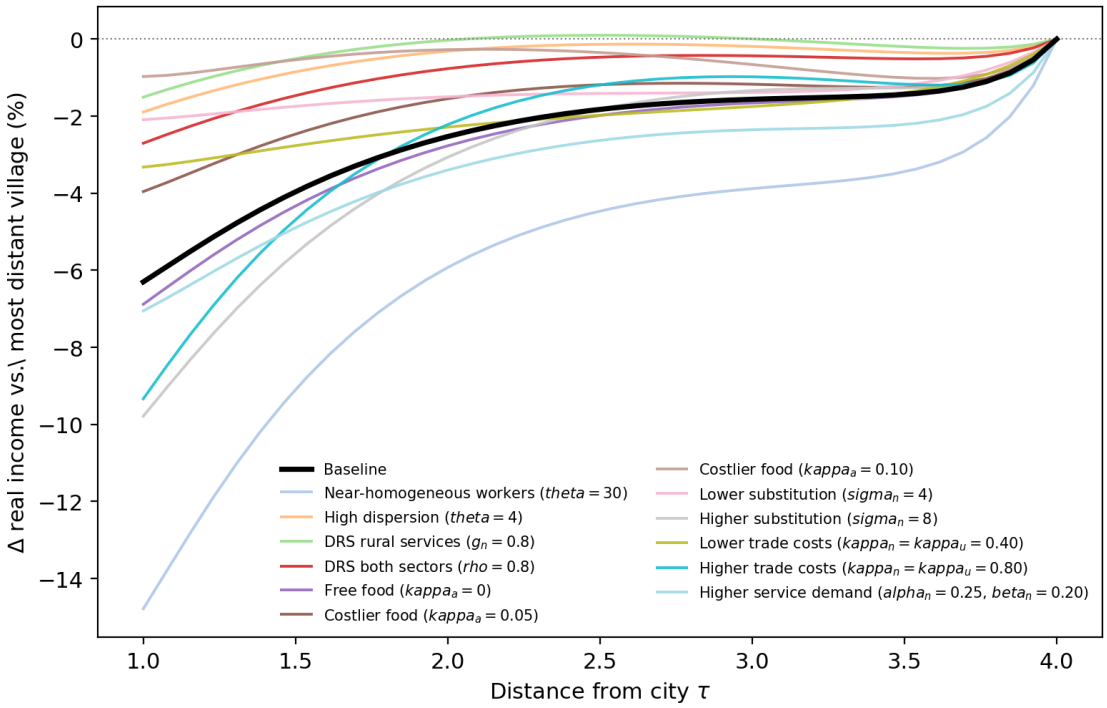
Third, the sector-specific trade frictions are key to this mechanism. From a theoretical perspective, the cross-sectional pattern requires services to be effectively harder to trade than food, $\kappa_n(\sigma_n - 1)/\sigma_n > \kappa_a$ (Appendix G); at equal frictions the relative service price would rise with distance and the reallocation would run the other way. The key driving force that generates the

Table F1: Stress test: close-versus-far outcomes by variant

Variant	Δreal (%)	$\Delta\pi_n$ (pp)	$\Delta\text{real}_{\varphi=0}$ (%)	φ^*	avg π_a	Net food
Baseline	-6.30	+16.36	+7.14	0.45	0.652	> 0
Near-homogeneous workers ($\theta = 30$)	-14.78	+24.16	+6.99	0.33	0.652	> 0
High dispersion ($\theta = 4$)	-1.89	+12.54	+7.26	0.59	0.652	> 0
DRS rural services ($g_n = 0.8$)	-1.51	+11.29	+7.27	0.61	0.700	> 0
DRS both sectors ($\rho = 0.8$)	-2.70	+12.09	+7.31	0.56	0.652	> 0
Free food ($\kappa_a = 0$)	-6.88	+17.01	+7.06	0.43	0.652	> 0
Costlier food ($\kappa_a = 0.05$)	-3.96	+13.77	+7.47	0.52	0.652	> 0
Costlier food ($\kappa_a = 0.10$)	-0.97	+10.54	+7.88	0.65	0.652	> 0
Lower substitution ($\sigma_n = 4$)	-2.09	+10.75	+6.90	0.58	0.651	> 0
Higher substitution ($\sigma_n = 8$)	-9.79	+20.99	+7.30	0.38	0.652	> 0
Lower trade costs ($\kappa_n = \kappa_u = 0.40$)	-3.32	+9.91	+4.77	0.49	0.651	> 0
Higher trade costs ($\kappa_n = \kappa_u = 0.80$)	-9.33	+22.76	+9.57	0.43	0.652	> 0
Higher service demand ($\alpha_n = 0.25$, $\beta_n = 0.20$)	-7.05	+14.25	+8.06	0.49	0.407	> 0

Note: Each row re-solves the simple model at $\varphi = 0.70$ under one deviation from the baseline calibration of Section F.2, by continuation from $\varphi = 0$. Δreal is the real-income gap between the closest and most distant village. $\Delta\pi_n$ is the corresponding gap in the non-agricultural employment share. $\Delta\text{real}_{\varphi=0}$ is the income gap with the externality shut off. φ^* is the externality value at which the income gap changes sign, interpolated along the continuation path. Avg π_a is the average agricultural employment share, and the final column reports the sign of aggregate village net food exports.

Figure F2: Simple Model: Real-Income Profiles Across Stress-Test Variants



Note: Each curve plots real income relative to the most distant village against distance from the city, solved at $\varphi = 0.70$ under one deviation from the baseline calibration of Section F.2. The baseline is plotted as the heavy black line. The corresponding summary statistics are reported in Table F1.

cross-sectional relationship between connectivity and non-agricultural employment share is that the demand for non-agricultural goods concentrates nearby the city; without the differences in trade costs, that never occurs. From a quantitative perspective, the size of the curse shrinks as food becomes more costly to trade. At $\kappa_a = 0.10$ the gap nearly closes, while at $\kappa_a = 0$ the curse deepens to -6.9 percent. Given the importance the values of κ have for this model, I also include robustness tests that vary the values of κ_a in the full model (Appendix N.7).

Fourth, the forces that further concentrate city service demand spatially amplify the curse. Raising the substitution elasticity from $\sigma_n = 4$ to 8 moves the gap from -2.1 to -9.8 percent, and raising the trade-cost elasticities from 0.40 to 0.80 moves it from -3.3 to -9.3 percent. Consistent with Proposition 1, the sign-reversal threshold in the last column moves with the same forces (between 0.33 and 0.65 across variants) and never exceeds the calibrated $\varphi = 0.70$, itself below every 2SLS point estimate of Section 8 and inside every weak-instrument-robust confidence set. The final row returns to the demand weights that ignore the employment-level target; the curse is essentially unchanged, so the level calibration of Section F.2 is not what generates it.

Fifth, agriculture in the model uses no capital, while the estimate of Section 8 is the return to labor at given land and capital. Adding capital rented from the city, $Y_{a,v} = \bar{A}_a Z_{a,v}^{\gamma_a+\varphi} K_v^{\gamma_K} H_v^{1-\gamma_a-\gamma_K}$, with $\gamma_a + \varphi$ held at its calibrated value, deepens the curse and lowers the threshold. At a capital share of $\gamma_K = 0.08$ the income gap between the closest and most distant village more than doubles and φ^* falls by a fifth, and the same holds when capital is shipped from the city at the measured goods friction. As labor leaves agriculture, capital leaves with it, so a given externality does more damage. The model without capital is therefore conservative.

The magnitudes in this appendix are illustrative (they depend on a stylized geometry and on levels the toy is not asked to match) and the quantitative statements of the paper rest on the estimated model of Sections 9–11. What the stress test establishes is that the curse's existence needs two key ingredients: the externality itself, and a trade structure in which proximity concentrates service demand without handing food producers a large enough offsetting price premium. Both ingredients are measured in the data.

G Derivation of the Sign-Reversal Threshold

This appendix derives the closed-form connectivity threshold stated in Proposition 1 of Section 6.2. The environment is the simple model of Appendix F with one additional modification that is important for reaching closed-form solutions. The independent Fréchet ability draws are replaced by a system where agricultural ability is common across workers and non-agricultural ability drawn from a Pareto distribution. The Pareto tail makes sectoral labor supplies closed-form in the service employment share, and every response to a connectivity shock then follows by applying the chain rule. The trade structure is still like the full model's. The mechanism is identical to the quantitative

model's, and only the sorting algebra differs.[23]

G.1 The Village Block

A village has a unit mass of workers that consumes three goods (food, village non-agricultural services, and a city good) and the technology of the main text. Food is produced with labor and land and features a positive external scale elasticity, and services are produced with a constant returns to scale labor only technology,

$$Y_a = \bar{A}_a \, Z_a^{\gamma_a+\varphi} H^{1-\gamma_a}, \qquad Y_n = A_n \, Z_n^{\gamma_n}, \qquad \gamma_n \in (0,1], \tag{G1}$$

where $\gamma_n = 1$ is the constant-returns baseline and $\gamma_n < 1$ the decreasing-returns variant of Appendix F.[24] The village faces three prices: a farm-gate food price p_a, a village-gate service price p_n, and a delivered city-good price p_u^d; Section G.2 ties all three to distance. Village income is the value of production, $I = p_a Y_a + p_n Y_n$, and the city good enters only through the budget. The occupational margin depends on the two producer prices only through their ratio,

$$\rho \equiv \frac{p_n}{p_a},$$

the relative price of the two village goods.

Each worker has the same agricultural ability and draws a service ability z from a Pareto distribution with shape $k > 1$ and minimum $\underline{z}$. A worker chooses services when $w_n z > w_a$, so the marginal worker sits at the threshold $z^* = w_a/w_n$. The service employment share π_n is then given by $\pi_n = (\underline{z}/z^*)^k$. The Pareto partial expectation then gives effective labor in both sectors:

$$Z_a = 1 - \pi_n, \qquad Z_n = \frac{k}{k-1} \, \underline{z} \, \pi_n^{\frac{k-1}{k}}. \tag{G2}$$

Substituting the competitive wage conditions from the FOCs $w_a = \gamma_a p_a \bar{A}_a Z_a^{\gamma_a+\varphi-1} H^{1-\gamma_a}$ and $w_n = \gamma_n p_n A_n Z_n^{\gamma_n-1}$ into the threshold and collecting powers of π_n reduces the equilibrium to one scalar equation in the relative price,

$$\frac{(1-\pi_n)^B}{\pi_n^A} = \frac{\tilde{C}}{\rho}, \qquad A = \frac{1+(1-\gamma_n)(k-1)}{k}, \qquad B = 1-\gamma_a-\varphi, \tag{G3}$$

with $\tilde{C}$ a constant collecting the technology levels. Under the stability condition $\gamma_a + \varphi < 1$ (so $B > 0$) the left side falls monotonically from $+\infty$ to 0 on $(0,1)$, so an interior equilibrium exists, is unique, and π_n rises with ρ. The intuition is that better terms for services pull workers out of

[23] The one-village threshold model is developed in full in a companion derivation, where every elasticity below is confirmed against numerical solutions to 10^{-9} or better. Under independent Fréchet ability sorting the threshold still exists, but I cannot solve for it in closed-form. The numerical φ^* column of Table F1 confirms the crossing in the Fréchet simple model.

[24] Under $\gamma_n < 1$ the residual $(1-\gamma_n)$ share of service revenue is a profit rebated to the village.

agriculture. At the calibrated values $\gamma_a + \varphi = 0.98$, so $B = 0.02 > 0$ and the condition holds. Log-differentiating (G3) gives the structural transformation elasticity

$$\eta \equiv \frac{d\ln \pi_n}{d\ln \rho} = \frac{1}{A + B\,\dfrac{\pi_n}{1-\pi_n}} > 0, \tag{G4}$$

and the chain rule delivers the responses of TFP and output: $d\ln A_a/d\ln\rho = -\varphi\frac{\pi_n}{1-\pi_n}\eta < 0$ and $d\ln Y_n/d\ln\rho = \gamma_n\frac{k-1}{k}\eta > 0$.

The income response collapses by an envelope argument. Competitive sorting maximizes the private value of village production at given A_a, so the labor-reallocation terms cancel by the marginal worker's indifference, leaving only the direct price effects (Hotelling's lemma, $\partial I/\partial p_k = Y_k$) and the externality feedback:

$$d\ln I = s_a\, d\ln p_a + s_n\, d\ln p_n \;-\; \mathcal{E}\; d\ln\rho, \qquad \mathcal{E} \equiv s_a\,\varphi\,\frac{\pi_n}{1-\pi_n}\,\eta, \tag{G5}$$

where $s_a = p_a Y_a/I$ and $s_n = p_n Y_n/I = 1 - s_a$ are the sectoral income shares. The composite $\mathcal{E}$ is the externality drag. A 1 percent increase in the relative service price reduces effective labor in agriculture by approximately $\frac{\pi_n}{1-\pi_n}\,\eta$ percent. Because of the externality, this lowers agricultural TFP by φ times that reallocation. Finally, the resulting income loss is weighted by agriculture's share of village income s_a.

The village here consumes only its own service variety; the full model's rural CES over all origins adds cross-village service demand, from which the one-village block abstracts. With Cobb–Douglas consumption the exact price index is $P = p_a^{\alpha_a} p_n^{\alpha_n} (p_u^d)^{\alpha_u}$, welfare equals real income $V = I/P$, and

$$d\ln V = (s_a - \alpha_a)\, d\ln p_a + (s_n - \alpha_n)\, d\ln p_n - \alpha_u\, d\ln p_u^d \;-\; \mathcal{E}\; d\ln\rho. \tag{G6}$$

Each price enters with the village's net exposure to that good.[25] Net food exports, net service exports, and city-good imports sum to zero by the budget identity. At $\varphi = 0$ equation (G6) is the textbook terms-of-trade calculus; the externality contributes the single extra term $-\mathcal{E}\, d\ln\rho$.

G.2 The Three Frictions

Now place a continuum of such villages at distances τ from a single city, the arrangement of Appendix F. Food arbitrage sets the farm-gate price at the market price net of transport, $p_a = \tau^{-\kappa_a}$ in units of the numéraire, so selling food is worth less far from the market. The city good is produced at a price p_u set in the city and delivered at $p_u^d = \tau^{\kappa_u} p_u$, so buying city goods costs more far from the market. And the city allocates a service budget X_n across village varieties by CES with elasticity

[25] Here net exposure measures the production share (share of income that comes from a given sector) minus the consumption share for that same good.

$\sigma_n > 1$ and iceberg cost τ^{κ_n}, so the free-on-board demand reaching a village at distance τ is

$$D_c(\tau, p_n) = D\, \tau^{\kappa_n(1-\sigma_n)}\, p_n^{-\sigma_n}, \qquad D \equiv X_n/\Phi, \tag{G7}$$

with Φ the city's CES price index over origins. Each village is atomistic and takes D and p_u as given. The village's service market clears,

$$Y_n = \alpha_n \frac{I}{p_n} + D_c(\tau, p_n), \tag{G8}$$

one equation in the local service price $p_n(\tau)$.

Log-differentiating (G8) in $\ln \tau$, writing θ_c and $\theta_l = 1-\theta_c$ for the city and local shares of the village's service sales and using the one-village elasticities of Section G.1 (supply $\varepsilon_S \equiv d\ln Y_n/d\ln\rho = \gamma_n \frac{k-1}{k}\eta$ and the income response (G5), with $d\ln p_a/d\ln\tau = -\kappa_a$ substituted throughout), the service-price and relative-price gradients solve out to

$$\frac{d\ln\rho}{d\ln\tau} = -\frac{\theta_c\left[\kappa_n(\sigma_n-1)-\kappa_a\sigma_n\right]}{\varepsilon_S + \theta_c\,\sigma_n + \theta_l\,(s_a+\mathcal{E})}, \qquad \frac{d\ln p_n}{d\ln\tau} = \frac{d\ln\rho}{d\ln\tau} - \kappa_a. \tag{G9}$$

The denominator captures how much the service price must adjust for the market to clear, and it is always positive. The numerator is what distance does to city demand for the village's services.

Distance increases the cost of delivering services to the city, which shrinks city demand and pushes the village's service price down. But distance also lowers its farm-gate food price, and since ρ measures services against food, a lower food price pushes ρ up. The relative service price falls with distance, and the mechanism operates, if and only if

$$\kappa_n \frac{\sigma_n - 1}{\sigma_n} \;>\; \kappa_a : \tag{G10}$$

The service friction shrinks city demand at rate $\kappa_n(\sigma_n-1)$ rather than $\kappa_n\sigma_n$, because shipping τ^{κ_n} units to deliver one raises the village's sales at the same time as the higher delivered price cuts them. The food price carries no such offset, so it enters at the full σ_n. That gap is why the threshold is $\kappa_n(\sigma_n-1)/\sigma_n$ and not κ_n. Services must be effectively harder to trade than food. This is the friction asymmetry of Section 9.1 appearing as an explicit condition; at the measured frictions ($\kappa_n = 0.90$ against $\kappa_a = 0.05$) it holds by an order of magnitude.

Because distance affects sectoral sorting only through ρ, the employment gradient $d\ln\pi_n/d\ln\tau = \eta\,(d\ln\rho/d\ln\tau) < 0$ and the TFP gradient $d\ln A_a/d\ln\tau > 0$ reproduce the reduced-form patterns of Section 5: villages closer to the city do more of their work in services and have lower agricultural productivity.

G.3 The Welfare Gradient and Its Decomposition

Substituting the three price gradients into (G6) decomposes the welfare gradient into four named channels:

$$\frac{d\ln V}{d\ln\tau} = \underbrace{(s_n-\alpha_n)\,\frac{d\ln p_n}{d\ln\tau}}_{\substack{\text{service terms of trade}\\(<0\text{: proximity gain})}} \underbrace{-\,(s_a-\alpha_a)\,\kappa_a}_{\substack{\text{farm-gate food price}\\(<0\text{: proximity gain})}} \underbrace{-\,\alpha_u\,\kappa_u}_{\substack{\text{delivered city goods}\\(<0\text{: proximity gain})}} \underbrace{-\,\mathcal{E}\,\frac{d\ln\rho}{d\ln\tau}}_{\substack{\text{externality drag}\\(>0\text{: proximity curse})}} . \tag{G11}$$

The first three terms are the standard gains from market access: a connected village sells its services and food for more (for a net food exporter, $s_a > \alpha_a$ and $s_n > \alpha_n$, the empirically relevant case), and buys its city goods cheaper. All three are losses when distance increases. In contrast is the single term the externality adds. At $\varphi = 0$ the relationship between distance and income is unambiguously negative, so the externality is the only potential source of any curse.

Using the trade balance $(s_a-\alpha_a)+(s_n-\alpha_n) = \alpha_u$ to eliminate the food term, the four channels collapse to:

$$\frac{d\ln V}{d\ln\tau} = \left[(s_n-\alpha_n) - \mathcal{E}\right]\frac{d\ln\rho}{d\ln\tau} \;-\; \alpha_u\,(\kappa_a+\kappa_u). \tag{G12}$$

Distance hurts the village through exactly two channels. The first works through the relative price ρ. The bracket in front of it is the village's own response, its terms-of-trade gain net of the externality drag, and it multiplies the equilibrium price gradient in (G9). The second is the cost of trading with the city at all. Distance lowers the price the village receives for what it sells, at rate κ_a, and raises the price it pays for what it buys, at rate κ_u, and neither wedge depends on anything the village does. By the trade balance its net exports are exactly what it spends on city goods, α_u, so both rates apply to that same share and the loss is $\alpha_u(\kappa_a+\kappa_u)$ per log point of distance.

The service friction is missing from that second channel because it never sets the village's service price. It weakens the demand the city sends, and the price that follows is the relative-price gradient, so κ_n is already inside the first channel. That second channel involves no externality at all, and it is the ordinary gain from market access that any trade model delivers. It exists only because food and city goods carry frictions of their own. With a single friction, κ_a and κ_u would both be zero, distance would work through the relative price alone, and this level effect would disappear.

G.4 Proof of Proposition 1

Under condition (G10) the relative-price gradient is strictly negative, so (G12) turns into a condition where real income increases with distance if and only if

$$\underbrace{\mathcal{E}\left|\frac{d\ln\rho}{d\ln\tau}\right|}_{\text{externality loss}} \;>\; \underbrace{(s_n-\alpha_n)\left|\frac{d\ln\rho}{d\ln\tau}\right|}_{\text{terms-of-trade gain}} \;+\; \underbrace{\alpha_u\,(\kappa_a+\kappa_u)}_{\text{market-access gain}} . \tag{G13}$$

Condition (G13) is a threshold in φ. Both $\mathcal{E}$ and the denominator of (G9) are linear in η, and

$1/\eta = A + (1-\gamma_a)\frac{\pi_n}{1-\pi_n} - \varphi\frac{\pi_n}{1-\pi_n}$ is linear in φ, so holding the equilibrium objects $(\pi_n, s_a, s_n, \theta_c)$ fixed at their values in a given equilibrium, (G13) holds at equality at exactly one externality value, and rearranging (multiplying through by $1/\eta > 0$ and collecting φ) gives the closed form solution for the sign reversal threshold. Because those objects themselves vary with φ in equilibrium, the threshold is a pointwise characterization evaluated at each village's own equilibrium rather than a proof of a unique global crossing; the numerical sweeps below and in Appendix F find a single sign change along the φ path. Define the market-access-to-terms-of-trade ratio and the gain composite

$$b \equiv \frac{\alpha_u(\kappa_a + \kappa_u)}{\theta_c\big[\kappa_n(\sigma_n - 1) - \kappa_a\sigma_n\big]}, \qquad S \equiv (s_n - \alpha_n) + b\left(\theta_c\sigma_n + \theta_l\, s_a\right);$$

then proximity is a curse if and only if $\varphi > \varphi^*$, with

$$\varphi^* = \frac{S\left[(1-\gamma_a) + A\,\dfrac{1-\pi_n}{\pi_n}\right] \;+\; b\,\gamma_n\,\dfrac{k-1}{k}\,\dfrac{1-\pi_n}{\pi_n}}{S + s_a\,(1 - b\,\theta_l)}, \tag{G14}$$

provided the denominator is positive, which holds whenever b is small ($b \approx 0.01$ at the measured frictions and shares). With frictionless food and city goods, $\kappa_a = \kappa_u = 0$ so $b = 0$ and $S = s_n - \alpha_n$, and (G14) reduces, using $s_a + s_n = 1$, to

$$\varphi^*\big|_{\kappa_a=\kappa_u=0} = \frac{(s_n - \alpha_n)(1-\pi_n)}{s_a\,\pi_n\,\eta} = \frac{s_n - \alpha_n}{1 - \alpha_n}\left[(1-\gamma_a) + A\,\frac{1-\pi_n}{\pi_n}\right], \tag{G15}$$

the frictionless special case of the threshold in Proposition 1. The threshold is a local object, evaluated at each village's own shares, but the sign-reversal logic is exact at every village, which establishes the proposition.

G.5 Numerical Confirmation

The analytical gradients are confirmed against full numerical solutions of the clearing condition (G8) on a distance grid at an illustrative calibration sharing the quantitative model's agricultural labor share ($\gamma_a = 0.28$, $\gamma_n = 0.92$, $k = 4$, $\sigma_n = 4$, $\kappa_n = 0.65$, $\kappa_a = 0.06$, $\kappa_u = 0.10$, $(\alpha_a, \alpha_n, \alpha_u) = (0.85, 0.05, 0.10)$, $\varphi = 0.65$). The analytic gradients (G9) and (G12) match the numerical solutions at every grid point:

	$d\ln\rho/d\ln\tau$ analytic / finite diff.	$d\ln\pi_n/d\ln\tau$ analytic / finite diff.	$d\ln V/d\ln\tau$ analytic / finite diff.
$\tau \approx 1.6$	-0.2566 / -0.2566	-0.7166 / -0.7166	$+0.2493$ / $+0.2493$
$\tau \approx 2.8$	-0.2307 / -0.2307	-0.6905 / -0.6905	$+0.1433$ / $+0.1433$

At the model's demand shares (0.667, 0.103, 0.230) and the measured food and city-good frictions ($\kappa_a = 0.05$, $\kappa_u = 0.203$), with the illustrative $\kappa_n = 0.65$, $\sigma_n = 4$, $k = 4$, and $\gamma_n = 0.92$ retained,

the welfare gradient crosses zero at $\varphi = 0.376$, and the closed form (G14), evaluated at that equilibrium's own shares, returns the same value; shutting off κ_a and κ_u reproduces the frictionless threshold (G15) (about 0.16). The content of the check is the agreement between the analytic gradients and the finite differences and between the crossing and the closed form; the threshold itself depends on the illustrative sorting and service parameters, and the quantitative model's own reversal is reported in Section 10.2.

H Additional Empirical Results

Table H1: Effects of Market Access on Housing

	ln(Housing Expenditure)	ln(Housing Expenditure)	ln(House Area)	ln(Price Per Sq Meter)
ln(MA)	0.0130*	0.0118*	0.0030	0.0093
	(0.0074)	(0.0070)	(0.0029)	(0.0074)
Household Size		0.0510***	0.0756***	0.0078
		(0.0082)	(0.0095)	(0.0105)
House Area		0.0094***		
		(0.0018)		
House Area, Sq		-0.0000		
		(0.0000)		
Expected MA Control	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes
District Controls	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes
Island FE	Yes	Yes	Yes	Yes
Unique Villages	70	70	70	70
Unique Households	1086	1086	1086	1086

Standard errors in parentheses

Standard errors clustered at the village level

* $p < .1$, ** $p < .05$, *** $p < .01$

Note: This table presents results from estimating the effect of market access on housing price and expenditure according to Equation (2). Columns 1-4 use data from the 2002 SUSENAS household expenditure module.

Table H2: Effects of Market Access on Land Use Dynamics

	ln(Ag Land)	ln(Housing + Firm Land)	Change in Ag Land	Change in Housing + Firm Land	New Housing Development
ln(Market Access)	-0.0237***	-0.0155**	-2.7412**	-1.7831*	-0.0000
	(0.0056)	(0.0071)	(1.1285)	(1.0409)	(0.0009)
Expected MA Control	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes
District Controls	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes
Island FE	Yes	Yes	Yes	Yes	Yes
Year FE	No	No	No	No	No
Unique Villages	940	935	944	944	1147

Standard errors in parentheses

Standard errors clustered at the village level

* $p < .1$, ** $p < .05$, *** $p < .01$

Note: This table presents results from estimating the effect of market access on land use dynamics according to Equation (2). Columns 1-4 use data from the 2003 PODES. Column 5 uses data from the 2000 PODES. Cultivated land declines with market access while land devoted to housing and firm activity does not expand; see Section 7.

Table H3: Market Access and Agricultural Potential

	(1) ln(GDP per Capita)	(2) ln(GDP per Capita)
ln(Market Access)	-0.0051***	-0.0050***
	(0.0017)	(0.0016)
ln(Market Access) × Potential Yield Index	0.0028**	
	(0.0012)	
ln(Market Access) × Rice Potential Yield		0.0037***
		(0.0012)
Expected MA Control	Closest 20	Closest 20
Local Controls	No potential yields	Yes
District Controls	Yes	Yes
Cohort FE	Yes	Yes
Island FE	Yes	Yes
Year FE	Yes	Yes
Std. Errors	Conley 100 km	Conley 100 km
Unique Villages	1151	1151

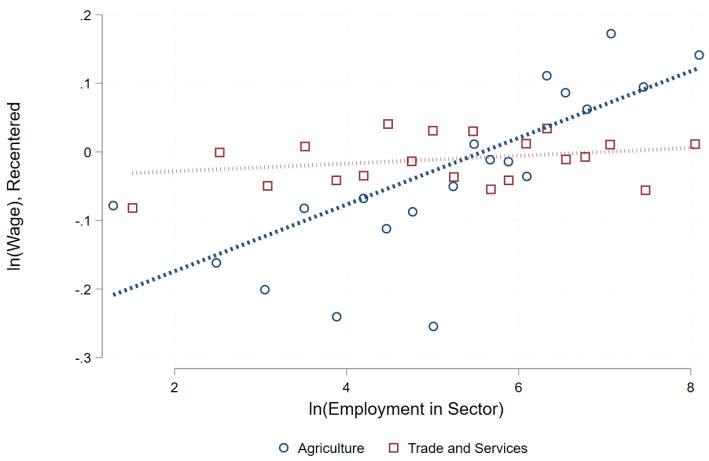
Note: This table presents results from estimating Equation (2) with log initial market access interacted with a village's agricultural potential. Column 1 interacts it with an index of the potential yields of wetland rice, dryland rice, palm oil, coffee, and cassava, the standardized sum of the standardized yields, and drops the potential-yield controls from which the index is built. Column 2 interacts it with wetland rice potential yield and keeps all controls. Both measures are standardized over the regression sample, so each interaction is the change in the coefficient on log market access per standard deviation of potential yield. Standard errors, in parentheses, allow for spatial correlation following Conley (1999), with a Bartlett kernel and a 100 kilometer cutoff. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I The Agricultural Externality: Additional Results

Figure I2 plots the first stage of the instrumental variables design of Section 8. Table I2 reports the balance of initial settlement size with respect to observable village and district characteristics: initial settlement size in this sample is uncorrelated with village-level characteristics, and with both market access and expected market access, so the instrument captures variation in village scale that is orthogonal to the connectivity variation exploited in Section 5. In the full sample of Table 1, larger settlements sit in slightly less connected places, and every specification of Section 8 controls for market access and expected market access. Several of the district-level characteristics from 1980 do show statistically significant correlations with settlement size, concentrated among measures of infrastructure and urbanization; the omnibus F-statistic of 1.32 fails to reject the null of joint insignificance, and the specifications control for these district-level measures directly.

Figure I1 and Table I1 report the wage evidence cited in Section 6.1. The figure is a binscatter of rural workers' log wages against the size of the worker's own sector in the village, separately for agriculture and retail; the table adds village and demographic controls, and the agricultural slope survives while the retail and service slope remains flat.

Figure I1: Sector Size vs. Individual Wages



Note: Binscatter of log wages (demeaned) against log sector size. The data here contains all formally employed rural individuals on the Outer Islands that appear in the pooled SUSENAS data.

Table I3 examines specific threats to exclusion. Larger settlements may develop in different ways that could affect agricultural productivity beyond the size of the sector itself. The table reports the association of initial settlement size with candidate downstream channels such as market presence, market access, financial institutions, and schooling infrastructure. Because these are themselves post-settlement outcomes, I read this table as suggestive evidence that exclusion is not violated. Most of these associations are small and insignificant, but I also report specifications that condition on these outcomes. When I control for them, the estimated scale externality increases in magnitude

Table I1: Relationship between Sector Size and Wages

	Agriculture		Trade		Services	
	ln(Hourly Wage)	ln(Hourly Wage)	ln(Hourly Wage)	ln(Hourly Wage)	ln(Hourly Wage)	ln(Hourly Wage)
ln(Employment in Sector)	0.0296***	0.0173***	0.0070	-0.0164	0.0216***	0.0091
	(0.0056)	(0.0052)	(0.0184)	(0.0199)	(0.0060)	(0.0056)
Village Controls	Yes	Yes	Yes	Yes	Yes	Yes
HH Controls	Yes	Yes	Yes	Yes	Yes	Yes
FE Type	Island	Province X Year	Island	Province X Year	Island	Province X Year
Unique Villages	3177	3177	1015	1007	6233	6233
Obs	16646	16645	1687	1678	26864	26864

Standard errors in parentheses

Standard errors clustered at the village level

* $p < .1$, ** $p < .05$, *** $p < .01$

Note: Each column estimates the relationship between log sector size and log hourly wages for all formally employed rural individuals on the Outer Islands that appear in the pooled SUSENAS data. Village controls are the potential yield variables described in the primary specification along with the overall village population, and household controls are gender, age, and years of education.

rather than decreases. Larger settlements may also shift crop choices, which the crop fixed effects and the composition test of Section 8 address. While it is impossible to definitively rule out violations of the exclusion restriction, the patterns I can examine do not point toward one.

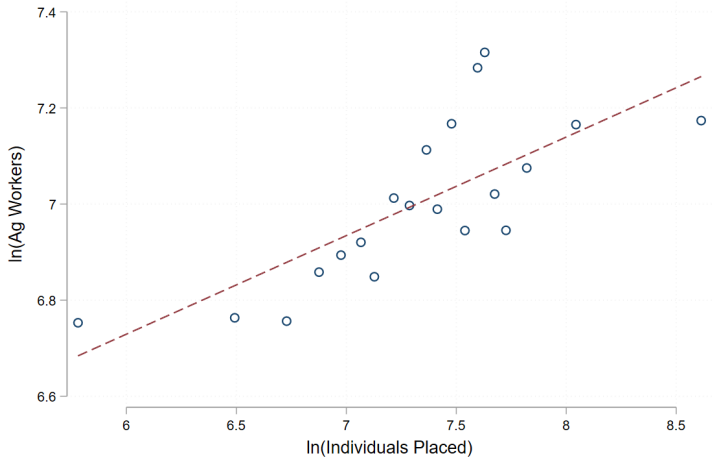
Planted area is chosen in the same season as output, so unlike the predetermined village and district controls it is the control most likely to respond to the same shocks. It also rises with settlement size: in the specification of column 2 of Table 6 without the area control, a log point of individuals initially placed raises the area planted to a crop by 0.29 log points (s.e. 0.10). That response is what the program's design predicts, since land was allocated per household and larger settlements therefore received more of it. Omitting the area control loads this land allocation onto the labor coefficient, which falls from 1.41 to 1.09 (Anderson–Rubin set $[0.17, 5.24]$), and the implied externality, 0.81, remains large and positive. The baseline includes planted area because β is the labor coefficient at given land. The estimator run on model-generated data shows the same mechanism: with the recorded land allocations and no land control, the land response loads onto the labor coefficient (Appendix N.1).

I.1 The Elasticity at Larger Scales

Section 8.3 argues that an externality with a minimum viable scale should weaken as the farm workforce grows. The village sample cannot measure this. With the nonlinear terms in $\ln L_v^{ag}$ instrumented by the same terms in $\ln W_v$, the Kleibergen-Paap F-statistic is about 2, and the village estimates in column 1 of Table I4 are uninformative. I therefore estimate Equation (11) at two larger levels: contiguous groups of program villages, and subdistricts. Aggregation strengthens the first stage, to F-statistics of 143 and 154 in the linear specification, and widens the range of agricultural workforces. Between the 5th and 95th percentiles they span a factor of about 11 across villages and 24 to 30 across the larger units.

In both larger units the elasticity falls with size (columns 2 and 3). With a quadratic in the log agricultural workforce, the slope falls from 0.58 at the 5th percentile to 0.33 at the 95th across

Figure I2: First Stage: Ag Working Population vs. Initial Settlement Population



Note: Binscatter of log agricultural working population against log initial settlement population, controlling for island and cohort fixed effects.

Table I2: Village Characteristics Across Initial Settlement Size

Dependent variable (std): ln(Individuals Placed) (std)	
Wetland Rice Potential Yield	-0.012 (0.032)
Palm Oil Potential Yield	0.075* (0.043)
Coffee Potential Yield	-0.013 (0.040)
Cassava Potential Yield	0.025 (0.030)
Vector Ruggedness Measure	0.042 (0.043)
Agrosimilarity Score	0.002 (0.036)
Share of Land with Poor Drainage	0.067 (0.046)
Share of Land with Good Drainage	-0.059 (0.045)
Organic Carbon (%)	0.050 (0.039)
Topsoil Salinity (Elco) (dS/m)	-0.098 (0.075)
Topsoil Base Saturation (%)	-0.027 (0.056)
Avg. rainfall, 1948–1978	-0.061 (0.043)
Avg. temp (Celsius), 1948–1978	-0.027 (0.044)
ln(MA)	-0.023 (0.039)
ln(Expected MA)	-0.032 (0.033)
ln(District Population, 1980)	-0.024 (0.037)
District Own Electricity, 1980	-0.098** (0.046)
District Own Radio, 1980	-0.030 (0.046)
District Own TV, 1980	-0.108** (0.049)
District Literacy, 1980	-0.062 (0.042)
District Avg Years Schooling, 1980	-0.062 (0.044)
District Agricultural Employment Share, 1980	0.062 (0.046)
District Wetland Rice Potential Yield, 1980	0.016 (0.039)
District Palmoil Potential Yield, 1980	0.012 (0.040)
District Coffee Potential Yield, 1980	0.052 (0.042)
District Cassava Potential Yield, 1980	0.018 (0.039)
District Mining Employment Share, 1980	0.102** (0.040)
District Manufacturing Employment Share, 1980	0.007 (0.038)
District Trade Employment Share, 1980	-0.069 (0.046)
District Service Employment Share, 1980	-0.085* (0.048)
District Wage Worker Share, 1980	-0.052 (0.039)
District Share HHs with Piped Water, 1980	-0.119** (0.049)
District Share HHs with Sewage, 1980	-0.080* (0.046)
District Share HHs with Modern Fuel, 1980	-0.130*** (0.049)
District Share HHs with Modern Floor, 1980	-0.053 (0.045)
District Share HHs with Modern Roof, 1980	0.039 (0.043)
F-stat of omnibus test	1.319
Observations	719

Note: Each row reports the coefficient from a regression of the standardized dependent variable listed in the row on the standardized log number of individuals initially placed in a transmigration village, controlling for island fixed effects and cohort fixed effects. Standard errors, clustered at the subdistrict (kecamatan) level, are reported in parentheses. The omnibus F-statistic tests joint significance across all dependent variables. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I3: Effects of Initial Settlement Size on Downstream Outcomes

	Presence of Permanent Market	MA, recentered	Total Banks	Banks per Capita	Elementary School Density (per 1000)	Middle School Density (per 1000)	High School Density (per 1000)
ln(Individuals Placed)	0.0619	-0.7123	0.0252	0.0000	-0.0636	-0.0222	-0.0216**
	(0.0412)	(0.5626)	(0.0383)	(0.0000)	(0.0465)	(0.0240)	(0.0105)
MA Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Local Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Island FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Unique Villages	601	601	601	601	601	601	601

Standard errors in parentheses

Standard errors clustered at the village level

* $p < .1$, ** $p < .05$, *** $p < .01$

Note: This table presents results from estimating the effect of initial settlement size on a variety of outcomes that could indicate a violation of the exclusion restriction in the 2SLS specification of Section 8. All columns except column 2 use data from the 2003 PODES wave. Column 2 uses the constructed measure of market access, recentered by the expected market access measure described in Section 4.

contiguous groups, and from 0.84 to 0.00 across subdistricts. With separate slopes below and above the median, it falls from 0.77 to 0.29 and from 0.69 to 0.08. Per log point of the agricultural workforce, these are declines of 0.07 to 0.39. None is statistically significant at conventional levels, with p-values between 0.13 and 0.47, so I treat the evidence as suggestive. Appendix N.10 solves the model with the externality declining at rates in this range.

The aggregated elasticities are also lower in level than the village estimate, 0.42 and 0.38 against 1.25. Several things besides scale can produce that gap. The village estimate is imprecise, and both aggregated estimates lie inside its weak-IV-robust confidence set, $[0.31, 6.65]$. Aggregation also changes what is estimated: output, labor, and the controls are summed or averaged over villages that differ in crops and conditions, and part of the variation across the larger units comes from how many program villages they contain rather than from the size of each. Size alone is unlikely to explain the gap, since the median contiguous group's agricultural workforce is only about a third larger than the median village's. The model exercise therefore draws on the decline within each level rather than on the difference in levels across them.

Table I4: The Agricultural Scale Elasticity at Larger Scales

	Village	Contiguous Group	Subdistrict
A. Linear			
β	1.248**	0.421***	0.375***
	(0.636)	(0.101)	(0.118)
B. Quadratic			
Slope at 5th percentile	-0.977	0.579**	0.843**
	(1.199)	(0.263)	(0.358)
Slope at 95th percentile	3.751*	0.325**	-0.000
	(2.184)	(0.144)	(0.246)
C. Median spline			
Slope below median	-0.523	0.774**	0.692**
	(1.066)	(0.322)	(0.276)
Slope above median	3.502	0.286**	0.078
	(2.133)	(0.122)	(0.221)
KP F, linear	6.8	143.5	153.7
KP F, quadratic	2.3	15.6	11.7
p-value: no curvature	0.127	0.470	0.132
KP F, spline	2.1	12.5	30.2
p-value: equal slopes	0.167	0.203	0.167
Agricultural workforce, 5th percentile	306	360	569
Agricultural workforce, median	1,185	1,615	3,190
Agricultural workforce, 95th percentile	3,258	10,860	13,462
Observations	1,486	835	610
Units	601	307	202

Note: 2SLS estimates of Equation (11) for program villages (column 1), contiguous groups of program villages (column 2), and subdistricts (column 3). In columns 2 and 3, production and area planted of the five main crops, agricultural workers, tractors, and individuals initially placed are summed within each unit, the village controls are averaged, and the cohort effect is the placement year in which the unit received the most settlers. Panel A instruments the log agricultural workforce with the log of individuals initially placed. Panel B adds the square of the log agricultural workforce, instrumented by the square of the log of individuals placed, and reports the slope at the 5th and 95th percentiles of the agricultural workforce across units; its p-value tests the squared term. Panel C lets the slope change at the median, with the hinge instrumented by the same hinge in the log of individuals placed; its p-value tests equal slopes. KP F is the Kleibergen-Paap rk Wald statistic of each specification. Controls and fixed effects are those of column 1 of Table 6. Standard errors, in parentheses, are clustered by village, group, or subdistrict. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Measuring the Private Labor Share

The externality in Section 8 is the gap between the coefficient on agricultural labor and the private return to labor, γ_a. In the model γ_a is labor's share of the value of output, and land, which stands in for every input that is not labor, receives the rest. Under competitive input pricing that share equals the elasticity of output with respect to labor at given land and capital, the object the 2SLS estimates, so I measure γ_a as a share of output rather than of cost. This appendix describes the measurement, compares it with aggregate estimates for Indonesia, and reports household-panel estimates that bound it from below.

J.1 Farm Cost Structure

The measurement uses BPS's 2017 farm cost-structure survey for paddy in South Sumatra, a transmigration province (BPS-Statistics of Sumatera Selatan Province, 2018). For one plot and one growing season, the survey records the days and wages of hired workers, the days of family workers valued at the local wage, fees paid to contractors for tasks such as tractor land preparation, transplanting, and harvesting, purchased inputs, land rent, imputed for owned land, and the quantity and value of output. The published tables cover 4,925 wetland-paddy households. Panel A of Table J1 reports each cost component as a share of the value of output. Hired workers receive 7 percent of output and family workers 13 percent. Contracted field services take a further 16 percent, and they are part labor and part machinery: two thirds of households prepare their land with a tractor, most of them hiring it as a service with an operator, while harvesting and transplanting are contracted to work gangs.

Panel B reports the labor share under alternative treatments of that line. Counting half of the services fee as labor gives $\gamma_a = 0.28$, the value used in the paper. Counting a third gives 0.25 and two thirds 0.31, and valuing family labor at the household-panel elasticity below rather than at the local wage gives 0.24. The rainy and dry seasons bracket the annual figure. I take 0.25 to 0.31 as the range.

Two aggregate estimates for Indonesia agree. Fuglie (2003) estimates a Cobb–Douglas production function on national data for 1961–2000 and finds a labor elasticity of 0.38, or 0.30 once the livestock coefficient is fixed at a conventional value; the confidence interval is wide. Fuglie (2010) reports labor's share of total agricultural cost at 0.42. Costs are 72 percent of the value of output in the South Sumatra survey, so that share corresponds to about 0.30 of output.

J.2 Household Panel Estimates

The Indonesia Family Life Survey (Strauss et al., 2016) follows households over time and allows the elasticity to be estimated directly. I use farm households in the 2000, 2007, and 2014 waves, which record farm revenue, total farm expenses, cultivated area, the value of equipment and livestock, and the hours each member works on the family farm. Regressing log revenue on log family hours,

Table J1: Labor's share of paddy output, South Sumatra 2017

	All seasons	Rainy season	Dry season
A. Cost components as shares of the value of output			
Hired workers	0.07	0.06	0.09
Family workers, valued at the local wage	0.13	0.11	0.21
Contracted field services	0.16	0.18	0.12
Land rent, actual and imputed	0.18	0.19	0.17
Seed, fertilizer, pesticide	0.11	0.11	0.11
Equipment rental, depreciation, fuel	0.04	0.04	0.06
Taxes, interest, other	0.02	0.02	0.03
Residual	0.28	0.30	0.22
B. Labor share of output, by treatment of contracted services and family labor			
Services excluded	0.20	0.16	0.30
One third of services as labor	0.25	0.22	0.34
Half of services as labor (preferred)	0.28	0.25	0.36
Two thirds of services as labor	0.31	0.28	0.38
All services as labor	0.36	0.34	0.42
Half of services, family labor at the panel elasticity	0.24	0.24	0.24

Note: Source: BPS, *Hasil Survei Struktur Ongkos Usaha Tanaman Padi 2017 Provinsi Sumatera Selatan*, Tables 29 and 30; wetland paddy, 4,925 households. Costs are per hectare per planting season and include imputed rent on owned land, imputed rent on owned equipment, and imputed wages for family workers at the local wage. Panel B counts the stated fraction of contracted field services as labor; the last row values family labor at the household-panel elasticity of Table J2 instead of the local wage.

capital, area, and expenses gives the elasticity of output with respect to family labor. Table J2 reports OLS with wave and province effects, household fixed effects, and the control-function estimator of Gandhi et al. (2020), which allows input choices to respond to productivity shocks, along with two fixed-effects estimates that instrument the hours measure with a second measure of the same input from the survey, hours worked in the previous week and the number of household members working the farm, to reduce the effect of recall error. The elasticity is 0.05 to 0.09 across estimators. It does not vary with the main crop grown, between Java and the outer islands, or with local land scarcity once households are compared with themselves over time. Across communities it is somewhat higher where land is cheap and forest is still being cleared for cultivation, which suggests the return to labor may be a little larger in outer-island settings like the transmigration villages, though the within-household comparisons cannot confirm it.

The panel estimate is a lower bound on γ_a for two reasons. First, the survey records a single farm expense total, so hired and contracted labor cannot be separated from purchased inputs, and the estimate covers family labor only. In the cost-structure survey family labor is 48 percent of labor cost, and dividing by that share gives the all-labor equivalents in the last column of Table J2, 0.11 to 0.18. Second, annual recall hours summed over household members are noisy, which biases the coefficient toward zero. The same family component measured from a single plot-season in the cost-structure survey is 0.135, and the same step turns it into 0.28. A private share at the lower

end of these values would raise the estimated externality one for one, since $\varphi = \beta - \gamma_a$, and leaves the paper's conclusions unchanged.

Table J2: The family-labor elasticity in the IFLS farm panel

Estimator	Family-labor elasticity	Observations	All-labor equivalent
OLS, wave and province effects	0.051 (0.011)	8,144	0.11
Household fixed effects	0.078 (0.024)	5,943	0.16
Control function (Gandhi, Navarro, and Rivers 2020)	0.073 (0.013)	7,980	0.15
Household fixed effects, hours instrumented by hours last week	0.059 (0.017)	5,200	0.12
Household fixed effects, hours instrumented by workers on the farm	0.087 (0.033)	5,200	0.18

Note: IFLS farm households in the 2000, 2007, and 2014 waves. The dependent variable is log farm revenue; every specification controls for log capital, log cultivated area, and log farm expenses, with wave effects. Standard errors in parentheses, clustered by community for OLS and by household otherwise. The all-labor equivalent divides the family-labor elasticity by family labor's share of labor cost in the South Sumatra cost structure, 0.48.

K Where the Scale Externality Could Come From

Section 8 estimates the externality as a reduced form: agricultural TFP rises with the size of the agricultural workforce with elasticity φ, and the data say how large φ is but not why. This appendix is an illustration of why. It writes down the simplest village in which a larger workforce raises everyone's productivity, for the three reasons the literature on agglomeration usually separates (Duranton and Puga, 2004): indivisible inputs get shared, markets get thicker, and people learn from one another. The same settlement-size variation that identifies φ also moves things these mechanisms should move, namely whether a village has a tractor or a rice mill, how concentrated its land has become, and how educated its largest farmers are. Where a mechanism leaves a footprint I can see, I use it to size the mechanism's part of φ; where it does not, I say so. The measured parts add up to about 0.165; the rest of the 0.70 the model carries is a residual that the data I have cannot split further. Table K1 collects the facts and the arithmetic.

Throughout, a fact is the coefficient from a regression of an outcome on log individuals initially placed, with the controls of Section 8.1. I divide each by the first stage of 0.200 so that everything is per log point of the agricultural workforce, the units of $\beta = \gamma_a + \varphi$.

K.1 The Village

A village has L farmers and land H. Farmer i works their own farm, so land has diminishing returns at the farm level, and produces

$$y_i = \omega\, \tilde{a}_i\, h_i^{\rho}, \tag{K1}$$

where h_i is the land the farmer operates, $\tilde{a}_i$ is the farmer's productivity, and ω collects the services the farm buys locally. When farmers are alike and hold equal plots, village output is $Y = \omega\tilde{a}\, L^{1-\rho}H^{\rho}$, the technology of the main text with $\gamma_a = 1-\rho$; with farmers of equal ability, the farmer count L is the main text's effective agricultural labor Z_a. I therefore set $\rho = 1-\gamma_a = 0.720$, and the division of output between farmers and land is the same as in Section 6.

Three things change when the village is larger. More farmers support more local services, so ω rises. More farmers make it easier to trade land, so land ends up closer to where it is most productive. And more farmers make it more likely that one of them is unusually good, and that the others copy what they do. Each of the three multiplies village TFP, so their elasticities add:

$$\ln Y = \ln A(L) + \gamma_a \ln L + (1-\gamma_a)\ln H, \qquad \varphi = \frac{d\ln A}{d\ln L} = \varphi_S + \varphi_M + \varphi_R, \tag{K2}$$

with φ_S the shared-input part, φ_M the land-market part, and φ_R everything the data do not reach.

K.2 Shared Inputs: Tractors and Rice Mills

A tractor rental business or a rice mill has a fixed cost and needs enough customers to cover it, so a larger village is more likely to have one. Larger settlements are indeed more likely to have a tractor, by 0.35 per log point of the workforce, and a rice mill, by 0.30. If having the service available raises farm TFP by a factor λ_k,

$$\varphi_S = \sum_k \ln \lambda_k \, \frac{d \Pr(\text{service } k \text{ present})}{d \ln L}. \tag{K3}$$

The gains are the one input I take from outside. I set $\lambda_t = 1.10$ for tractors, because mechanised land preparation mostly saves labor and affects output through timeliness and cropping intensity rather than yields (Pingali, 2007), and $\lambda_m = 1.08$ for mills, because a mill changes what the paddy fetches and how much is lost after harvest rather than what the field produces. This gives $\varphi_S = 0.056$. Moving the gains across 1.05–1.20 for tractors and 1.00–1.15 for mills moves it between 0.017 and 0.106.

Two remarks. The numbers of tractors, mills, and formal agricultural firms also rise with settlement size (Table K1), as free entry with a fixed cost predicts, but the median village has none of them, so the counts mostly restate the presence margin and I do not use them separately. And tractors and mills are simply the two shared inputs I observe. Threshers, pumps, drying floors, storage, seasonal work gangs, and the upkeep of irrigation channels (Ostrom and Gardner, 1993) have the same logic and no data, so φ_S is a floor for this mechanism rather than an estimate of it. The externality is that a farmer who leaves takes demand with them, and the provider who then exits, or never enters, was serving everyone (Bassi et al., 2022; Caunedo et al., 2026).

K.3 The Land Market

Every household in a program village started with the same two hectares, so whatever concentration of holdings exists today came from trades, and a trade needs a counterparty. A household that wants to expand has to find one that wants to sell, and in a larger village that is easier (Diamond, 1982). Let m be the share of households that have reached the land market. Those that have not still hold their original plot; among those that have, I assume land has moved to where it is most productive, that is, in proportion to $\tilde{a}_i^{1/(1-\rho)}$.

Under these assumptions village TFP is linear in participation,

$$\frac{Y(m)}{Y(0)} = 1 + m\,(g-1), \qquad \text{so} \qquad \varphi_M = \frac{g-1}{1+\bar{m}\,(g-1)} \, \frac{dm}{d \ln L}, \tag{K4}$$

where g is the factor by which TFP would rise if all of the village's land were reallocated efficiently. Three numbers are needed. For g I take a doubling, in the middle of the range the misallocation literature finds, from about 1.4 in Ethiopia (Chen et al., 2022) to 3.6 in Malawi (Restuccia and

Santaeulàlia-Llopis, 2023). The level of concentration then says how far the village has got. With land among the households that trade as dispersed as a doubling implies, a top-decile share of 0.30 means $\bar{m} = 0.25$, so a quarter of households have traded; the same numbers imply a Gini of 0.36 against 0.38 observed, a check the setup was not fitted to. Finally, the top-decile share rises by 0.115 per log point of the workforce, which the same setup converts into $dm/d\ln L = 0.135$. Together, $\varphi_M = 0.108$; with g at the ends of the literature's range it is 0.070 or 0.197.

As with shared inputs, the land market is the one market I can see. Hired labor at planting and harvest, credit, and the traders who buy the crop all work better with more participants, and none of them is measured here.

K.4 What Is Left

The two measured parts sum to $\varphi_S + \varphi_M = 0.165$, and to between 0.09 and 0.30 across the ranges above. Against the $\varphi = 0.70$ the model carries that leaves $\varphi_R = 0.54$, and against the point estimates of Section 8 more. I cannot split the residual with the data I have, so I say what it plausibly contains. Part of it is the unmeasured shared inputs and thicker markets just listed, which belong to the two channels above and would raise their floors. The rest is knowledge. A larger village is more likely to contain an exceptional farmer, and farmers copy the practices of their most successful neighbors (Foster and Rosenzweig, 1995; Conley and Udry, 2010). The census offers a hint that scale does operate at the top: the agricultural employers of larger settlements, the farmers who run the larger operations, have 0.23 more years of schooling per log point of settlers, while the schooling of self-employed farmers and of farm employees does not move (Table K1). To give a sense of magnitude, suppose farmers' practices converge a fraction λ of the way to the best farmer's and productivity has a Pareto upper tail, so that the best of L draws improves with L at a rate set by the tail. At the tail I measure in land holdings (1.8, the rank-minus-one-half slope of Gabaix and Ibragimov, 2011), this channel would fill the residual on its own at $\lambda \approx 0.77$, that is, if most of the gap to the best farmer closes.[26] Over the decades these villages have existed, on soils the settlers did not know, that is not implausible, but it is not something these data establish, and I read the residual as the sum of all of the above rather than as evidence for any one of them.

One property of the measured parts is worth carrying back to the main text. A tractor or a mill matters most where a village is on the margin of having one at all, and participation in the land market cannot exceed one, so both measured channels are strongest among small villages and fade in large ones. This is a reason to treat the constant φ that the aggregate experiments of Section 11.2 apply to the crowded inner islands as an upper bound.

[26] With $\tilde{a}_i = a_i^{1-\lambda} a_{(1)}^{\lambda}$ and land in proportion to $\tilde{a}^{1/(1-\rho)}$, the channel's part of φ is $\gamma_a \lambda / ((1-\lambda)\zeta_h)$.

Table K1: Settlement size, the mechanism outcomes, and the parts of φ they size

Outcome	Mean	Coef. on $\ln Z_v$	Per log point of L	Part of φ
Any tractor (0/1)	–	0.070	0.35	0.033
Any rice mill (0/1)	–	0.060	0.30	0.023
Top-decile share of owned land	0.30	0.023	0.115	0.108
Gini of owned land	0.38	0.013	0.065	(check)
Number of tractors	2.24	1.020	2.28	(entry)
Number of rice mills	1.20	0.220	0.92	(entry)
Number of formal agricultural firms	0.33	$0.180^{\dagger}$	2.73	(entry)
Years of schooling, agricultural employers	–	0.230	1.15	(hint)
Years of schooling, self-employed and employees	–	≈ 0	≈ 0	(hint)
Shared inputs, φ_S				0.056
Land market, φ_M				0.108
Measured, $\varphi_S + \varphi_M$				0.165
Residual to $\varphi = 0.70$, φ_R				0.535

Note: Coefficients are from regressions of each outcome on log individuals initially placed with the controls and fixed effects of Section 8.1; counts are in levels, presence indicators are linear-probability coefficients, and schooling is years among agricultural workers by employment status. The response per log point of the workforce divides by the first stage 0.200 and, for counts, by the sample mean. The parts of φ use (K3) at $\lambda_t = 1.10$ and $\lambda_m = 1.08$, and (K4) at a reallocation gain of a factor of two; the Gini is the check described in the text, the counts are consistent with free entry but not used, and the schooling rows are the hint about the residual. † not significant at the 10 percent level.

L Measurement of Model Inputs

This appendix describes how I construct each measured input to the model of Section 9. I describe the value-added demand shares, the three trade frictions, the income anchors, the land and urban endowments, the test of the top-level substitution elasticity, and the Engel slopes used in the extension of Section 11.2.

L.1 Value-Added Expenditure Shares and the Consumption Crosswalk

The model's demand shares are measured in value-added terms, while the survey measures spending, so I develop a crosswalk that translates one into the other. Because the rural households the shares should describe are those of the transmigration villages, I use the 2002 SUSENAS consumption module and restrict the sample to rural households on the Outer Islands, dividing their spending between food, services, and goods.

The difficulty in converting this to a value-added measure is that the local services bundle service labor together with purchased intermediates. I measure the split from the Economic Census cost schedules of "household-scale food-service producers." Raw and auxiliary materials, which I assume are predominantly locally supplied food, account for 85 percent of intermediate spending, and fuel, transport, and utilities make up the rest. The labor value-added share of output, ℓ, runs from 0.42 to 0.54 across regions and establishment scales, and I take 0.48 as the baseline.

The crosswalk assigns the labor content ℓ of own-village service spending to service demand and moves the remainder into the food share. That remainder also contains the capital value added of

the service producer and non-food intermediates such as fuel, transport, and utilities, so assigning all of it to food overstates α_a and understates α_n relative to a complete value-added decomposition; the crosswalk preserves total spending but reclassifies those components toward food. Pricing those components at the village-gate food price rather than at delivered city-good prices strengthens the negative effect of market access on income slightly, since the farm-gate food price falls with distance while delivered city-good prices rise. Appendix N.9 bounds how much this matters: repricing the measured non-food portion at delivered city-good prices attenuates the income gradient by under 10 percent, and moving ℓ changes the results little.

Classifying every service not explicitly consumed in a city as locally produced gives an own-village share of 0.904, but the survey cannot tell where a service was produced, so this treats every locally purchased service as locally produced and is an upper bound. At $\ell = 0.48$ the crosswalk evaluated at that bound puts the outer-island rural shares at $(\alpha_a, \alpha_n, \alpha_u) = (0.694, 0.076, 0.230)$. At the estimated $s^{\text{own}} = 0.533$ they are (0.667, 0.103, 0.230), and these are the values the model carries. A higher sourcing share shifts spending from services toward food, so it lowers α_n and raises α_a, while α_u is untouched because city goods do not pass through the crosswalk.

In the baseline model I measure the value-added shares on outer-island rural households and apply them to every rural location, inner islands included. Appendix N re-measures the entire demand block described above in the 2012 SUSENAS on both a national sample and the outer-island rural sample. Re-estimating the sourcing share on each of these leaves the untargeted income gradient close to its baseline value, so the negative relationship between connectivity and income is not driven by a particular snapshot of household preferences.

The urban budget shares are not measured from the survey; the aggregate accounting identities imply them (Section 9.2), so the survey provides an overidentifying check. I measure the urban food share two ways. Running the same crosswalk on outer-urban households gives 0.522, which is what urban households actually spend. Taking the rural shares instead and moving them to urban income levels, using the relationship between budget shares and expenditure estimated in Appendix L.7, gives 0.560. The second is the object the model's assumption predicts, since each location's shares are fixed at its own income level, and it sits within roughly 3 percentage points of the accounting-implied $\beta_a = 0.595$.

L.2 Trade Frictions

I measure the three sector-specific trade frictions separately. The food and rural service frictions come from an accounting exercise and from the relationship between prices and travel times, and the city-good friction is built as a combination of the other two.

I measure κ_a and κ_n from the 1993 wave of the Indonesia Family Life Survey community and facility module. For each community surveyed, I observe one-way travel times and fares to 9 destinations.[27] I also observe local wages, doctors' fees, and food prices. The median community

[27] The destinations are the nearest bus terminal, market, public telephone, post office, health facility, and bank,

daily wage is Rp 2,800 and the median rice price Rp 625 per kilogram. Travel cost enters throughout as $\tau = 1 + t$, with t the one-way time in hours, which is the model's own geometry convention and makes the iceberg factor equal one at zero distance. Every exponent below is specific to this metric and to hours as the unit.

Person-embodied services, $\kappa_n = 0.90$. A unit of the rural service is a unit of labor, so delivering it means a person travels, and the friction is that trip's cost as a share of the value the trip carries. The survey lets me price this two ways, and Table L1 reports both.

Route 1 prices a day of service labor delivered to the city. For a round trip of one-way time t and one-way fare f against a daily wage w, the share of a day's gross value consumed in transit is

$$\text{cost share} = \underbrace{2t/8}_{\text{time}} + \underbrace{2f/w}_{\text{fare}}, \tag{L1}$$

with the workday set at 8 hours, so an hour in transit costs $w/8$ and the wage cancels from the time term. Where the cost share reaches one the day trip is uneconomic, and a weekly variant amortizes one round trip over 6 workdays instead. Route 2 prices an urban service consumed by a villager. For the doctor and hospital rows of the facility roster, the cost share is the round-trip travel cost, fare plus transit time valued at the village wage, divided by the transaction value, with the private doctor's posted consultation charge of Rp 3,750 as the denominator.

The two give the same picture. A day trip to the district capital consumes about 40 percent of a day's value and is uneconomic for a quarter of villages. The province capital is infeasible as a day trip for 72 percent of villages, and even a 6-day stay leaves it uneconomic for 38 percent. A doctor visit costs 15 percent of a consultation's value at the median and a hospital visit 40 percent.

I estimate the exponent by pooling every destination pair within a community and fitting $\ln(\text{iceberg}) = \kappa_n \ln(1+t)$ with community fixed effects and standard errors clustered by community. Route 1 gives $\kappa_n = 0.94$ (s.e. 0.02) across all 1,812 delivery pairs, and Route 2 gives 0.89 (s.e. 0.05) across the 995 doctor and hospital pairs. These are conceptually distinct measurements, one denominated in a day's wage and borne by the provider who travels out, the other in a service fee and borne by the buyer who travels in, and they agree. Because there is no spatial price gradient to discipline κ_n, that agreement is the discipline instead. I take $\kappa_n = 0.90$.

Two things bound how much weight the exact figure carries. First, the mapping from a trip's cost share c to the model's iceberg factor is not unique. The headline 0.94 uses $\ln(1 + c)$, which is defined for every pair including the infeasible ones; the mapping implied by the delivered-share normalization, $-\ln(1 - c)$, is defined only on feasible trips, and because the trip cost is a fixed fare plus a time cost linear in t rather than a power law, on those trips the fitted exponent runs between 1.6 and 2.1 without an intercept and returns toward 0.9 once an intercept absorbs the fare, while the semi-log form that matches the fare-plus-time structure most closely gives 0.88 per hour. Second, because the mechanism runs on village-to-city service trade, matching the friction's level

and the subdistrict, district, and province capitals.

at the trips that carry that trade gives a range of roughly 0.4 to 1.2. Both regression estimates and the baseline sit inside it.

Table L1: Person-embodied service frictions from trips (IFLS 1993)

Trip (community medians)	t (h)	fare (Rp)	cost share	pairs	κ_n (s.e.)
Route 1: delivering a day of service labor (cost share $= 2t/8 + 2f/w$)					
Kabupaten capital, day trip	0.50	600	0.40		
Province capital, day trip	2.50	2,375	infeasible		
Province capital, weekly stay	2.50	2,375	0.38		
Pooled fit				1,812	0.94 (0.02)
Route 2: consuming an urban service (cost share = round-trip cost / fee)					
Doctor visit	0.17	200	0.15		
Hospital visit	0.42	500	0.40		
Pooled fit				995	0.89 (0.05)
Baseline					0.90

Note: Trip rows are community medians across the 312 communities. Cost shares are the trip's cost as a share of the value it carries, computed village by village and reported as medians over the villages for which the trip is feasible. Route 1 values travel time at the median daily wage of Rp 2,800; the province-capital day trip is infeasible for 72 percent of villages, and the weekly row amortizes one round trip over 6 workdays. Route 2 divides the round-trip cost by the private doctor's consultation fee; at the seventy-fifth percentile the doctor and hospital shares are 0.28 and 0.81. The pooled fits regress ln(iceberg) on $\ln(1+t)$ across every destination pair, feasible or not, with community fixed effects and standard errors clustered by community. Across alternative fee denominators the Route 2 estimate ranges from 0.69 to 1.27.

Food, $\kappa_a = 0.05$. Food is homogeneous and the villages are net exporters, so spatial arbitrage sets the farm-gate price at the market price net of transport, $p_{a,v} = \tau_v^{-\kappa_a} p_a$. The survey brackets the friction from two sides (Table L2).

The upper side comes from freight. The same round trip that delivers a day of service labor can instead carry a 50-kilogram rice load, and the round-trip passenger fare over the load's value at the measured rice price is 4 percent to the district capital and 16 percent to the province capital. Those are an order of magnitude below the person-embodied frictions from the identical trips. Converting them to the model's exponent, $\kappa_a = \ln(1+\text{cost})/\ln(1+t)$ per village, gives medians of 0.10 to the district capital and 0.12 to the province capital. Both are upper bounds, for two reasons. The 50-kilogram headload is the smallest commercially meaningful lot, since a motorbike carries several times that and a truck far more, spreading the same trip over more value. And cargo tariffs run below the passenger fares I use here.

The lower side comes from prices. Regressing village rice prices on $\ln(1+t)$ with province fixed effects and standard errors clustered by district gives gradients of -0.006 (s.e. 0.051) to the nearest market, -0.035 (s.e. 0.022) to the district capital, and -0.004 (s.e. 0.015) to the province capital. All are statistically zero and, if anything, faintly negative. That is the farm-gate pattern for a cheap-to-ship staple that villages produce themselves, and it is inconsistent with a friction anywhere near the freight bound.

The baseline $\kappa_a = 0.05$ sits inside this interval, in the lower half toward which the null price

gradients point and below the 0.10 to 0.12 freight bound. The moment system disciplines it from both sides within [0.024, 0.12] (Appendix N.7).

City goods, $\kappa_u = 0.203$. The city good is a composite, neither a single shipped box nor a single carried service, so rather than give it a friction of its own I build κ_u from the two frictions I do measure. It is a bundle of physical goods such as fuel, vehicles, clothing, durables, toiletries, and medicines, which ship as freight at the food friction κ_a, together with a smaller set of person-embodied town services, formal health and education, which move with people at the service friction κ_n. Under Cobb–Douglas within the bundle the delivered-price index is $\tau^{s_g\kappa_a + s_s\kappa_n}$, so the representative iceberg exponent is the consumption-share-weighted average of the two. In SUSENAS rural households the bundle is 82 percent physical goods and 18 percent person-embodied services, which gives $\kappa_u = 0.82 \times 0.05 + 0.18 \times 0.90 = 0.203$ (Table L2). It ranges from roughly 0.18 to 0.22 across the service-definition and network-utility scenarios.

Building it this way has three consequences. It uses only the two frictions I measure directly. It makes the goods side a mild upper bound, since manufactured imports have higher value-to-weight than the rice on which κ_a is measured. And because it comes from these primitives, κ_u does not depend on σ_u. In the model κ_u prices a real trade channel rather than only the cost of living, because city goods are supplied by the endogenous urban block. The gap between the person friction of 0.90 and this composite goods friction of 0.203 is the asymmetry the mechanism runs on, and Appendix N.8 shows what breaks when either friction is set to the other's value.

Table L2: Goods frictions from prices and freight

Measurement	Statistic	Value
Food, κ_a: homogeneous, village-exported; farm-gate $p_{a,v} = \tau_v^{-\kappa_a} p_a$		
Freight accounting, kabupaten capital	round-trip fare / 50 kg rice value	0.04
Freight accounting, province capital	round-trip fare / 50 kg rice value	0.16
Rice-price gradient, nearest market	$d \ln p_a / d \ln(1+t)$, province FE	−0.006 (0.051)
Rice-price gradient, kabupaten capital	”	−0.035 (0.022)
Rice-price gradient, province capital	”	−0.004 (0.015)
Baseline		$\kappa_a = 0.05$
City goods, κ_u: a bundle of physical goods and person-embodied services		
Physical goods (fuel, vehicles, clothing, durables)	ship as freight, $\kappa_a = 0.06$	share 0.82
Person-embodied services (health, education)	move with people, $\kappa_n = 0.90$	share 0.18
Representative city good	$s_g\kappa_a + s_s\kappa_n$	$\kappa_u = 0.21$

Note: The freight accounting prices the round-trip passenger fare against a 50-kilogram rice load valued at the community's measured rice price. Rice-price gradients are estimated at the village level, in the same $\tau = 1 + t$ units as the service frictions. The city-good friction is a consumption-share-weighted mix of the goods and service frictions, $\kappa_u = s_g\kappa_a + s_s\kappa_n$, with the goods and service shares (s_g, s_s) measured on SUSENAS rural households. Across the service-definition and network-utility scenarios κ_u ranges from 0.19 to 0.23.

L.3 Income Anchors

I measure the rural and urban per-capita income anchors on the same outer-island samples as the demand shares, and the urban-to-rural ratio is 1.75. Location-level incomes are observed inputs to the inversion, so the anchors do only one job. The rural anchor converts the village-GDP series into the model's wage units, which is the normalization of Section 9.2.

L.4 Land Endowments and the Household-Holdings Gap

Transmigration villages carry their program land allocations. In the baseline, every program village is assigned the mean recorded labor and land endowment of the program: any village-level noise in the endowments would appear in the model as an endowment cross-section the data do not contain, and a (statistically insignificant) negative relationship between market access and the amount of land a program village has can magnify the negative relationship between market access and income by a noticeable amount. In the data both program moments are *invariant* to controls for recorded placement size and land per worker, so the common scale is the convention the data support. Outer villages carry agricultural land from the 2003 agricultural census collapsed to the village level, matched to the geometry for 74 percent of locations (85 percent population-weighted), with district-median land-per-worker imputed to the remainder and an upper cap at 1.5 times polygon area. The census concept is *household* landholdings, which misses estate, plantation, and fishery agriculture. The gap appears as populated locations, with median population around 2,400, that have near-zero recorded household land, and they are concentrated in the Sumatran and Sulawesi plantation belts and the fishery provinces.

These locations would become spuriously cheap service suppliers in the model. With no measured land, agriculture cannot absorb their labor, so service wages fall toward zero, and because a CES competition aggregate is set by its cheap tail they would dominate the service market far beyond their size. Lower-winsorizing land-per-worker at 0.02 hectares per person, which touches 7,178 locations or 12 percent of the outer sample, removes the artifact and moves the outer-island aggregate agricultural share onto its census value. I report floors of 0.05 and 0.10 as robustness checks. The transmigration villages themselves are land-rich under the household concept, at a median of 0.50 hectares per person against 0.30 for outer locations, which is consistent with the program allocation.

L.5 Urban Endowments

The endogenous urban block requires the same endowment pair as the villages. Urban workforces are location populations from the settlement layer, floored at 10 workers for a small number of tiny administrative units. Urban agricultural land comes from the same 2003 agricultural census. About 7,200 of the 8,759 administratively urban locations match directly, with the district median imputed to the remainder, and the functionally urban locations of Section 9.2 are ordinary location polygons

that carry their own census land. The same land-per-worker floor applies. Urban agricultural shares are far from zero, at 0.26 for the median administrative city and 0.36 for the functionally urban locations, which is why the census land matters. It is what lets the model's cities produce the food their measured employment says they produce, and lets the national food account close without phantom imports (Appendix M.4).

L.6 The Top-Level Substitution Elasticity

The Cobb–Douglas assumption across the three consumption groups is a testable restriction, and the same survey that measures the shares tests it. The test conditions on total expenditure throughout, so it disciplines the price margin of demand and not the Engel slopes, which are measured from the expenditure variation the test holds fixed (Appendix L.7). Under CES(σ_c) over food, services, and city goods, the log ratio of the city-good to the food expenditure share is taste $-(\sigma_c - 1)\ln(P_u/P_a)$. City goods carry a higher trade friction than food, so their price rises relative to food in more isolated villages, and that makes the restriction's sign sharp. Under Cobb–Douglas the share ratio does not respond to isolation at all. Under price-elastic demand it tilts away from city goods as they grow dearer, sharply so at $\sigma_c = 2$, the value at which the road counterfactuals' rural gains would largely vanish.

In the 2002 SUSENAS I match household shares over the (a, n, u) groups to travel times at the ten-digit village code, and regress the log share ratio on travel time to the nearest city with log total-expenditure controls, district fixed effects, and survey weights. Table L3 reports the results. The coefficient is statistically indistinguishable from zero and, if anything, of the opposite sign to the price-elastic prediction, and the implied σ_c sits at or just below one on every specification. Cobb–Douglas is therefore not rejected, while the price-elastic value $\sigma_c = 2$, which predicts $b_\tau = -0.05$ per hour, is rejected at roughly four standard errors. The result survives restricting the city-good group to core tradables, which removes administered-price health, education, and telecommunication services, and it passes a placebo on those administered services. The design's contribution is to test the top-level restriction using within-country variation in isolation in a developing economy.

Table L3: The top-level substitution elasticity σ_c

Specification	b_τ (s.e.)	implied σ_c (s.e.)	N
$\ln(s_u/s_a)$, all rural	+0.0087 (0.0144)	0.83 (0.29)	33,462
$\ln(s_u/s_a)$, outer islands	+0.0107 (0.0150)	0.79 (0.30)	18,982
core goods/food, excl. durables	+0.0021 (0.0183)	0.96 (0.37)	33,449
core goods/food, incl. durables	+0.0062 (0.0190)	0.88 (0.38)	33,449
placebo: administered services/food	−0.0268 (0.0349)	n/a	26,092

Note: 2002 SUSENAS households, 97 percent of them rural, matched to travel times at the ten-digit village code. Each row regresses the stated expenditure-share object on travel time to the nearest city, controlling for log total expenditure, with district fixed effects and survey weights. b_τ is the travel-time coefficient and σ_c is its conversion through the measured delivered-goods price gradient. A value of $\sigma_c = 2$ implies $b_\tau = -0.05$ per hour. Regressing each share level separately rather than the ratio gives implied values of σ_c of 0.95, 1.00, and 1.05 for city goods, food, and services. The placebo row uses administered-price services, whose delivered price does not vary with isolation.

L.7 The Engel Slopes and the Extended Demand System

This section describes the demand system used in the extension of Section 11.2. I take in turn the measured Engel slopes, the estimated curvature, the preferences behind them, the anchoring convention, and the evidence on functional form.

Slopes. The slopes come from the same 2002 SUSENAS sample as the share levels. For each of the three consumption groups I regress the household's budget share on log per-capita expenditure with desa fixed effects, survey weights, and district-clustered standard errors, which is the Working-Leser form. The slope is the level response of the share at the rural mean, and I measure the curvature of the Engel curve separately below. Table L4 reports the estimates. The raw slopes are -0.166 for food, $+0.037$ for local services, and $+0.129$ for the city bundle per log point of expenditure, and they sum to zero by construction.

Table L4: The measured Engel slopes

	Food	Services	City bundle
Budget share	0.586	0.179	0.235
Engel slope b_g^{raw}	-0.1655	$+0.0367$	$+0.1288$
	(0.0041)	(0.0029)	(0.0042)
Value-added slope b_g	-0.1553	$+0.0265$	$+0.1288$
Observations		34,423	

Note: 2002 SUSENAS rural households; desa fixed effects, survey weights, district-clustered standard errors. The value-added row applies the crosswalk transform of equation (L2).

The extension's goods are value-added concepts while the survey measures spending on final goods, so the raw slopes pass through the same crosswalk as the levels (Appendix L.1). This is the empirical counterpart of the mapping described below, under which PIGL demand over final goods produced with Cobb–Douglas intermediates implies PIGL demand over value added with the same curvature, so the crosswalk moves the slopes between the two systems and leaves ε alone. A fraction f of marginal service spending is service labor and the remainder is food ingredients, with $f = \alpha_n/e_n = 0.72$ the value-added retained fraction at the estimated sourcing share and $e_n = 0.143$ the raw service expenditure share, so

$$b_n = f\, b_n^{\text{raw}}, \qquad b_a = b_a^{\text{raw}} + (1-f)\, b_n^{\text{raw}}, \qquad b_u = b_u^{\text{raw}}, \tag{L2}$$

which gives the slopes the extension carries, $(b_a, b_n, b_u) = (-0.155,\ 0.027,\ 0.129)$. The slopes are measured across households spanning several log points of expenditure, so the income movements the counterfactuals generate stay far inside the measured support, and the shares stay far from their bounds at every reported solution. The urban comparison of Appendix L.1 provides an overidentifying check, since extrapolating the rural anchors along the estimated curve across the measured urban-to-rural income gap reproduces the accounting-implied urban food share within roughly 3 percentage points; the type intercepts nearly lie on one common curve.

The curvature ε. I estimate the curvature with the estimator of Fan et al. (2023), regressing the log budget share on log per-capita expenditure with desa fixed effects and district-clustered standard errors. Under the PIGL form with the asymptotic food share set to zero, the convention of Fan et al. (2023), the log food share is linear in log expenditure and this recovers ε as the negative of its slope. With a positive asymptotic food share the same regression identifies $\varepsilon(\bar{s}_a - \omega_a)/\bar{s}_a$ instead, which at the anchoring described below equals the Working-Leser slope divided by the share and carries no separate information about ε; the curvature is then identified only from how the slope changes with expenditure, which is the functional-form evidence at the end of this section. On the same sample it gives -0.340 (s.e. 0.011) for food, so $\varepsilon = 0.34$, with luxury elasticities of $+0.19$ for services and $+0.51$ for the city bundle (Table L5). Because the estimator is identical, the comparison with the literature is direct. Fan et al. (2023) obtain 0.33 by least squares and 0.395 instrumenting expenditure on Indian household data, Boppart (2014) obtains 0.18 to 0.25 on United States household data, and Alder et al. (2022) obtain 0.34 to 0.90 on long-run sectoral shares for four rich countries. Indonesian rural households in 2002 sit almost exactly on the Indian value.

Table L5: The PIGL curvature and the functional-form evidence

	Food	Services	City bundle
Panel A. Share elasticities and the curvature			
$d\ln s_g/d\ln e$	-0.340	$+0.188$	$+0.511$
	(0.011)	(0.020)	(0.014)
Implied ε		0.340 (0.011)	
Panel B. Functional form			
Quadratic term	-0.022	-0.003	$+0.025$
	(0.004)	(0.004)	(0.004)
Per-equation exponent κ_g	$+0.28$	-0.14	$+0.43$
Common κ, unrestricted		$+0.31$ (0.042)	
Common κ, restricted to $(-1, 0]$		0.00 (the PIGLOG corner)	

Note: 2002 SUSENAS rural households, desa fixed effects, survey weights, district-clustered standard errors. The elasticity panel regresses log budget shares on log per-capita expenditure, the estimator of Fan et al. (2023); ε is the negative of the food coefficient. The functional-form panel reports the quadratic Working-Leser coefficients and the profiled NLS exponent described in the text.

Preferences. The extension's budget shares follow from the PIGL indirect utility

$$V_\ell(p,e) = \frac{1}{\varepsilon}\Big(\frac{e}{A_\ell(p)}\Big)^{\varepsilon} - \sum_g \nu_g \ln p_{g,\ell}, \qquad A_\ell(p) = \prod_g p_{g,\ell}^{\omega_{g,\ell}}, \tag{L3}$$

by Roy's identity, with the composite CES price indices of Section 9.1 standing in for the service and city-good prices by two-stage budgeting. Two properties of this class are the reason I use it rather than another non-homothetic system. It aggregates, so a representative household per location is still a legitimate object even though demand is non-homothetic at the household level. And combined with Cobb–Douglas intermediates it maps demand over final goods into demand over

sectoral value added with the same curvature ε. The second property is what lets me estimate ε on household spending over final goods and then use it in a model whose sectors are value added, and it is not a generic feature of non-homothetic demand (Herrendorf et al., 2013). The aggregation property is established in Boppart (2014). The value-added mapping is not: Boppart (2014) is explicit that moving from final goods to sectoral value added requires a model of the production process, and Herrendorf et al. (2013) show that the two perspectives differ in general. What delivers it here is the Cobb–Douglas intermediate structure of the crosswalk (Appendix L.1), under which the value-added shares are affine in the same $x^{-\varepsilon}$ as the final-good shares, the same route taken by Fan et al. (2023) and Peters et al. (2026). The share equations are

$$s_{g,\ell} = \omega_{g,\ell} + \nu_g \, x_\ell^{-\varepsilon}, \qquad \textstyle\sum_g \omega_{g,\ell} = 1, \quad \sum_g \nu_g = 0, \tag{L4}$$

with x_ℓ real expenditure per capita. The logarithmic price term in (L3) is the $\sigma_c \to 1$ case of Boppart (2014)'s price aggregator, the case the price test of Appendix L.6 supports, so the system is Cobb–Douglas in prices and PIGL in income, and the main model's homothetic demand is the case with the loadings ν_g set to zero. The loadings are pinned by the measured slopes at the rural mean, so the Engel slope at income x is $b_g \, (x/\bar{x}_r)^{-\varepsilon}$ and flattens as households grow richer; at the urban mean the implied slopes are about 14 percent flatter than the rural ones. The intercepts $\omega_{g,\ell}$ anchor each location at its observed baseline budget and are the asymptotic budget shares; at the rural mean they are (0.21, 0.18, 0.61), summing to one automatically because the slopes sum to zero. The PIGLOG/Working-Leser system is the $\varepsilon \to 0$ limit.

The anchoring convention and what identifies it. The invariance of the baseline to the demand parameters rests on this convention, so I state it precisely. The slopes are identified within desa, where the fixed effects compare richer and poorer households in the same village facing the same prices. The estimate is therefore a pure income response, uncontaminated by the spatial variation in delivered prices that the model prices through its own trade structure. The fixed effects absorb the cross-village variation in share levels, and the location-specific intercepts $\omega_{g,\ell}$ are the model counterpart of those fixed effects, since every location sits at its observed budget at its own baseline income and the Engel curve governs movements away from that point. Applying the within-village slope to counterfactual changes in a location's income is the experiment the estimate matches. Applying it to baseline level differences across locations would extrapolate onto the margin the design deliberately excludes.

Two things follow. First, the baseline equilibrium, the inversion, and every targeted moment are invariant to (b_g, ε), which operate only when counterfactuals move real incomes. Second, the extension says nothing about whether baseline budget composition varies with income across locations. What that omits is bounded by the curse's income spread times the food slope, roughly a percentage point of budget shares across program villages, and the 3-percentage-point urban check above bounds it across types.

This construction involves one approximation. Because shares are non-linear in expenditure, the aggregate share of a population with dispersed incomes is not the share of the household at mean income, and the exact aggregate carries a term in the dispersion of income within the location (Boppart, 2014; Peters et al., 2026). The model has one household per location while the slopes applied to it are estimated across households, so the income margin here is the response of a household at the location's mean income rather than the exact aggregate response. The two differ by a factor below one that falls with within-location income dispersion, and at the estimated ε that factor is roughly 3 to 5 percent for log income standard deviations of 0.5 to 0.7. Because the anchoring already fixes each location's baseline budget at its observed value, the approximation affects the slope and not the level, and it scales the income margin uniformly rather than distorting comparisons across experiments or across values of φ. At the magnitudes the counterfactuals produce it moves the reported aggregates by well under a hundredth of a percentage point.

Functional form. Write κ for the exponent on expenditure, so that shares are affine in x^κ. The Boppart (2014) class is $\kappa = -\varepsilon \in (-1, 0]$ and PIGLOG is $\kappa = 0$. Three pieces of evidence discipline the choice.

A quadratic term added to the Working-Leser regressions is statistically significant for food (-0.022, $t = -5.7$) and the city bundle ($+0.025$, $t = 6.5$) and zero for services. That says the magnitude of the food-share elasticity rises with income, from 0.20 at the rural tenth percentile to 0.37 at the ninetieth. The Boppart (2014) form with a positive asymptotic food share implies the opposite direction, a magnitude that falls as the share approaches ω_a, so the disagreement is with the form's curvature rather than with a constant-elasticity restriction. Profiled nonlinear least squares over the full share system puts the unrestricted exponent at $\kappa = +0.31$ (s.e. 0.04), and constrained to the admissible region of the Boppart (2014) class the estimate is the PIGLOG corner $\kappa = 0$. The three candidate fits differ by two tenths of a percent of residual variance end to end, with per-equation exponents of $+0.28$ for food, -0.14 for services, and $+0.43$ for the city bundle, so no one-parameter member fits the curvature exactly.

The choice is second order. Predicted demand responses coincide at every income move the counterfactuals in the paper generate; the three fits separate only at income changes an order of magnitude larger, where for a change of roughly 60 percent in a location's income the predicted food-share response is $+0.17$ under the operative form, $+0.15$ under PIGLOG, and $+0.12$ under the unrestricted exponent. I carry the Boppart (2014) form because it is the standard in this literature and because its estimator delivers the cross-country comparison above. Its shares are bounded only locally, as for any system affine in x^κ, but they stay far from their bounds over the measured support and at every reported solution. Because the system is anchored at each location's observed budget and the loadings are pinned by the measured level slopes, the income response the counterfactuals run on is the measured slope at baseline whatever the curvature; ε governs only how that slope flattens as income rises, which is why the three fits coincide at counterfactual-sized income moves.

Welfare. With (L3) in hand welfare is exact, but the tables need a deflator and there are two candidates that sit far apart. The price aggregator $A_\ell(p)$ in (L3) weights prices by the asymptotic shares $\omega_{g,\ell}$, while the reported real incomes deflate by the observed baseline shares $\bar{s}_{g,\ell}$. I therefore set out which one the tables use. Write $\Delta x_\ell = \Delta \ln e_\ell - \sum_g \omega_{g,\ell} \Delta \ln p_{g,\ell}$ for the change in the model's own real expenditure. Equating utility across price vectors, the equivalent variation at baseline prices is

$$\Delta_\ell^{EV} = \frac{1}{\varepsilon} \ln\Big[e^{\varepsilon\, \Delta x_\ell} - \varepsilon \sum_g \left(\bar{s}_{g,\ell} - \omega_{g,\ell}\right) \Delta \ln p_{g,\ell}\Big], \tag{L5}$$

using $\nu_g x_\ell^{-\varepsilon} = \bar{s}_{g,\ell} - \omega_{g,\ell}$ at the baseline. Expanding to first order, the ω terms cancel exactly and (L5) collapses to $\Delta \ln e_\ell - \sum_g \bar{s}_{g,\ell} \Delta \ln p_{g,\ell}$, which is the object the tables report. The asymptotic shares therefore never enter the welfare measure, and the familiar result that the first-order welfare effect of a price change is its baseline expenditure share survives the non-homotheticity intact. What the curvature adds is the second-order term $\varepsilon\big(\Delta x_\ell B_\ell - \frac{1}{2} B_\ell^2\big)$, with $B_\ell \equiv \sum_g (\bar{s}_{g,\ell} - \omega_{g,\ell}) \Delta \ln p_{g,\ell}$, which at the estimated ε and the price movements these counterfactuals generate is on the order of thousandths of a percentage point, against reported effects of tenths of a percent.

M Computation

This appendix documents how the 69,225-location joint equilibrium is represented, solved, selected, and audited, and how the estimation is organized around the solver.

M.1 Exact Dense Representation

The service demand system requires, at every price vector, each buyer's competition index over all rural origins and each origin's demand sum over all buyers, bilateral objects on a 69,225 × 60,466 travel-cost matrix: all 69,225 locations as buyers against the 60,466 locations as potential service origins. The stored matrix keeps every location column regardless of classification; under the baseline functional-urban classification the 11,868 reclassified location polygons are removed from the rural supplier pool by a mask (their variety weight is set to zero and their price left inert), so the active rural-origin set is the 48,598 rural villages while the matrix dimensions are classification-invariant.

The natural way to economize would be sparsity: truncate each buyer's sourcing at a travel-time cutoff and bound the discarded tail. That shortcut fails here, and measurably so: computing the exact tail share of every buyer's competition index at baseline prices shows that at every cutoff up to 8 hours one-way, population mass at distance grows faster than the $\tau^{-\kappa_n(\sigma_n-1)}$ kernel decays, leaving median tail shares above 1 percent and, for isolated buyers, near 100 percent. The implementation therefore uses the exact dense system: the two required orientations of the travel-cost matrix are stored as single-precision arrays of about 17 gigabytes each, the kernel powers are built in memory, and a full evaluation of all demand aggregates is two matrix-vector products taking 1 to 2 seconds

on a workstation. There is no truncation error to defend.

Two conventions matter for exactness. The fast-marching travel-time matrix is numerically asymmetric (median 0.2 percent, ninety-ninth percentile 3.3 percent), so each location's statistics are computed from its own matrix row (buyers' competition indices from buyer rows, origins' demand sums from origin rows), which is why both orientations are stored; this makes the general-equilibrium solver reproduce the per-village estimation battery exactly when the rest of the world is frozen. And transmigration-village terms are carried in dedicated double-precision blocks outside the single-precision products, so the objects the moments are built on never see single-precision rounding.

M.2 Fixed Point, Continuation, and Audits

Given the kernel operators, the equilibrium is computed by a damped two-operator iteration: service prices imply competition indices and demand sums; each location's two-equation system (sorting, agricultural first-order condition, service clearing) then solves in closed form up to a scalar per location, vectorized across all locations; updated wages imply updated prices. Adaptive damping starts at one half. The convergence tolerance of 5×10^{-8} in maximum log-wage change is set just above the single-precision noise floor of the matrix products (about 1.5×10^{-6} relative, which propagates to roughly 3×10^{-8} in the fixed-point residual); tightening beyond it wastes iterations wandering the noise floor. A full solve from a warm start converges in 50 to 90 iterations, about 11 minutes.

Selection follows the protocol of Section 9.1: continuation in φ from the externality-free equilibrium, in 13 steps refined near the top of the path, warm-starting each step. At the endpoint the audit battery runs: a perturbation restart (log-wage noise injected and re-solved) and a full cold start with no continuation. At the calibrated point both return the identical equilibrium, with maximum log-wage discrepancies of 4×10^{-8} for the perturbation and 3×10^{-7} for the cold start and no entry of the rural wage vectors or the urban price and wage vectors differing beyond 10^{-6}; the audit covers both blocks of the joint system. Uniqueness at system scale cannot be certified in general; what is certified is that the reported equilibrium is not an artifact of its construction path, and the audit battery re-runs at every reported point, including every counterfactual.

M.3 Estimation and the Inversion

The estimation never solves for the fundamentals inside its loops; it exploits the architecture instead. The inversion does not involve θ, but it does involve the demand shares, so each trial of the sourcing share s updates $(\alpha_a, \alpha_n, \beta_n)$ through the crosswalk and the adding-up identity and re-inverts the non-program world once (one fixed point over competition indices at observed incomes), cached for the trial. The composition moment is then evaluated in the joint equilibrium of Appendix M.4: the agricultural scale T_a concentrates out by bisection on the program agricultural-employment level, every joint solve warm-starting from its predecessor so that an evaluation costs a handful of

damped iterations rather than a cold start, and the outer root closes the structural transformation moment in s. The two-parameter variant of Appendix N.2 adds an inner root that sets θ on the composition moment before the outer root closes the income moment.

A small-open alternative evaluates the program side as a *battery*: each of the 1,147 program villages solves as its own two-equation system (sorting, agricultural first-order condition, service clearing) against the fixed world objects it faces (its demand access, its competition index, its farm-gate price, its city-good price index), which takes seconds and remains the frame for the single-village experiment of Section 11.1. The program's general-equilibrium footprint is visible in the estimate itself: the battery moment is exact at a sourcing share of 0.67 rather than the joint frame's 0.53, which is why the baseline moment is evaluated in the joint equilibrium, the frame every counterfactual solves in. Table M1 reports the targeted moments and the estimated parameters.

Table M1: Targeted moments and estimated parameters

Moment	Data (s.e.)	Model
Agricultural employment share, TM	0.648	0.648
b_{ST}, reduced-form measure, full sample	+0.0053 (0.0009)	+0.0053
Imposed or set:	$\theta = 12.5$ (Bryan and Morten 2019); $\sigma_n = \sigma_u = 5$	
Estimated:	$s_{\text{own}} = 0.533$ (T_a concentrated out)	

Note: Moments computed identically in data and model: non-agricultural employment share (b_{ST}) on the reduced-form measure of market access, island and cohort fixed effects, district-clustered standard errors, 1,147 transmigration villages; the model moment is computed in the joint equilibrium (Appendix M.4). Estimation on the functional-urban inverted world with the city-good price index from the inverted urban block.

M.4 The Joint Equilibrium, Certification, and the Food Account

Counterfactuals solve the full two-block system: rural service prices and city variety prices update jointly, with the city-good gravity system over 20,627 suppliers stored in half precision (about 6 gigabytes) and applied through single-precision chunked products; the whole solve holds in about 45 gigabytes and converges in 30 to 70 damped iterations. Certification re-solves the entire economy forward from the recovered fundamentals and compares to observed incomes: mean absolute error 0.6 percent across rural locations and 0.6 percent across urban ones. The food market, closed by Walras' law in the solver, is audited by direct accounting at the certified baseline: total food spending matches total farm-gate revenue to 0.2 percent (within the audit's own reconstruction tolerance), iceberg losses on shipped food are 0.45 percent of production, and the sell-side farm-gate convention, exact for net exporters, covers locations holding 89 percent of the rural population. The net-importing remainder, concentrated among the most transformed, best-connected locations, including a third of the program villages, is mispriced by at most the arbitrage band $2\kappa_a \ln \tau$, under 4 percent where the importers cluster.

N Quantitative Robustness

This appendix collects the robustness of the quantitative results to every input that could reasonably be measured differently: the mapping of the externality estimate from the data into the model, the elasticity pair (θ, σ_n), the variety weighting, the demand block's vintage and sample, the food friction, the friction asymmetry, the bundle labor share, the outer land floor, the externality outside the program villages, and the food-market closure. The organizing finding is that the model's untargeted income gradient is an *envelope*: most inputs move offsetting margins, since a change in a measured share moves both village income and the cost-of-living weights and the recalibration of the internal parameters absorbs the remainder, so the gradient moves by far less than the inputs do.

N.1 The Estimator Run on Model-Generated Data

The model imposes $\varphi = 0.70$ through the mapping $\varphi = \hat{\beta} - \gamma_a$, which reads the 2SLS coefficient on agricultural headcount as the model's coefficient on effective agricultural labor. Because workers sort on comparative advantage, instrumented headcount and effective labor need not move one for one, so I validate the mapping by simulation. Each of the 1,147 program villages is solved in the per-village battery frame with its *recorded* settlement size W_v in place of the homogenized mean, under two land conventions (the common mean $\bar{H}$ held fixed, and the recorded two-hectare-per-household allocation), in the baseline world and in the re-pinned $\varphi = 0$ world; the sorting scale T_a is held at its baseline value throughout, since a common constant cannot absorb the cross-village variation that identifies the regressions. On the model-generated data I then run the estimator of Section 8 exactly: log agricultural output on log agricultural headcount $\pi_{a,v} L_v$, instrumented by $\ln W_v$, island and cohort fixed effects, standard errors clustered by village, with and without the land control.

Table N1 reports the results. The benchmark column, output regressed on effective labor, returns $\gamma_a + \varphi$ exactly, a mechanical identity that certifies the construction. The paper's estimator returns 0.979 and 0.968 against the true $\gamma_a + \varphi$, with Anderson–Rubin sets [0.977, 0.983] and [0.964, 0.974], and in the externality-free world returns 0.316 and 0.318 against the true γ_a: with the land control in place the estimator recovers the structural coefficient to within 2 to 4 percent in both worlds, and does not manufacture an externality where none exists. The one exception is instructive: with recorded land and no land control, the externality-free world returns 0.482, with an Anderson–Rubin set that excludes the true γ_a, because recorded land co-moves with settlement size (0.17 log points per log point of persons placed) and the omitted land loads onto the labor coefficient. That is the model's version of why the empirical specification controls for planted area.

The deviation from the truth is the Roy-selection wedge itself, and the table names it. Because $\ln Q_a = (\gamma_a + \varphi) \ln Z_a + (1 - \gamma_a) \ln H_v$ holds identically, the estimates satisfy $\hat{\beta} = (\gamma_a + \varphi)$ times the 2SLS of $\ln Z_a$ on headcount once the land control is partialled out, and the passthrough column

shows effective labor moving 0.96 to 0.98 for one with instrumented headcount at the calibrated externality, and 1.04 to 1.05 for one at $\varphi = 0$. The sign tracks the sorting response: at $\varphi = 0.70$ agricultural returns are nearly constant in labor, larger villages sort *into* agriculture (first stages above one), and the marginal entrants dilute average farming ability, while at $\varphi = 0$ diminishing returns push the marginal workers out and the sector concentrates on those best suited to it. The wedge is small because composition scales with $1/\theta$; at $\theta = 12.5$ even sizable sorting responses move average ability by a few percent. Its direction means the empirical $\hat{\beta}$, if anything, understates $\gamma_a + \varphi$, so the calibrated $\varphi = 0.70$ is conservative on this margin as well.

Two features of the table deserve interpretation. First, the model's first stages (1.44 to 1.80 at the calibrated externality, 0.63 to 0.67 at $\varphi = 0$) are far from the empirical 0.18 (s.e. 0.07): in the model the settlement size is the village's permanent labor endowment, while in the data five decades of demographic churn separate placement from the observed workforce, so only part of the initial size differences survives. Because 2SLS divides the reduced form by the first stage, the recovered coefficient is invariant to this attenuation; the exercise validates the estimand mapping, not the persistence process, and whether the churn is itself selective is an exclusion question outside the model, treated by the balance and mediation evidence of Section 8. Second, dropping the land control moves the records-row estimate from 0.968 to 1.006, the model's version of the with-and-without-area bracket of Section 8, though the two pairs are not the same experiment: the model's land object is the endowment allocation, which in the settlement records moves only 0.17 log points (s.e. 0.03) per log point of persons placed within fixed-effect cells, while the empirical control is chosen planted area.

Table N1: The paper's estimator run on model-generated data

	Truth	Benchmark	IV, land control	IV, no control	First stage	Z_a passthrough
$\varphi = 0.70$, H fixed	1.004	1.004	0.979	0.979	1.439	0.976
			(0.002)	(0.002)	(0.039)	(0.001)
			[0.977, 0.983]	[0.977, 0.983]		
$\varphi = 0.70$, H records	1.004	1.004	0.968	1.006	1.801	0.964
			(0.003)	(0.006)	(0.103)	(0.003)
			[0.964, 0.974]	[0.995, 1.017]		
$\varphi = 0$, H fixed	0.304	0.304	0.316	0.316	0.672	1.039
			(0.001)	(0.001)	(0.011)	(0.002)
			[0.315, 0.317]	[0.315, 0.317]		
$\varphi = 0$, H records	0.304	0.304	0.318	0.482	0.627	1.048
			(0.001)	(0.026)	(0.016)	(0.003)
			[0.317, 0.320]	[0.429, 0.531]		

Note: Each row solves the 1,147 program villages with their recorded settlement sizes W_v in place of the homogenized mean, under the stated land convention and externality, and then runs the 2SLS specification of Section 8 on the resulting model-generated data. That specification regresses log agricultural output on log agricultural headcount, instrumented by $\ln W_v$, with island and cohort fixed effects and standard errors clustered by village reported in parentheses. Brackets report Anderson–Rubin 95 percent confidence sets. Truth is the model's labor coefficient $\gamma_a + \varphi$. The benchmark column regresses output on effective labor Z_a. First stage is the coefficient of log headcount on $\ln W_v$ within the fixed effects. The Z_a passthrough column runs the same 2SLS with $\ln Z_a$ as the outcome. The sorting scale T_a is held at its baseline value in every row.

N.2 Estimating Both Elasticities

The main text imposes $\theta = 12.5$ from Bryan and Morten (2019) and estimates the sourcing share s_{own} alone on the structural transformation slope. Freeing θ as a second parameter in the per-village battery frame of Table N2, with σ_n held at its set value and the income-access slope on the reduced-form measure as the second moment, moves the estimate to $\theta = 16.3$ and $s_{\text{own}} = 0.84$; the composition slope stays matched throughout, with the sourcing share climbing from 0.67 as θ rises. The data's full-sample income slope, -0.0066, is steeper than the model reaches at any θ up to 30, where it attains only -0.0047: the sorting margin does not reproduce the extreme isolation tail, so the second moment is the interior slope that drops the 5 percent most isolated villages, -0.0047, which the model matches at $\theta = 16.3$. Two things follow. First, the freed elasticity lies above the imposed one, just past the upper end of the Bryan and Morten (2019) confidence interval (9.8 to 15.2), so imposing the migration estimate is conservative for the sorting margin and the curse it drives, not generous. Second, the untargeted predictions barely move: the model's own value-added home share is 0.30 at both the imposed and the freed point. The income moment carries a standard error in this appendix and nowhere else; in the main text it is a prediction, never a target.

Table N2: Calibration perturbations

Perturbation	s^*_{own}	b_{ST}	b_Y (full)	b_Y (trim)	χ
Panel A. Own-share re-estimated at each perturbation					
baseline	0.673	+0.0053	−0.0070	−0.0084	0.30
unit weights $\omega_o = 1$	0.904†	+0.0060	−0.0076	−0.0083	0.41
$\kappa_a \times 0.5$	0.904†	+0.0054	−0.0089	−0.0104	0.31
$\kappa_a \times 2$	0.471†	+0.0043	−0.0020	−0.0031	0.30
$\ell = 0.42$	0.603	+0.0053	−0.0070	−0.0084	0.30
$\ell = 0.54$	0.761	+0.0053	−0.0070	−0.0084	0.30
Panel B. Own-share held at 0.673*; friction symmetry broken*					
κ_n set to goods friction	0.673	−0.0009	+0.0045	+0.0045	0.001
κ_u set to person friction	0.673	+0.0053	+0.0040	+0.0021	0.30

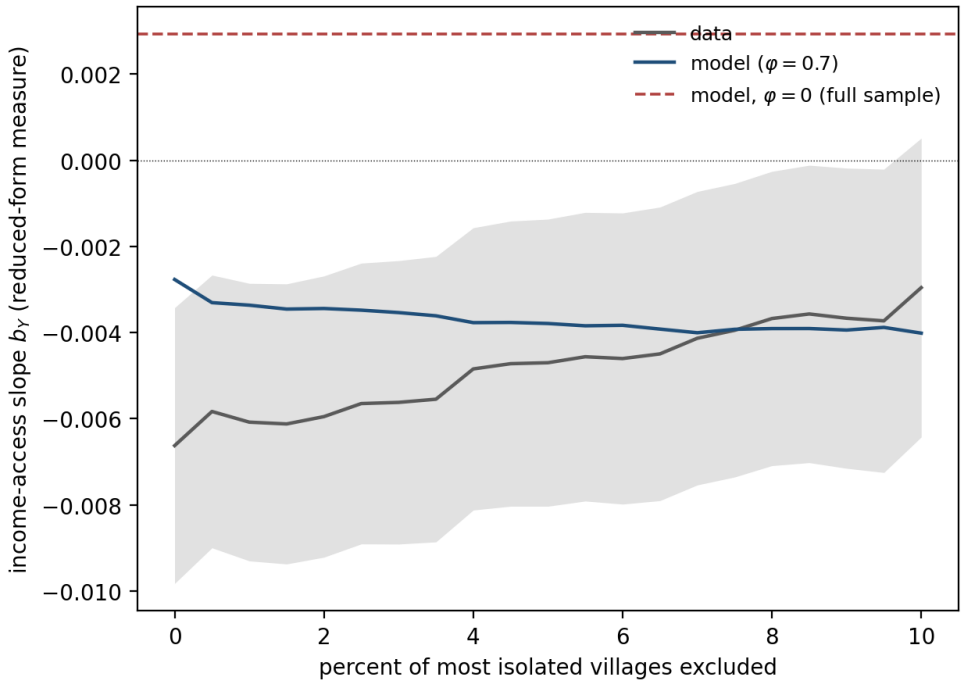
Note: Each row re-solves the model under the stated change to a calibration input. All rows use the per-village frame, in which the own-share estimate is $s^*_{\text{own}} = 0.673$, while the joint-equilibrium estimator of the main text gives 0.533 on its own moment. b_{ST} is the structural transformation slope on the reduced-form access measure, which is the estimation target. b_Y is the income-access gradient on the frozen-kernel measure, reported for the full sample and with the bottom decile trimmed. χ is the model's value-added home share. In Panel A the own-share is re-estimated to hold the target at each perturbation, and rows marked † are those where the estimate reaches a bound of its admissible range $[0.471, 0.904]$ and the target is left slightly off. Panel B holds the own-share at its baseline value and sets the person-embodied service friction κ_n and the goods friction κ_u to each other's values.

N.3 The Isolation Tail

Figure N1 traces how the income slopes in the data and in the model move as the most isolated villages are excluded, from 0 to 10 percent. The data slope flattens as the isolated tail is dropped, because the most remote program villages earn a premium the model does not capture, while the model slope steepens slightly; the two cross near the bottom decile. Section 10.3 shows what the

inversion makes of the premium: it loads into the recovered fundamentals of the most isolated villages, which is where the model's null balance result carries its largest residuals.

Figure N1: Income-Access Relationship as Isolated Villages Are Excluded



Note: The figure plots the relationship between income and market access for transmigration villages in the data and in the model as the villages lowest in market access are excluded, from 0 to 10 percent, with 95 percent confidence intervals for the data estimates. The red horizontal line marks the full-sample model slope at $\varphi = 0$.

N.4 Variety Weights

The variety weights $\omega_o = N_o/\bar{s}$ make a large location a proportionally larger mass of service varieties rather than one oversized variety, which keeps the inversion invariant to administrative aggregation: splitting a location into two polygons changes no equilibrium object. Dropping the weights and giving every location one variety is the unit-weights row of Table N2: re-estimating under unit weights pushes the own-share to the top of its range and leaves the income gradient essentially where it was ($b_Y = -0.0076$ against the baseline -0.0070). In the two-parameter variant of Appendix N.2 the same change moves only θ (to 17.5) along the weakly identified dimension, with σ_n and every untargeted income slope unchanged. The weighting is a reporting convention, disciplined by replication invariance, not a quantitative driver.

N.5 The Urban Classification

Section 9.2 classifies as urban the administratively urban locations plus every location whose centroid falls inside a satellite-defined functional urban area (the GHS-FUA layer). The administrative classification alone produces an implausibly low urban share for Indonesia, 44 percent against roughly 60 percent in aggregate statistics, because it assigns many peri-urban and urban areas on the outskirts of cities to the rural side; the functional definition brings the share to just under 65 percent. Table N3 reports the estimated point under alternative classifications.

Table N3: The estimated point under alternative urban classifications

Classification	Urban pop. share	b_{ST}	b_Y (estimation)	b_Y (reduced form)
GHS-FUA (baseline)	0.65	+0.0136	−0.0128	−0.0050
outcome-based	0.62	+0.0141	−0.0129	−0.0050
FUA + 10 minutes	0.73	+0.0136	−0.0127	−0.0049
administrative +	0.48	+0.0076	−0.0084	−0.0037

Note: Model slopes are computed at $(\theta, \sigma_n) = (12.5, 5)$. For each classification the service world and the urban block are re-inverted at that point, with the city-good friction held at its consumption-weighted value of $\kappa_u = 0.203$ across classifications. Slopes are computed in the per-village estimation frame of Appendix M.3: b_{ST} and b_Y (estimation) are on the model-consistent access measure, and b_Y (reduced form) is on the reduced-form measure of Section 5. The targeted moment of Table M1 is the reduced-form b_{ST} in the joint equilibrium, which is why the b_{ST} here differs from 0.0053; the baseline row's reduced-form joint-equilibrium counterpart appears in Table 8. "Outcome-based" reclassifies near-city locations whose agricultural share falls below the urban 75th percentile. "Administrative +" applies a minimal correction to the administrative classification, reclassifying locations below the urban median agricultural share and within 30 minutes of a city. Data slopes are estimated as in Table 8, with island and cohort fixed effects and district-clustered standard errors.

Every functional definition of urban, satellite-based, outcome-based, or widened by a commuting buffer, delivers slopes of the same sign and similar size, with the structural transformation slope between 0.0136 and 0.0159 and the reduced-form income gradient between −0.0050 and −0.0062: nothing qualitative hinges on where exactly the boundary is drawn, provided functionally urban locations are not counted as rural service suppliers. Under the administrative classification the same qualitative patterns appear at smaller magnitudes: with functionally urban locations left in the rural supplier pool, they absorb part of the city service demand before it reaches the countryside, and the structural transformation slope roughly halves. One caveat applies to all functional layers: the satellite footprint reflects the contemporary built environment rather than a pre-program one, so the classification conditions the model on the observed urban system; the administrative row, whose definition is closest to predetermined, is the relevant bound on what this choice contributes.

N.6 The Demand-Block Envelope

Every survey-based demand parameter can be re-measured at a different vintage or sample. Re-running the inversion and the own-share estimation on the value-added crosswalk of the 2012 SUSENAS, for outer-island rural households and for the national sample, matches the composition slope at every block and leaves the untargeted income gradient and own-village share close to their

baseline values: a decade of preference drift and a change of sample move nothing the paper uses.

The crosswalk itself is not optional, and the accounting shows why. Feeding the model raw expenditure shares instead of their value-added conversion roughly doubles rural own-spending on services, and the economy-wide adding-up identity, which sets the urban service budget from total measured service revenue net of rural own-demand, pays the price. At the raw 2002 shares the implied urban service share collapses from 0.037 to 0.005 and the urban composition gradient inverts (-0.026 against the measured $+0.0147$); at the raw 2012 shares the implied share is negative and the inversion's fixed point does not exist. Raw expenditure weights assign the countryside more service spending than the economy's measured service income can absorb; the crosswalk is an existence condition, not a preference.

N.7 The Food Friction

The trip-cost accounting puts κ_a at 0.05, and the moment system disciplines it from both sides (the κ_a rows of Table N2). At half the measured value the own-share estimate rises to the top of its admissible range and the untargeted gradient steepens slightly ($b_Y = -0.0089$); at double the value the estimate falls to the bottom of the range, the target can no longer be held exactly, and the gradient flattens toward zero (-0.0020). A food friction far from the measured value forces the sourcing share against a bound rather than reproducing the data. The simple model shows what is at stake without this discipline: at stylized parameters, pushing the food friction to 0.10 can extinguish that model's curse entirely (Appendix F), which is why κ_a is measured and its measurement cross-checked against the null staple-price slope.

N.8 The Friction Asymmetry

Section 9.1 states that the gap between the person-embodied service friction ($\kappa_n = 0.90$) and the goods friction ($\kappa_u = 0.203$) is central to the results; Panel B of Table N2 quantifies how, holding the own-share at its baseline value. Setting κ_n down to the goods friction, so that services ship as cheaply as goods, destroys the mechanism's spatial structure: the own-village share collapses toward zero, the structural transformation slope changes sign, and the income gradient turns positive, because geography no longer concentrates city demand on nearby villages. Setting κ_u up to the person friction instead leaves the composition slope untouched but turns the income gradient positive on its own ($b_Y = +0.0040$ against the baseline -0.0070): expensive goods make proximity valuable on the consumption side, and the gains-from-trade channel overwhelms the externality. The measured flatness of spatial goods prices rules the second world out, and the measured cost of moving people makes the first counterfactual; the size of the curse rests on both measurements.

N.9 The Bundle Labor Share and the Own-Consumption Crosswalk

The Economic Census cost structures put the bundle labor share in a band from 0.42 to 0.54, and the value-added crosswalk that turns measured expenditure shares into the model's demand block depends on it. Re-estimating the own-share at the band endpoints (the ℓ rows of Table N2) moves the sourcing estimate as the crosswalked shares shift, from 0.60 to 0.76 in the per-village frame of that table, whose baseline estimate is 0.673 rather than the joint-equilibrium 0.533 of the main text. Every reported moment, targeted and untargeted, is unchanged to the stated precision: the income gradient stays at -0.0070 throughout. The moment that identifies the own-share pins the service value-added share α_n, so the band feeds the estimate and not the predictions. The crosswalk's composition assumption (pricing the ingredients content at the village-gate food price) is the curse-favorable case; repricing the measured non-food portion of the intermediates basket at delivered city-good prices attenuates the real gradient by under 10 percent, the relevant cost-of-living weight being about 0.06.

N.10 The Externality Outside the Program Villages

The outer land floor of Appendix L.4 and the externality outside the program villages do not affect the baseline. Both enter only through the recovered fundamentals, which adjust so that the model reproduces observed outcomes whatever floor or externality is imposed, so no baseline moment, targeted or untargeted, can move. They matter only for how strongly non-program agriculture responds to shocks. Appendix N.12 varies the externality everywhere at once. This subsection instead lets it decline with the size of a location's farm workforce, as the estimates of Appendix I.1 suggest.

Let L_ℓ^{ag} be location ℓ's baseline farm workforce, its population times its observed agricultural employment share, and let $L_0 = 1{,}293$ be that of a program village. The externality is

$$\varphi_\ell = \max\Big\{0,\ \varphi - \lambda \max\big\{0,\ \ln\big(L_\ell^{ag}/L_0\big)\big\}\Big\}, \tag{N1}$$

so it equals the calibrated $\varphi = 0.70$ up to the size of a program village and falls by λ per log point above it, down to zero. Program villages keep $\varphi = 0.70$, the value estimated on villages of their size. I consider λ from 0 to 0.4, which spans the estimated declines of 0.07 to 0.39. At $\lambda = 0.4$ the externality reaches zero at 5.8 times a program village's farm workforce, about 7,400 farm workers, and at $\lambda = 0.3$ it does so at 10 times.

Each value of λ defines a new model, which I recalibrate as I do the baseline. The agricultural productivity of every non-program location is recovered again at that location's own externality, so each economy reproduces observed incomes and agricultural shares. The estimated parameters do not change, because they are estimated on the program villages, whose externality does not change, against recovered price fields that do not depend on the externality. The re-solved baselines stay as close to the data as the calibrated model, with a population-weighted mean absolute gap in

log income across non-program rural locations of 0.5 to 0.7 percent, and the program villages' agricultural employment share stays within 0.0005 of its target. At $\lambda = 0$ the procedure reproduces the results of Section 11.2 exactly. Each location's externality is set at its baseline farm workforce and held there in the counterfactual. This is exact to first order, and the omitted second-order term is under half a percent of the externality's first-order effect on national income.

Table N4 reports the results. The road's national gain stays between 0.73 and 0.74 percent, while the gain from rural productivity growth rises from 0.74 to 0.91 percent, so ignoring the externality overstates that gain by 33 rather than 64 percent. The difference lies in where each shock moves farm labor, because a weaker externality in large locations matters only through the farm labor that moves there. Locations with more than ten times a program village's farm workforce produce 35 percent of farm output. With a constant externality, the road raises their farm labor by 0.18 percent, mostly in large rural locations, at the expense of smaller ones, and those gains offset part of the cost elsewhere. Decay removes these offsetting gains about as fast as it removes costs. Rural productivity growth instead lowers their farm labor by 0.80 percent, because it shifts farming from urban to rural locations and most of their farm output is urban, so decay removes a cost without removing any offsetting gain. One effect works the other way. Locations no larger than a program village keep the full externality, and under decay they shed more farm labor, 2.2 rather than 1.3 percent under productivity growth at $\lambda = 0.3$, which limits how much the gain rises.

Table N4: Aggregate programs with an externality that declines with scale

	Externality		Uniform road	Rural agricultural TFP +5%	
Decline per log point (λ)	Mean φ	Share at zero	National	National	Overstatement
0 (calibrated)	0.70	0%	+0.73	+0.74	64%
0.1	0.54	0%	+0.73	+0.82	48%
0.2	0.41	15%	+0.73	+0.87	39%
0.3	0.34	30%	+0.73	+0.90	35%
0.4	0.30	39%	+0.74	+0.91	33%
Externality off ($\varphi = 0$)	0	100%	+0.86	+1.21	

Note: Changes in national real income, in percent, under non-homothetic demand, from the uniform 10 percent cut in travel times of Table 10 and the 5 percent rise in rural agricultural TFP of Table 11, when the externality declines with the size of a location's farm workforce as in Equation (N1). Each row recalibrates the model at its value of λ. Mean φ is the average externality across locations, weighted by baseline farm workforce, and share at zero is the share of the farm workforce in locations with no externality. Overstatement is the gain without the externality divided by the gain in the row, minus one. The last row re-solves each experiment in the re-inverted externality-free world.

N.11 The Food-Market Closure

The joint model is closed: food is the numéraire, the national food market clears by Walras' law, and there is no open-economy bound left to report. The closure question that the earlier exogenous-city architecture had to bracket is settled by construction and audited by direct accounting (Appendix M.4): the value balance holds to 0.3 percent and iceberg losses are 0.45 percent of

production. What remains of the closure as a modeling choice is the sell-side farm-gate convention, whose scope the same audit quantifies: it is exact for the locations holding 89 percent of the rural population, and for the net-importing remainder it errs by under 4 percent, within the arbitrage band.

N.12 The Road Across Externality Values

Section 10.2 solves the whole economy across the externality under homothetic demand, which isolates the relative-access margin, and Figure 4 there shows the cross-sectional reversal near $\varphi = 0.55$. Two further patterns hold across the range. First, the relative-access margin's aggregate is flat in φ: under homothetic demand the uniform 10 percent road raises national income by about 0.8 percent at every φ, so the externality never taxes this component of the aggregate anywhere in the range. Under the extended demand system of Section 11.2 the aggregate does fall with the externality, from 0.86 percent at $\varphi = 0$ to 0.73 at the calibrated value, and the decline is entirely the income margin. Second, what the externality moves is incidence: the program villages' share of the uniform gains falls with φ, from +0.95 percent at $\varphi = 0$ to +0.64 at the calibrated value (+0.65 at $\varphi = 0.74$), the redistribution that Section 11 documents experiment by experiment.

Table N5: Counterfactual robustness: demand elasticities

	uniform road (national)	program villages
baseline ($\sigma_u = 5$, Cobb–Douglas)	+0.83	+0.64
Panel A. City-good elasticity σ_u		
$\sigma_u = 8$	+0.79	+0.61
$\sigma_u = 12$	+0.75	+0.58
Panel B. Demand margins ($\sigma_u = 5$)		
measured Engel curves	+0.75	+0.53
$\sigma_c = 2$ (rejected)	+0.37	−0.45

Note: Each row re-solves the 10 percent uniform road of Section 11.2 under homothetic demand. The baseline row reproduces the homothetic figure of Section 11.2 and the calibrated-φ values reported in Appendix N.12. Columns report the change in national real income and in the program villages' real income, in percent. Panel A varies the city-good substitution elasticity σ_u, holding the city-good friction at its consumption-weighted value. Panel B sets the top-level substitution elasticity to $\sigma_c = 2$, the value the test of Appendix L.6 rejects. Results under non-homothetic demand are reported in Appendix Table O1.

N.13 The Income Gradient Above the Calibrated Externality

Figure 4 extends the income gradient to $\varphi = 1$. Above roughly $\varphi = 0.8$ the village block changes character. With $g = \gamma_a + \varphi - 1 > 0$, adding farm workers raises the farm wage, and once $g(\theta - 1) > 1$, which at the calibrated values holds for $\varphi > 1 - \gamma_a + 1/(\theta - 1) \approx 0.8$, the farm wage responds to farm employment more strongly than sorting responds to the farm wage. Two things follow in every village. A village with no farmers stays that way, since with no farmers farm productivity is

zero. And in a village with few farmers, losing some lowers farm productivity enough to push out the rest, so a mixed equilibrium with a small farm share is unstable.

The model selects equilibria by continuation in φ from $\varphi = 0$, so each program village stays on the branch it starts on for as long as that branch exists. Above the threshold the branch of a village with almost no farming ends, and the village moves to the equilibrium without farming. The standard solver cannot represent that equilibrium, at which the farm wage is zero, so I solve this range by writing each village's equations as a single equation in its farm share and locating every equilibrium directly; below the threshold this reproduces the standard solver to numerical precision. The number of program villages without farming rises from 83 of 1,147 at $\varphi = 0.8$ to 361 at $\varphi = 1$. They are the best-connected villages, which is why the gradient steepens quickly above 0.8.

Outside the program, the inversion makes each location's observed outcomes an equilibrium at every φ, but above the threshold, for locations with small farm shares, an unstable one. As the program villages change and incomes adjust across the network, those locations move away from their observed outcomes, and the re-solved economy drifts from the observed baseline. The population-weighted mean absolute gap between model and observed log income across non-program rural locations is under 1 percent for $\varphi \leq 0.74$, and 1.4, 4.6, and 12 percent at $\varphi = 0.8$, 0.9, and 1. The points above 0.8 in Figure 4 therefore describe economies that no longer reproduce the observed geography. They place the calibrated value on the curve rather than offer alternative calibrations.

N.14 The Urban-Differentiation Sweep

The city-good elasticity σ_u is set, not estimated, so the results are reported across $\sigma_u \in \{5, 8, 12\}$ with the friction held fixed at its consumption-weighted value ($\kappa_u = 0.203$, Appendix L.2), a construction that does not depend on σ_u. The sweep runs under homothetic demand, where the city-good channel it varies operates. Panel A of Table N5 reports the 10 percent uniform road across the sweep: the national gain falls modestly from +0.83 to +0.75 percent of national income and the program villages' share from +0.64 to +0.58, the entire movement carried by the city-good channel, which alone runs +0.59, +0.53, and +0.48 percent as σ_u rises. No sign changes anywhere in the sweep.

N.15 The Rejected Price Elasticity

Price substitution is the demand margin that could move the road results from the other side, and it is disciplined by measurement (Panel B of Table N5). At the rejected $\sigma_c = 2$ the 10 percent uniform road's national gain roughly halves (to +0.37 percent) and the program villages turn from a gain to a loss (−0.45 percent), because price-elastic demand triggers the same reallocation the externality penalizes; the test of Appendix L.6 rejects the price-elastic case in direction, so at $\sigma_c \approx 1$ this margin is shut. The exercise is evaluated in the homothetic frame, the case its null hypothesis describes. The income margin, the demand margin the data do support, is part of the extended

model of Section 11.2; shutting it there raises the road's national gain from 0.73 to 0.83 percent and the program villages' share from 0.55 to 0.64, and at $\varphi = 0$ shutting it moves the aggregate by almost nothing, which confirms that the tax comes from the externality rather than from the demand system itself.

N.16 The Service-Price Slope

The spatial slope of village service prices is +0.006, estimated from a specification that strips out the delivered-goods and composition content that inflates the broad non-food index to +0.01, and it enters the general-equilibrium calibration (Section 9.2) as a held-out check, not a targeted moment. The comparison between the model's held-out slope and this estimate is informative about structure: the model's village-gate service price is a pure labor cost, so any component of the measured slope that reflects premises costs is a channel the model deliberately omits, and the model's slope should undershoot the data by roughly that component. The housing evidence of Section 7 suggests that such a component exists: housing expenditure rises with market access, with the point estimates attributing most of the rise to prices, though residential prices are an imperfect proxy for commercial premises costs and the price component is imprecisely estimated.

O Additional Counterfactuals

This appendix holds the exhibits that support Section 11 without belonging in it: the distribution of the single-village road effects, the channel decomposition of the uniform road, a connectivity experiment that lowers the cost of distance itself rather than travel times, and the costing of the road network.

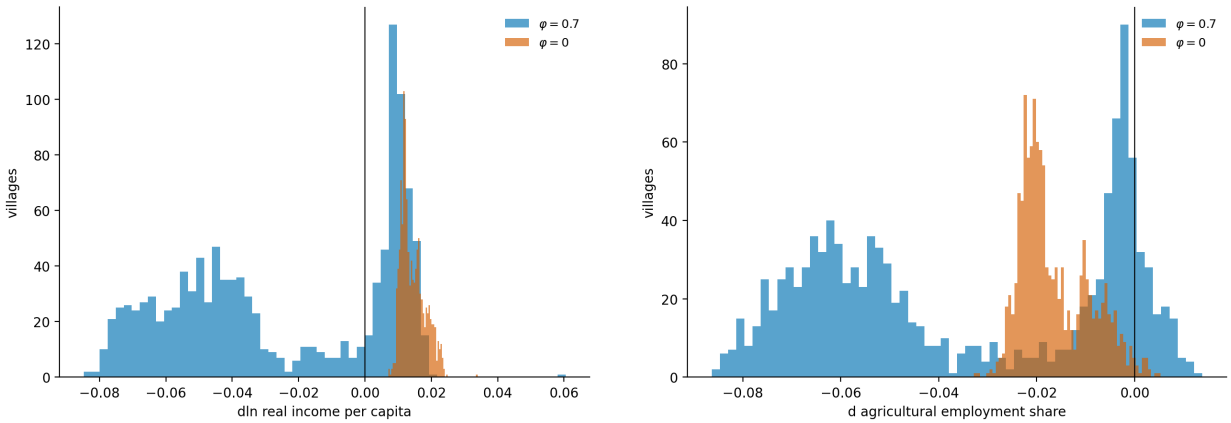
O.1 The Single-Village Road: Distribution of Effects

Figure O1 plots the distribution of the effects of the single-village road of Section 11.1 on income and on the agricultural employment share across the 1,147 experiments, with and without the externality.

O.2 The Uniform Road by Channel and by Group

Table O1 decomposes the uniform 10 percent travel-time reduction of Section 11.2 by trade friction and by group, under both demand systems and with and without the externality. Without the externality the policy is close to distributionally neutral, with the program villages, other rural villages, and the cities all gaining about 0.9 percent, close to the national figure. With the externality the other rural villages keep their full gain at 0.9 percent, while the program villages keep only 0.6 percent and the cities 0.6 percent. A policy that improves every link in the country by the same proportion still has unequal incidence once the externality is in place, and the locations that

Figure O1: Effects of a Road Built to a Single Program Village



Note: The figure plots the distribution of effects across the 1,147 separate experiments, each of which cuts one village's travel times by 10 percent and re-solves that village against the unchanged rest of the economy. The left panel plots the change in log real income per capita and the right panel the change in the agricultural employment share, with the externality ($\varphi = 0.7$) and without it ($\varphi = 0$).

give up the most are the isolated settlements the program created. That redistribution holds at every value of φ the model can solve (Appendix N.12) and across different values of the elasticity of substitution across city varieties (Appendix N.14).

Table O1: A uniform 10 percent travel-time reduction, by channel and by group

	$\varphi = 0.7$ (measured)				$\varphi = 0$			
	TM	Outer rural	Urban	National	TM	Outer rural	Urban	National
Non-homothetic demand								
farm-gate channel (κ_a)	+0.1%	+0.1%	0.0%	0.00%	+0.1%	0.0%	0.0%	+0.01%
service channel (κ_n)	0.0%	+0.5%	+0.1%	+0.22%	+0.4%	+0.5%	+0.2%	+0.25%
urban-variety channel (κ_u)	+0.4%	+0.4%	+0.5%	+0.52%	+0.4%	+0.4%	+0.7%	+0.59%
all channels	**+0.6%**	**+0.9%**	**+0.6%**	**+0.73%**	**+0.9%**	**+0.9%**	**+0.9%**	**+0.86%**
Homothetic demand								
all channels	+0.6%	+1.0%	+0.7%	+0.83%	+1.0%	+1.0%	+0.8%	+0.85%

Note: Population-weighted changes in log real income, in percent, from a uniform 10 percent cut in all travel times. Other rural is every non-program rural village, on the inner and outer islands. Urban real income is deflated by the city block's own consumption index. The upper block solves the model under non-homothetic demand and the lower row under homothetic demand. Each channel row re-solves the equilibrium with one friction's kernel system shocked at a time. National is the income-weighted total across every location, gross of construction costs, which are applied in Appendix O.4. The $\varphi = 0$ columns re-solve the identical experiments in the re-inverted externality-free world.

O.3 Cutting the Friction Parameters

The experiments of Section 11 all cut travel times; a natural closing question asks what a different kind of progress does. Roads scale travel times inside a fixed cost-of-distance technology, whereas cutting each friction parameter κ by 10 percent rotates the technology itself: it lowers delivered costs in proportion to $\ln \tau$, so the reduction grows with distance and the experiment is isolation-

targeted by construction, what containerization, logistics, and telecommunication do rather than what asphalt does. Table O2 reports the results.

Table O2: Cutting the friction parameters ($\kappa \times 0.9$)

	$\varphi = 0.7$ (measured)				$\varphi = 0$	
Friction cut	TM	Outer rural	Urban	National	TM	National
Non-homothetic demand						
κ_a (farm-gate)	+0.2%	+0.1%	0.0%	0.00%	+0.1%	+0.02%
κ_n (services)	−0.4%	+1.0%	+0.2%	+0.51%	+0.7%	+0.61%
κ_u (goods)	+1.3%	+1.1%	+1.3%	+1.49%	+1.3%	+1.66%
all three	+1.1%	+2.2%	+1.4%	+2.01%	+2.1%	+2.30%

Note: Population-weighted changes in log real income, in percent, from cutting each friction parameter by 10 percent. Each row cuts one parameter, and the final row cuts all three. All rows are solved under non-homothetic demand. The $\varphi = 0$ columns re-solve the identical experiments in the re-inverted externality-free world.

Both friction cuts deliver large gains. The goods-friction cut is worth +1.5 percent of national income, because the rotation concentrates its cost reductions on long-haul inter-island trade, exactly where the value lies; this is the margin that logistics, shipping, and port technology operate on. The service-friction cut is worth +0.5 percent, delivered by any technology that lets services reach buyers with less person-travel, from communications to cheaper passenger transport. These experiments carry no resource cost and rotate the cost-of-distance schedule rather than cutting travel times, so their gross gains are not directly comparable with the costed road program of Appendix O.4. Cutting all three frictions together raises national income by 2.0 percent against 2.3 percent in the externality-free economy, the same overstatement the other aggregate experiments display.

The service-friction cut also has the most distinctive incidence of any experiment in the paper: it is the one that harms the program villages on average while the other groups gain on average. Other rural villages and the cities gain 0.2 to 1.0 percent while the transmigration villages lose 0.4 percent, their isolation protection stripped as remote service delivery spreads exposure to city demand everywhere. The externality is what makes them lose, since without it the same cut turns their loss into a gain of 0.7 percent. It is also the one experiment that visibly compresses the cross-sectional curse, with the trimmed gradient flattening by 6 percent; the group averages and the gradient change are consistent with the curse shrinking by pulling the protected down rather than lifting the exposed up, though they do not by themselves establish that no exposed village is lifted.

O.4 The Cost of the Road Network

The gains reported in Section 11.2 are annual flows, while roads are stocks that depreciate. Pricing the 10 percent experiment against Indonesian cost data therefore requires three inputs: a per-kilometer cost for the works that raise speeds, a mapping from those works to the 10 percent travel-time reduction, and an annualization. The natural anchor for the first two is the Gertler

et al. (2024) study of Indonesian road maintenance. For costs, they price periodic maintenance at the ROCKS overlay benchmark of roughly $93,000 per kilometer, and they measure pavement deterioration of 6.3 percent per year, which implies an overlay is needed about every 8 years. For the mapping, their estimates of how speed falls with roughness imply that resurfacing a rough road raises achievable speeds on the order of 10 to 20 percent; I read the 10 percent experiment as the conservative end of what clearing the roughness backlog delivers, applied across the network. The annualization then prices resurfacing the whole network on its replacement cycle. Indonesia's road network is roughly 540,000 kilometers across national, provincial, and district classes, and resurfacing it at $93,000 per kilometer, spread over the 8-year cycle, comes to about $6.3 billion per year, or 0.68 percent of Indonesia's roughly $920 billion GDP. Against the model's +0.73 percent gross flow at the measured externality this is a benefit-cost ratio just above one. That gross flow is itself generous to the road program, because the uniform experiment improves every link in the country, including the city-to-city connections that a maintenance budget of this size would not purchase, so the benefits it delivers are an upper bound on what the costed program buys. The same arithmetic at the realistic works mix documented by Collier et al. (2015), whose unit costs for developing-country road works run well above the overlay benchmark, gives a ratio below one, and at the betterment costs actually incurred on eastern Indonesian national roads, roughly $850,000 per kilometer under EINRIP for the widening and realignment that raises design speeds once pavements are already smooth, the ratio falls well below one half.

The published claim that road maintenance in Indonesia pays two to three times its cost sits well above any of these ratios. Part of that is the cost side, since that claim prices the cheapest category of work. The rest is that the two exercises measure different quantities. A local estimate asks what happens to a district when its own roads improve and its neighbors' do not, which includes any activity it draws away from them; the experiment here improves every road at once, so activity that simply moves between districts nets out of the national total. Establishing how much of the gap that accounts for would require running the single-district experiment across all districts and comparing the sum of those local effects with the aggregate reported here. I do not do that, and I therefore do not attribute the gap to either source.